Foundations of Cat Theory

Objects, Morphisms, and Feline Structures

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Abstract

Let $\mathcal{C}$ be a category with an initial object. A *$\mathcal{C}$-cat* is a tuple $K = (H, T, (L_\alpha)_{\alpha \in A})$ consisting of an initial object H (the *head*), a terminal object T (the *tail*), and a small family of objects L_α (the *legs*), connected by the canonical morphisms $H \to L_\alpha \to T$. We define morphisms of $\mathcal{C}$-cats by commutative diagrams and develop the resulting theory systematically: categories of cats, kittens, and headless, tailless and legless cats; feline functors and natural transformations; limits and colimits; adjunctions and equivalences; a feline Yoneda embedding; monoidal and enriched cats; iterated and two-dimensional cats; feline fibrations; and representation and decomposition theorems. We introduce the Chloe Condition, Margaret Compatibility and Gideon Decompositions, and prove a CMG Correspondence Theorem relating them.

Every commutativity condition imposed on a morphism of $\mathcal{C}$-cats is satisfied automatically, because each such condition concerns a pair of parallel morphisms with initial domain or terminal codomain. Consequently the category $\mathsf{Fel}(\mathcal{C})$ of $\mathcal{C}$-cats is equivalent to the classical category $\mathsf{Fam}(\mathcal{C})$ of small families of objects of $\mathcal{C}$, that is, to the free completion of $\mathcal{C}$ under small coproducts. We make this reduction explicit at the outset and trace its consequences throughout. All results are proved in full. None of them is new in substance: each is identified, where possible, with the standard statement of which it is a special case or a reformulation.

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Preface

This monograph is concerned with a single definition and its consequences. The definition (Definition 3.2) concerns tuples consisting of an initial object, a terminal object, and a family of objects of a fixed category, together with the morphisms that join them. We call such a tuple a *cat*, and the subject developed from it *cat theory*.

The reader is entitled to know at the outset how much is at stake. The answer is: very little. The head of a cat is an initial object, and the tail is a terminal object. Initial and terminal objects are unique up to unique isomorphism, and any two parallel morphisms out of an initial object, or into a terminal object, coincide. It follows that the head and the tail of a cat carry no information beyond their existence, that the morphisms joining them to the legs are uniquely determined, and that every commutative diagram we impose on these morphisms commutes for trivial reasons. Cat theory is therefore the theory of small families of objects of a category, a subject which has been understood for a long time. We prove this reduction in Theorem 3.13 and use it freely afterwards.

Why, then, write a monograph? The purpose of the present work is architectural rather than substantive. We have attempted to construct a complete, internally consistent theory, with its own terminology, its own named conditions, and its own correspondence theorem, while adhering throughout to the standards of rigour one would expect of a research monograph in category theory. Every result is stated with its hypotheses, every proof is given in full, every type signature has been checked, and every proposed generalization that fails is accompanied by a counterexample. Where a feline statement is a special case or a reformulation of a classical one, we say so explicitly. We do not claim novelty for any result in this book.

The reader who wishes to evaluate the theory quickly may consult the dictionary in Appendix A, which records for each feline notion the classical notion to which it is equivalent.

Organization. Chapter 1 gives an overview and states the principal results. Chapter 2 collects the categorical background. Chapter 3 introduces $\mathcal{C}$-cats, their morphisms, and the category $\mathsf{Fel}(\mathcal{C})$, and proves the Fundamental Theorem identifying $\mathsf{Fel}(\mathcal{C})$ with $\mathsf{Fam}(\mathcal{C})$. Chapter 4 treats kittens, litters, cats satisfying the Chloe Condition, and cats lacking a head, a tail or legs. Chapter 5 studies feline functors and natural transformations.

Chapters 6 to 8 concern limits and colimits, adjunctions and equivalences, and Yoneda-type results respectively; the last of these introduces Margaret Compatibility. Chapter 9 treats monoidal and enriched cats, Chapter 10 feline fibrations and several derived constructions, and Chapter 11 iterated, gaited and two-dimensional generalizations. Chapter 12 proves representation and decomposition theorems and introduces Gideon Decompositions. Chapter 13 proves the CMG Correspondence Theorem relating the Chloe Condition, Margaret Compatibility and Gideon Decompositions, together with a recognition theorem for categories of cats. Chapter 14 concludes.

Logical dependence. Chapters 2 and 3 are required for everything that follows. Chapters 4 to 7 may be read in order; later chapters depend on them. Chapter 13 uses Chapters 4, 6 to 8 and 12 and, in its last section, Chapter 9.

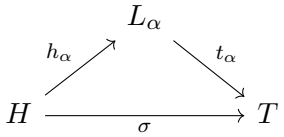
CHAPTER 1

Introduction

1. The basic definition

Let $\mathcal{C}$ be a category possessing an initial object. The central object of study in this monograph is the following.

Definition (informal version of Definition 3.2). A *$\mathcal{C}$-cat* is a tuple $K = (H, T, (L_\alpha)_{\alpha \in A})$ in which H is an initial object of $\mathcal{C}$, T is a terminal object of $\mathcal{C}$, A is a small set, and each L_α is an object of $\mathcal{C}$, together with the canonical morphisms



for $\alpha \in A$.

The object H is the *head*, T the *tail*, and the L_α the *legs* of K; the morphism σ is the *spine*. A morphism of $\mathcal{C}$-cats consists of a reindexing function between the sets of legs and morphisms between corresponding heads, legs and tails, subject to the commutativity of a family of squares (Definition 3.7). The resulting category is denoted $\mathsf{Fel}(\mathcal{C})$.

2. An immediate reduction

The first substantive observation of the theory, which we record here so that it is not obscured by what follows, is that the head and tail of a cat carry no information.

Theorem (Fundamental Theorem of Cat Theory; Theorem 3.13). *Let $\mathcal{C}$ be a category with an initial object. Every commutativity condition in the definition of a morphism of $\mathcal{C}$-cats is automatically satisfied. The functor $\mathsf{Fel}(\mathcal{C}) \to \mathsf{Fam}(\mathcal{C})$ which forgets the head and the tail is fully faithful, and it is an equivalence of categories if and only if $\mathcal{C}$ has a terminal object. It is an isomorphism of categories if and only if $\mathcal{C}$ has exactly one initial object and exactly one terminal object.*

Here $\mathsf{Fam}(\mathcal{C})$ is the category of small families of objects of $\mathcal{C}$, which is, by a classical observation, the free completion of $\mathcal{C}$ under small coproducts. The Fundamental Theorem converts every subsequent question about $\mathcal{C}$-cats into a question about families. We have nevertheless chosen to state and prove

all results in feline terms, partly in order to verify that the feline formalism is internally consistent and partly because certain features of families (for instance, the rôle played by the initial object of $\mathcal{C}$ in the existence of limits) become visible only when the head is recorded explicitly.

3. Principal results

We list the principal results. The labels refer to the body of the text.

THEOREM A (Limits and colimits; Theorems 6.8 and 6.14). *Let $\mathcal{C}$ have an initial and a terminal object. For every small category $\mathcal{J}$, the category $\mathsf{Fel}(\mathcal{C})$ has all limits of shape $\mathcal{J}$ if and only if $\mathcal{C}$ does. The category $\mathsf{Fel}(\mathcal{C})$ has all small colimits if and only if $\mathcal{C}$ has all small connected colimits. The hypothesis that $\mathcal{C}$ have an initial object is essential for the first statement.*

THEOREM B (The feline adjoint string; Theorem 7.3 and Propositions 7.5 and 7.6). *The leg functor $\pi\colon \mathsf{Fel}(\mathcal{C}) \to \mathbf{Set}$ sits in a string of adjunctions*

$$\Delta_0 \dashv \pi \dashv \Delta_1 \dashv \Gamma,$$

where Δ_0 (resp. Δ_1) sends a set S to the cat with S legs, each of which is a head (resp. a tail), and Γ sends a cat to the set of global elements of its legs. The functor Δ_0 has a left adjoint if and only if $\mathcal{C}$ is pointed, and Γ has a right adjoint if and only if $\mathcal{C}$ is pointed or the global-sections functor $\mathcal{C}(1,-)\colon \mathcal{C} \to \mathbf{Set}$ has a right adjoint. For $\mathcal{C} = \mathbf{Set}$ the string extends to five functors and cannot be extended further.

THEOREM C (Feline Yoneda; Theorems 8.5, 8.6 and 8.10). *The functor $\mathbb{Y}\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{PSh}(\mathcal{C})$ sending a cat to the coproduct of the presheaves represented by its legs is fully faithful. Its essential image consists of those presheaves whose category of elements is a disjoint union of categories with terminal objects. Under $\mathbb{Y}$, the leg functor is evaluation at the head and the points functor is evaluation at the tail.*

THEOREM D (Reconstruction; Theorem 7.17). *If $\mathsf{Fel}(\mathcal{C}) \simeq \mathsf{Fel}(\mathcal{D})$, then $\mathcal{C} \simeq \mathcal{D}$.*

THEOREM E (Computations; Theorems 7.21 and 11.4 and Corollary 7.23). *For every small category $\mathcal{J}$ there is an equivalence $\mathsf{Fel}(\mathbf{Set}^{\mathcal{J}}) \simeq \mathbf{Set}^{\mathcal{J}^{\triangleright}}$, where $\mathcal{J}^{\triangleright}$ is $\mathcal{J}$ with a terminal object adjoined. Consequently $\mathsf{Fel}(\mathsf{PSh}(\mathcal{D})) \simeq \mathsf{PSh}(\mathcal{D}^{\triangleleft})$ for every small $\mathcal{D}$, the n-fold iterated category of cats over the terminal category is equivalent to $\mathbf{Set}^{[n-1]}$, and in all cases the leg functor corresponds to evaluation at the adjoined object.*

THEOREM F (Feline toposes; Theorem 10.27). *If $\mathcal{E}$ is an elementary topos with small products, then $\mathsf{Fel}(\mathcal{E})$ is an elementary topos. Its subobject classifier is a two-legged cat whose legs are the subobject classifier of $\mathcal{E}$ and the terminal object.*

THEOREM G (CMG Correspondence; Theorem 13.2). *Let X be an object of $\mathcal{C}$. Every Gideon Decomposition of X satisfies the Chloe Condition and is Margaret compatible. Conversely, if X admits at least one Gideon Decomposition, then a cocone from a cat satisfying the Chloe Condition to X is a Gideon Decomposition if and only if it is Margaret compatible; Gideon Decompositions of X are unique up to unique isomorphism. Neither hypothesis can be omitted.*

THEOREM H (Recognition; Theorem 13.10). *A locally small category $\mathcal{E}$ is equivalent to $\mathsf{Fel}(\mathcal{D})$ for some category $\mathcal{D}$ with an initial and a terminal object if and only if $\mathcal{E}$ has small coproducts, every object of $\mathcal{E}$ admits a Gideon Decomposition, and the full subcategory of connected objects of $\mathcal{E}$ (equivalently, of objects satisfying the Chloe Condition) has an initial object. The existence of a terminal object in $\mathcal{E}$ is implied by these conditions, a Gideon Decomposition being a cat and therefore having a tail.*

4. The CMG framework

Three notions recur in the later chapters. The *Chloe Condition* (Section 5) is a condition on a cat: that each of its legs be a connected object of $\mathcal{C}$. *Margaret Compatibility* is a condition on a cocone from the legs of a cat to an object X: that every morphism from a connected object into X factor uniquely through exactly one leg (Section 6). A *Gideon Decomposition* of X (Section 7) is a cat satisfying the Chloe Condition together with a coproduct cocone from its legs to X. The CMG Correspondence Theorem (Theorem G), proved in Chapter 13, shows that, in the presence of at least one Gideon Decomposition, Margaret Compatibility characterizes Gideon Decompositions among cocones from cats satisfying the Chloe Condition, and that Gideon Decompositions are essentially unique. We identify these notions with familiar ones in Remark 13.7: the Chloe Condition is connectedness of legs, Margaret Compatibility is the invertibility of a comparison of restricted Yoneda presheaves, and Gideon Decompositions are decompositions into connected components. The three notions are introduced where they arise (in Chapters 4, 8 and 12 respectively), recur in Chapters 6, 7, 9 and 10, and are brought together in Chapter 13.

5. Relation to the literature

We make no claim of novelty. The category of small families $\mathsf{Fam}(\mathcal{C})$, its description as the free coproduct completion, the family fibration $\mathsf{Fam}(\mathcal{C}) \to \mathbf{Set}$, and the elementary facts concerning its limits, colimits, connected objects and Yoneda embedding are classical; general references are [**1**, **4**, **6**, **10**, **12**]. The characterization of coproduct completions by connected objects, which underlies Theorem H, is likewise well known and elementary. We have included complete proofs because the feline formulation of each statement differs superficially from the classical one and because we wished the account to be self-contained. Where we have not located a statement

in the literature in precisely the form given, we have nevertheless refrained from describing it as new: in every such case the statement is an elementary consequence of the Fundamental Theorem together with well-known facts.

The terminology of this monograph (heads, tails, legs, kittens, the Chloe Condition, Margaret Compatibility, Gideon Decompositions, and so on) is new, but new terminology does not constitute new mathematics. The reader is referred to Appendix A for a translation of every feline term into standard language.

6. Conventions

Our set-theoretic and categorical conventions are fixed in Section 1. We write composition of morphisms in applicative order, $g \circ f$ or gf for "first f, then g". Diagrams are said to commute if all composites of parallel paths between any two vertices agree. Throughout, *the* category $\mathcal{C}$ is a fixed ambient category; the standing hypotheses on it are stated at the beginning of each chapter.

CHAPTER 2

Categorical preliminaries

This chapter fixes conventions and collects the elementary facts on which the theory rests. Nothing in it is new; for general background we refer to [10, 12, 9, 1].

1. Conventions

Convention 2.1 (Size)**.** We fix a Grothendieck universe $\mathbb{U}$. A set is *small* if it is an element of $\mathbb{U}$. **Set** denotes the category of small sets. A category is *locally small* if each of its hom-sets is in bijection with a small set, and *small* if, in addition, its class of objects is small. Unless the contrary is stated, every category considered is locally small. We work in a set theory with the axiom of choice, which we use freely, for instance to choose limits and colimits, representatives of isomorphism classes, and elements of products of nonempty sets.

Convention 2.2 (Presheaves)**.** For a category $\mathcal{C}$ we write $\mathsf{PSh}(\mathcal{C})$ for the category of functors $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ and natural transformations between them. If $\mathcal{C}$ is not small, $\mathsf{PSh}(\mathcal{C})$ need not be locally small; we use it as a category with possibly large hom-sets, and every hom-set of $\mathsf{PSh}(\mathcal{C})$ that enters into a statement of this monograph will be shown to be small. We write $y\colon \mathcal{C} \to \mathsf{PSh}(\mathcal{C})$, $y(X) = \mathcal{C}(-, X)$, for the Yoneda embedding.

Convention 2.3 (Composition and diagrams)**.** Composition is written $g \circ f$ or gf. A diagram in $\mathcal{C}$ is said to *commute* if, for every pair of vertices, all composites along directed paths from the first vertex to the second are equal.

2. Initial and terminal objects

Definition 2.4. An object 0 of a category $\mathcal{C}$ is *initial* if for every object X the set $\mathcal{C}(0, X)$ has exactly one element, denoted $!_X\colon 0 \to X$. An object 1 is *terminal* if every $\mathcal{C}(X, 1)$ has exactly one element, denoted $!^X\colon X \to 1$. An object that is both initial and terminal is a *zero object*, and $\mathcal{C}$ is *pointed* if it has a zero object. An initial object 0 is *strict* if every morphism $X \to 0$ is an isomorphism.

The following lemma is the engine of cat theory; it will be invoked so often that we give it a name.

Lemma 2.5 (Parallel Morphism Lemma)**.** *Let $f, g\colon X \to Y$ be parallel morphisms in a category $\mathcal{C}$. If X is initial or Y is terminal, then $f = g$.*

PROOF. If X is initial then $\mathcal{C}(X,Y)$ has exactly one element, so $f=g$; the case of Y terminal is dual. □

Corollary 2.6. *Let D be a diagram in $\mathcal{C}$ in which every pair of parallel directed paths has an initial common source or a terminal common target. Then D commutes.*

PROOF. Two parallel paths have composites with the same domain and codomain; if the domain is initial or the codomain terminal, Lemma 2.5 shows the composites agree. □

Lemma 2.7. *Let $\mathcal{C}$ be a category.*

(i) *If H and H' are initial objects, the unique morphism $H \to H'$ is an isomorphism.*
(ii) *An object isomorphic to an initial object is initial.*
(iii) *If T and T' are terminal objects, the unique morphism $T \to T'$ is an isomorphism, and an object isomorphic to a terminal object is terminal.*

PROOF. (i) Let $u\colon H \to H'$ and $v\colon H' \to H$ be the unique morphisms. Then vu and id_H are endomorphisms of the initial object H, so $vu = \mathrm{id}_H$ by Lemma 2.5; similarly $uv = \mathrm{id}_{H'}$. (ii) If $\theta\colon H' \to H$ is an isomorphism and H is initial, then $f \mapsto f\theta$ is a bijection $\mathcal{C}(H,X) \to \mathcal{C}(H',X)$. (iii) is dual. □

Lemma 2.8. *Let $\mathcal{C}$ have an initial object 0 and a terminal object 1. The following are equivalent:*

(i) *$\mathcal{C}$ is pointed;*
(ii) *$0 \cong 1$;*
(iii) *$\mathcal{C}(1,0) \neq \varnothing$;*
(iv) *for every object X, $\mathcal{C}(X,0) \neq \varnothing$.*

PROOF. (i)⇒(ii): a zero object is isomorphic to both 0 and 1 by Lemma 2.7. (ii)⇒(iii) is clear. (iii)⇒(iv): compose $!^X\colon X \to 1$ with a morphism $1 \to 0$. (iv)⇒(i): let $u\colon 1 \to 0$. Then $u \circ !_1\colon 0 \to 0$ equals id_0 and $!_1 \circ u\colon 1 \to 1$ equals id_1 by Lemma 2.5, where $!_1\colon 0 \to 1$; so $0 \cong 1$ and 0 is a zero object by Lemma 2.7. □

3. Coproducts and connected objects

For a small family $(Y_i)_{i\in I}$ of objects of $\mathcal{C}$, a coproduct is an object $\coprod_i Y_i$ with morphisms $\mathrm{in}_i\colon Y_i \to \coprod_i Y_i$ such that $f \mapsto (f \circ \mathrm{in}_i)_i$ is a bijection $\mathcal{C}(\coprod_i Y_i, Z) \to \prod_i \mathcal{C}(Y_i, Z)$ for every Z. The coproduct of the empty family is an initial object.

Definition 2.9. An object X of $\mathcal{C}$ is *connected* if for every small family $(Y_i)_{i\in I}$ of objects of $\mathcal{C}$ that admits a coproduct in $\mathcal{C}$, the canonical map

$$\kappa\colon \coprod_{i\in I} \mathcal{C}(X,Y_i) \longrightarrow \mathcal{C}\Big(X, \coprod_{i\in I} Y_i\Big), \qquad (i,g) \longmapsto \mathrm{in}_i \circ g, \tag{2.1}$$

is a bijection. We write $\mathcal{C}^\circ$ for the full subcategory of $\mathcal{C}$ spanned by the connected objects.

Remark 2.10. If $\mathcal{C}$ has an initial object 0, then, applying (2.1) to the empty family, a connected object X satisfies $\mathcal{C}(X, 0) = \varnothing$. In particular an initial object is never connected, and by Lemma 2.8 a pointed category has no connected objects at all. Connectedness is invariant under isomorphism. One may also consider related notions, for example being non-initial and admitting no nontrivial binary coproduct decomposition; we use only Definition 2.9.

Example 2.11.

(i) In **Set** the connected objects are precisely the one-element sets. Indeed, a connected set X is nonempty by Remark 2.10; if X has two distinct elements, a surjection $X \to 1 \sqcup 1$ which separates them does not factor through either coprojection; and if $X = \{x\}$ then $\mathbf{Set}(X, \coprod_i Y_i) \cong \coprod_i Y_i \cong \coprod_i \mathbf{Set}(X, Y_i)$ compatibly with κ.

(ii) In the category **Top** of topological spaces and continuous maps, the connected objects are the nonempty connected spaces. A continuous map from a nonempty connected space into a disjoint union has connected image, which lies in exactly one summand because the summands are disjoint and clopen; since the coprojections are injective, the map factors uniquely through that summand, so κ is bijective. Conversely, a connected object is nonempty, and if it were a disjoint union of two nonempty clopen subsets $U \sqcup V$ the evident map to $1 \sqcup 1$ would not factor through a coprojection.

(iii) Let **Gph** be the category whose objects are pairs (V, E) consisting of a small set V and a symmetric relation $E \subseteq V \times V$, and whose morphisms are relation-preserving maps. The connected objects are the graphs that are nonempty and connected in the graph-theoretic sense, by the argument of (ii) with connected components in place of clopen subsets; here 1 is the terminal graph, a single vertex with a loop, so that the map from a disconnected graph to $1 \sqcup 1$ separating two components is a morphism.

(iv) In **Ab**, **Grp**, $\mathbf{Vect}_k$ or the category $\mathbf{Set}_*$ of pointed sets there are no connected objects, these categories being pointed.

4. Families

Definition 2.12. The category $\mathsf{Fam}(\mathcal{C})$ of *small families* in $\mathcal{C}$ has as objects pairs (A, X) where A is a small set and $X = (X_a)_{a \in A}$ is an A-indexed family of objects of $\mathcal{C}$. A morphism $(\varphi, f) \colon (A, X) \to (B, Y)$ consists of a function $\varphi \colon A \to B$ and a family of morphisms $f_a \colon X_a \to Y_{\varphi(a)}$ of $\mathcal{C}$. Composition and identities are

$$(\psi, g) \circ (\varphi, f) = (\psi\varphi, (g_{\varphi(a)} f_a)_{a \in A}), \qquad \mathrm{id}_{(A,X)} = (\mathrm{id}_A, (\mathrm{id}_{X_a})_{a \in A}).$$

Associativity and unitality are inherited from **Set** and $\mathcal{C}$.

Lemma 2.13. *For objects (A, X) and (B, Y) of $\mathsf{Fam}(\mathcal{C})$, the assignment $(\varphi, f) \mapsto ((\varphi(a), f_a))_{a \in A}$ is a bijection*

$$\mathsf{Fam}(\mathcal{C})((A, X), (B, Y)) \cong \prod_{a \in A} \coprod_{b \in B} \mathcal{C}(X_a, Y_b).$$

In particular $\mathsf{Fam}(\mathcal{C})$ is locally small.

Proof. An element of the right-hand side assigns to each a a pair (b_a, f_a) with $f_a \colon X_a \to Y_{b_a}$; setting $\varphi(a) = b_a$ recovers (φ, f) uniquely. The right-hand side is small since A, B and the hom-sets of $\mathcal{C}$ are. □

Remark 2.14. The category $\mathsf{Fam}(\mathcal{C})$ has small coproducts, given by

$$\textstyle\coprod_i (A_i, X^i) = \big(\coprod_i A_i, (X^i_a)_{(i,a)}\big),$$

and the functor $\mathcal{C} \to \mathsf{Fam}(\mathcal{C})$, $X \mapsto (1, X)$, exhibits it as the free completion of $\mathcal{C}$ under small coproducts. This is well known; the projection $\mathsf{Fam}(\mathcal{C}) \to \mathbf{Set}$, the *family fibration*, is treated for instance in [4]. We shall reprove everything we need in feline form (Theorem 12.4).

5. Adjoining extremal objects

Definition 2.15. Let $\mathcal{D}$ be a category. The category $\mathcal{D}^{\triangleleft}$ is obtained from $\mathcal{D}$ by adjoining a new object $\bot$ with $\mathcal{D}^{\triangleleft}(\bot, d) = \{*\}$ for all objects d (including $d = \bot$, where $* = \mathrm{id}_\bot$) and $\mathcal{D}^{\triangleleft}(d, \bot) = \varnothing$ for $d \neq \bot$. Dually $\mathcal{D}^{\triangleright}$ is obtained by adjoining a new object $\top$ with $\mathcal{D}^{\triangleright}(d, \top) = \{*\}$ and $\mathcal{D}^{\triangleright}(\top, d) = \varnothing$ for $d \neq \top$. Composition is forced.

Thus $\bot$ is initial in $\mathcal{D}^{\triangleleft}$ and $\top$ is terminal in $\mathcal{D}^{\triangleright}$, and $(\mathcal{D}^{\mathrm{op}})^{\triangleright} = (\mathcal{D}^{\triangleleft})^{\mathrm{op}}$. For a set A regarded as a discrete category, $(A^{\triangleleft})^{\triangleright}$ is the poset obtained from the discrete poset A by adjoining a bottom and a top element. For $n \geq 0$ we write $[n]$ for the totally ordered set $\{0 < 1 < \cdots < n\}$ regarded as a category; thus $[0] = \mathbb{1}$ and $[n]^{\triangleright} \cong [n+1]$.

6. Categories of elements

Definition 2.16. Let $P \colon \mathcal{D}^{\mathrm{op}} \to \mathbf{Set}$ be a presheaf. Its *category of elements* $\int P$ has objects the pairs (d, x) with $x \in P(d)$, and morphisms $u \colon (d, x) \to (d', x')$ the morphisms $u \colon d \to d'$ of $\mathcal{D}$ with $P(u)(x') = x$. For a covariant functor $Q \colon \mathcal{D} \to \mathbf{Set}$, $\int Q$ has objects (d, x) with $x \in Q(d)$ and morphisms $u \colon (d, x) \to (d', x')$ with $Q(u)(x) = x'$.

A category is *connected* if it is nonempty and any two objects are joined by a finite zigzag of morphisms. For a small category $\mathcal{E}$ we write $\pi_0(\mathcal{E})$ for its set of connected components.

Lemma 2.17. *Let $\mathcal{D}$ be small and $Q \colon \mathcal{D} \to \mathbf{Set}$ a functor. Then $\operatorname{colim} Q \cong \pi_0(\int Q)$, with coprojections $Q(d) \to \pi_0(\int Q)$, $x \mapsto [(d, x)]$. In particular, for the constant functor const_1 at a one-element set, $\operatorname{colim}_{\mathcal{D}} \mathrm{const}_1 \cong \pi_0(\mathcal{D})$, which is a singleton if and only if $\mathcal{D}$ is connected.*

PROOF. The colimit of Q in **Set** is the quotient of $\coprod_d Q(d)$ by the equivalence relation generated by $(d,x) \sim (d', Q(u)(x))$ for $u\colon d \to d'$, with the evident coprojections; the classes of this relation are by definition the connected components of $\int Q$. For $Q = \mathrm{const}_1$ one has $\int Q \cong \mathcal{D}$. □

Lemma 2.18. *Let $F\colon \mathcal{D} \to \mathcal{E}$ be a functor. If $\mathcal{D}$ has a terminal object t, then the cocone $(F(!^d)\colon F(d) \to F(t))_d$ is a colimit of F. Dually, if $\mathcal{D}$ has an initial object i, then $(F(!_d)\colon F(i) \to F(d))_d$ is a limit of F.*

PROOF. The family is a cocone because $!^{d'} \circ u = !^d$ for every $u\colon d \to d'$. Given a cocone $(c_d\colon F(d) \to Z)_d$, the morphism c_t satisfies $c_t \circ F(!^d) = c_d$, and any z with $z \circ F(!^d) = c_d$ for all d satisfies $z = z \circ F(\mathrm{id}_t) = c_t$. The second statement is dual. □

7. Fibrations

We recall the basic vocabulary of Grothendieck fibrations; standard references are [3, 4, 1].

Definition 2.19. Let $p\colon \mathcal{E} \to \mathcal{B}$ be a functor. A morphism $f\colon e' \to e$ of $\mathcal{E}$ is *cartesian* (for p) if for every morphism $g\colon e'' \to e$ and every $v\colon p(e'') \to p(e')$ with $p(f) \circ v = p(g)$ there is a unique $h\colon e'' \to e'$ with $p(h) = v$ and $f \circ h = g$. The functor p is a *fibration* if for every object e of $\mathcal{E}$ and every morphism $u\colon b \to p(e)$ there is a cartesian morphism f with codomain e and $p(f) = u$. Dually one defines *opcartesian* morphisms and *opfibrations*; a functor that is both is a *bifibration*. A morphism f is *vertical* if $p(f)$ is an identity, and the *fibre* $\mathcal{E}_b$ over b is the subcategory of objects over b and vertical morphisms between them.

Definition 2.20. Let $F\colon \mathcal{B}^{\mathrm{op}} \to \mathbf{CAT}$ be a functor. The *Grothendieck construction* $\int F$ is the category whose objects are pairs (b,x) with x an object of $F(b)$ and whose morphisms $(b,x) \to (b',x')$ are pairs (u,f) with $u\colon b \to b'$ in $\mathcal{B}$ and $f\colon x \to F(u)(x')$ in $F(b)$; composition is $(u',f') \circ (u,f) = (u'u, F(u)(f') \circ f)$.

The projection $\int F \to \mathcal{B}$ is a fibration in which $(u, \mathrm{id})\colon (b, F(u)(x')) \to (b', x')$ is a cartesian lift of u; this is standard [4, 1].

8. Further background

We use the following standard facts without further comment [10, 12]: right adjoints preserve all limits and left adjoints preserve all colimits; representable functors $\mathcal{C}(X,-)$ preserve all limits; fully faithful functors reflect limits and colimits; adjoints are unique up to unique natural isomorphism; and an equivalence of categories preserves and reflects every property defined by a universal property. Monoidal categories, monoidal functors, monoids, and enriched categories are as in [10, 7]. For the nerve and classifying space of a small category we refer to [13, 11]; we need only the fact that a natural transformation between functors $F, G\colon \mathcal{D} \to \mathcal{D}'$ induces a homotopy between the induced maps of classifying spaces.

CHAPTER 3

Cats and their morphisms

Standing Assumption 3.1. Throughout this chapter $\mathcal{C}$ denotes a category possessing an initial object. No further assumption is made; in particular $\mathcal{C}$ is not assumed to have a terminal object.

1. The definition

Definition 3.2. A *$\mathcal{C}$-cat* is a tuple

$$K = (H, T, (L_\alpha)_{\alpha \in A})$$

in which

(i) H is an initial object of $\mathcal{C}$, called the *head* of K;
(ii) T is a terminal object of $\mathcal{C}$, called the *tail* of K;
(iii) A is a small set, called the *leg set* of K and denoted $\pi(K)$;
(iv) for each $\alpha \in A$, L_α is an object of $\mathcal{C}$, called the *α-th leg* of K.

The *canonical morphisms* of K are the unique morphisms

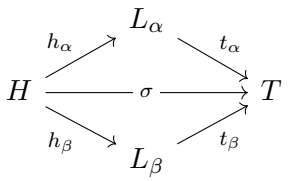
$$h_\alpha = h_\alpha^K \colon H \to L_\alpha, \qquad t_\alpha = t_\alpha^K \colon L_\alpha \to T, \qquad \sigma = \sigma^K \colon H \to T,$$

called respectively the *α-th shoulder*, the *α-th hip*, and the *spine* of K. If $|A| = n$ we say that K is *n-legged*; a 1-legged cat is also called *unipedal* and a 4-legged cat *quadrupedal*.

The canonical morphisms are not additional data: they are determined by H, T and the legs, because H is initial and T is terminal. We nevertheless name them, since they are the morphisms "connecting the head, legs and tail" and since they will appear in diagrams.

Proposition 3.3. *For every $\mathcal{C}$-cat K the diagram*

(3.1)



consisting of all shoulders, hips and the spine commutes. That is, $t_\alpha h_\alpha = \sigma$ for all $\alpha \in A$.

Proof. The only pairs of distinct parallel paths in the diagram run from H to T. Their source is initial, so they agree by Lemma 2.5. □

Remark 3.4. The pair $(H, (h_\alpha)_\alpha)$ is a cone over the discrete diagram $(L_\alpha)_{\alpha\in A}$, and $(T, (t_\alpha)_\alpha)$ is a cocone. Neither is a limit or colimit in general: $(H, (h_\alpha))$ is a product cone if and only if a product $\prod_\alpha L_\alpha$ exists and is initial, and $(T, (t_\alpha))$ is a coproduct cocone if and only if a coproduct $\coprod_\alpha L_\alpha$ exists and is terminal. In **Set**, for example, the head of K is a product of its legs if and only if $A \neq \varnothing$ and some leg is empty, and the tail is a coproduct of the legs if and only if exactly one leg is nonempty and that leg is a singleton.

2. Feline shapes

Definition 3.5. Let A be a small set, regarded as a discrete category. The *feline shape* on A is the category $\mathfrak{F}_A \coloneqq (A^\triangleleft)^\triangleright$ of Definition 2.15. We denote its objects by h (the adjoined initial object), t (the adjoined terminal object), and ℓ_α for $\alpha \in A$, and its non-identity morphisms by

$$\eta_\alpha \colon \mathsf{h} \to \ell_\alpha, \qquad \varepsilon_\alpha \colon \ell_\alpha \to \mathsf{t}, \qquad s \colon \mathsf{h} \to \mathsf{t},$$

subject to $\varepsilon_\alpha \eta_\alpha = s$ for all α.

Thus $\mathfrak{F}_A$ is the poset $\{\mathsf{h}\} \sqcup A \sqcup \{\mathsf{t}\}$ in which h is the least element, t is the greatest element, and distinct elements of A are incomparable. In particular $\mathfrak{F}_\varnothing \cong [1]$.

Proposition 3.6. *Let A be a small set. The assignment $K \mapsto (\mathsf{h} \mapsto H,\ \mathsf{t} \mapsto T,\ \ell_\alpha \mapsto L_\alpha,\ \eta_\alpha \mapsto h_\alpha,\ \varepsilon_\alpha \mapsto t_\alpha,\ s \mapsto \sigma)$ is a bijection between the class of $\mathcal{C}$-cats with leg set A and the class of functors $\mathfrak{F}_A \to \mathcal{C}$ sending h to an initial object and t to a terminal object.*

Proof. Given a $\mathcal{C}$-cat K, the assignment preserves identities by definition and preserves the only nontrivial composition relation $\varepsilon_\alpha \eta_\alpha = s$ by Proposition 3.3; hence it is a functor. Conversely, a functor $D \colon \mathfrak{F}_A \to \mathcal{C}$ with $D(\mathsf{h})$ initial and $D(\mathsf{t})$ terminal determines the cat $(D(\mathsf{h}), D(\mathsf{t}), (D(\ell_\alpha))_\alpha)$, and D is recovered from this cat because its values on η_α, ε_α and s are morphisms out of an initial object or into a terminal object and hence are the canonical morphisms. The two constructions are mutually inverse. □

3. Morphisms of cats

Definition 3.7. Let $K = (H, T, (L_\alpha)_{\alpha\in A})$ and $K' = (H', T', (L'_\beta)_{\beta\in B})$ be $\mathcal{C}$-cats. A *morphism of $\mathcal{C}$-cats* (or *feline morphism*) $F \colon K \to K'$ is a tuple

$$F = (\varphi, F_H, F_T, (F_\alpha)_{\alpha\in A})$$

consisting of a function $\varphi \colon A \to B$ (the *gait*), a morphism $F_H \colon H \to H'$, a morphism $F_T \colon T \to T'$, and morphisms $F_\alpha \colon L_\alpha \to L'_{\varphi(\alpha)}$ for $\alpha \in A$, such

that for every $\alpha \in A$ the diagram

$$\begin{array}{ccccc} H & \xrightarrow{h_\alpha} & L_\alpha & \xrightarrow{t_\alpha} & T \\ {\scriptstyle F_H}\downarrow & & \downarrow{\scriptstyle F_\alpha} & & \downarrow{\scriptstyle F_T} \\ H' & \xrightarrow[h'_{\varphi(\alpha)}]{} & L'_{\varphi(\alpha)} & \xrightarrow[t'_{\varphi(\alpha)}]{} & T' \end{array} \tag{3.2}$$

commutes, and the *spinal square* $F_T \circ \sigma = \sigma' \circ F_H$ commutes. The composite of $F\colon K \to K'$ and $G = (\psi, G_{H'}, G_{T'}, (G_\beta)_\beta)\colon K' \to K''$ is

$$G \circ F = (\psi\varphi,\ G_{H'}F_H,\ G_{T'}F_T,\ (G_{\varphi(\alpha)}F_\alpha)_{\alpha\in A}),$$

and the identity of K is $(\mathrm{id}_A, \mathrm{id}_H, \mathrm{id}_T, (\mathrm{id}_{L_\alpha})_\alpha)$.

The composite $G \circ F$ is again a morphism: each of its squares is obtained by pasting a square for F with a square for G. Composition is associative and unital because composition in **Set** and in $\mathcal{C}$ is.

Definition 3.8. The category of $\mathcal{C}$-cats and feline morphisms is denoted $\mathsf{Fel}(\mathcal{C})$.

We now record the observation on which the whole theory turns.

Proposition 3.9 (Automatic Commutativity)**.** *Let K and K' be $\mathcal{C}$-cats with leg sets A and B.*

(i) *For any morphism $F\colon K \to K'$, the component F_H is the unique morphism $H \to H'$, and it is an isomorphism.*
(ii) *For any morphism $F\colon K \to K'$, the component F_T is the unique morphism $T \to T'$, and it is an isomorphism.*
(iii) *Every square in (3.2) and the spinal square commute for* arbitrary *choices of φ, F_H, F_T and F_α of the correct types.*

Consequently, for every function $\varphi\colon A \to B$ and every family of morphisms $(F_\alpha\colon L_\alpha \to L'_{\varphi(\alpha)})_{\alpha\in A}$ there is exactly one morphism of $\mathcal{C}$-cats $K \to K'$ with gait φ and leg components (F_α).

Proof. (i) The object H is initial, so $\mathcal{C}(H, H')$ has exactly one element; it is an isomorphism by Lemma 2.7(i) since H' is also initial. (ii) is dual. (iii) The two composites around the left square of (3.2) are morphisms $H \to L'_{\varphi(\alpha)}$ with initial source; the two composites around the right square are morphisms $L_\alpha \to T'$ with terminal target; the two composites around the spinal square are morphisms $H \to T'$, whose source is initial. All three pairs agree by Lemma 2.5. The final assertion follows: F_H and F_T exist and are unique, and no condition remains on $(\varphi, (F_\alpha))$. □

We emphasize that Proposition 3.9 is not a technical convenience but a complete description of the situation: the commutative diagrams in Definition 3.7 impose no conditions whatsoever. The reader should keep this in mind whenever a commutative diagram involving a head or a tail appears below.

Notation 3.10. In view of Proposition 3.9 we write a feline morphism simply as $F = (\varphi, (F_\alpha)_{\alpha\in A})$ or (φ, F), suppressing F_H and F_T. In this notation composition reads $(\psi, G) \circ (\varphi, F) = (\psi\varphi, (G_{\varphi(\alpha)}F_\alpha)_\alpha)$.

Corollary 3.11. *For $\mathcal{C}$-cats K and K' as above there are bijections, natural in the evident sense,*

$$\mathsf{Fel}(\mathcal{C})(K, K') \cong \coprod_{\varphi\colon A\to B} \prod_{\alpha\in A} \mathcal{C}(L_\alpha, L'_{\varphi(\alpha)}) \cong \prod_{\alpha\in A} \coprod_{\beta\in B} \mathcal{C}(L_\alpha, L'_\beta).$$

In particular $\mathsf{Fel}(\mathcal{C})$ is locally small.

Proof. The first bijection is Proposition 3.9; the second is the distributivity of products over coproducts in **Set** (no choice is needed: an element of the right-hand side is a function $\alpha \mapsto (\beta_\alpha, f_\alpha)$). □

Remark 3.12 (Morphisms as natural transformations)**.** Under Proposition 3.6, a cat K with leg set A is a functor $K\colon \mathfrak{F}_A \to \mathcal{C}$. Call a functor $u\colon \mathfrak{F}_A \to \mathfrak{F}_B$ *leg-preserving* if $u(\mathsf{h}) = \mathsf{h}$, $u(\mathsf{t}) = \mathsf{t}$ and u maps each ℓ_α to some ℓ_β. Since the hom-sets of $\mathfrak{F}_B$ have at most one element, leg-preserving functors u correspond bijectively to functions $\varphi\colon A \to B$ via $u(\ell_\alpha) = \ell_{\varphi(\alpha)}$. A morphism $(\varphi, F_H, F_T, (F_\alpha))\colon K \to K'$ is then precisely a pair (u, θ) consisting of the leg-preserving functor u associated with φ and a natural transformation $\theta\colon K \Rightarrow K' \circ u$ with components $\theta_\mathsf{h} = F_H$, $\theta_\mathsf{t} = F_T$ and $\theta_{\ell_\alpha} = F_\alpha$: the naturality squares of θ at η_α, ε_α and s are exactly the squares of Definition 3.7. Thus $\mathsf{Fel}(\mathcal{C})$ is a (non-full) subcategory of the category of diagrams in $\mathcal{C}$ considered in Section 2. Functors $\mathfrak{F}_A \to \mathfrak{F}_B$ that fix h and t but send some leg to h or t are not leg-preserving. Those that send legs only to legs or to t give rise to the tail-absorbing morphisms of Section 2; those that send a leg to h are discussed in Remark 10.12.

4. The Fundamental Theorem

The following theorem identifies cat theory with the theory of families.

Theorem 3.13 (Fundamental Theorem of Cat Theory). *Let $\mathcal{C}$ be a category with an initial object, and let*

$$\begin{aligned} U\colon \mathsf{Fel}(\mathcal{C}) &\longrightarrow \mathsf{Fam}(\mathcal{C}),\\ (H, T, (L_\alpha)_{\alpha\in A}) &\longmapsto (A, (L_\alpha)_\alpha),\\ (\varphi, F) &\longmapsto (\varphi, F) \end{aligned}$$

be the functor forgetting the head and the tail.

(i) *U is fully faithful.*
(ii) *U is an equivalence of categories if and only if $\mathcal{C}$ has a terminal object.*
(iii) *U is an isomorphism of categories if and only if $\mathcal{C}$ has a terminal object, every initial object of $\mathcal{C}$ equals a fixed object 0, and every terminal object of $\mathcal{C}$ equals a fixed object 1.*

PROOF. U is a functor because composition in $\mathsf{Fel}(\mathcal{C})$, in the form of Notation 3.10, is given by the same formula as composition in $\mathsf{Fam}(\mathcal{C})$.

(i) By Proposition 3.9, the map $\mathsf{Fel}(\mathcal{C})(K, K') \to \mathsf{Fam}(\mathcal{C})(UK, UK')$ is a bijection.

(ii) A fully faithful functor is an equivalence if and only if it is essentially surjective. If $\mathcal{C}$ has a terminal object 1, then for every family (A, X) we have $(A, X) = U(0, 1, X)$ for an initial object 0, so U is surjective on objects. If $\mathcal{C}$ has no terminal object, then $\mathsf{Fel}(\mathcal{C})$ has no objects at all, whereas $\mathsf{Fam}(\mathcal{C})$ contains the empty family $(\varnothing, \varnothing)$; hence U is not essentially surjective.

(iii) A fully faithful functor is an isomorphism if and only if it is bijective on objects. The fibre of U over a family (A, X) is in bijection with the class of pairs (H, T) with H initial and T terminal. This class is nonempty if and only if $\mathcal{C}$ has a terminal object, and has exactly one element if and only if, in addition, $\mathcal{C}$ has exactly one initial object and exactly one terminal object. □

Remark 3.14. Theorem 3.13 has the following consequence, which governs the rest of this monograph. Every categorical property of $\mathsf{Fel}(\mathcal{C})$, that is, every property invariant under equivalence of categories, is a property of $\mathsf{Fam}(\mathcal{C})$ provided $\mathcal{C}$ has a terminal object, and every construction on $\mathsf{Fel}(\mathcal{C})$ may be computed in $\mathsf{Fam}(\mathcal{C})$. We nonetheless continue to work with $\mathsf{Fel}(\mathcal{C})$, and we indicate in each case the corresponding statement for families.

Remark 3.15. In most categories of interest the condition in Theorem 3.13(iii) fails. In **Set** there is one initial object but a large (non-small) collection of terminal objects, namely all singletons, so $U\colon \mathsf{Fel}(\mathbf{Set}) \to \mathsf{Fam}(\mathbf{Set})$ is an equivalence that is not an isomorphism. In a skeletal category with initial and terminal objects, U is an isomorphism.

Proposition 3.16. *The category $\mathsf{Fel}(\mathcal{C})$ is nonempty if and only if $\mathcal{C}$ has a terminal object. If $\mathcal{C}$ has no terminal object then $\mathsf{Fel}(\mathcal{C})$ is the empty category.*

PROOF. A $\mathcal{C}$-cat contains a terminal object. Conversely, if 1 is terminal and 0 initial, $(0, 1, \varnothing)$ is a $\mathcal{C}$-cat. □

Example 3.17. The poset $(\mathbb{N}, \leq)$, regarded as a category, has initial object 0 and no terminal object; hence $\mathsf{Fel}(\mathbb{N}, \leq) = \varnothing$. The category of fields and field homomorphisms has no initial object, so it does not satisfy Standing Assumption 3.1; nevertheless no tuple satisfies Definition 3.2 in it, and in Theorem 4.25 we adopt the convention that $\mathsf{Fel}(\mathcal{C}) = \varnothing$ for such $\mathcal{C}$. The categories of nonempty sets and of nonempty topological spaces have terminal objects but no initial object, so they have no cats; but the headless cats of Section 6 make sense there.

5. Normalization and standard notation

Standing Assumption 3.18. From now on, whenever $\mathcal{C}$ is assumed to have a terminal object, we fix once and for all an initial object $0 = 0_{\mathcal{C}}$ and a terminal object $1 = 1_{\mathcal{C}}$.

Definition 3.19. A $\mathcal{C}$-cat is *normalized* if its head is 0 and its tail is 1. We write $\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C})$ for the full subcategory of normalized cats.

Proposition 3.20. *Suppose $\mathcal{C}$ has a terminal object. Then U restricts to an isomorphism of categories $\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C}) \cong \mathsf{Fam}(\mathcal{C})$, the inclusion $\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C}) \hookrightarrow \mathsf{Fel}(\mathcal{C})$ is an equivalence, and the* normalization *functor*

$$N\colon \mathsf{Fel}(\mathcal{C}) \longrightarrow \mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C}), \qquad (H,T,L) \longmapsto (0,1,L), \qquad (\varphi,F) \longmapsto (\varphi,F)$$

is a quasi-inverse equivalence; the morphisms $(\mathrm{id}_A, (\mathrm{id}_{L_\alpha})_\alpha)\colon K \to NK$ form a natural isomorphism $\mathrm{id} \Rightarrow N$ (with inclusion suppressed).

Proof. On normalized cats U is bijective on objects and fully faithful. The morphism $(\mathrm{id}_A, \mathrm{id})\colon K \to NK$ exists by Proposition 3.9 and is an isomorphism with inverse $(\mathrm{id}_A, \mathrm{id})\colon NK \to K$. Naturality is the identity $(\varphi,F)\circ(\mathrm{id},\mathrm{id}) = (\mathrm{id},\mathrm{id})\circ(\varphi,F)$. □

Notation 3.21. Assume $\mathcal{C}$ has a terminal object. We use the following notation throughout.

(i) For a small set A and a family $L = (L_\alpha)_{\alpha\in A}$ we write $\langle L\rangle = \langle L_\alpha\rangle_{\alpha\in A}$ for the normalized cat $(0,1,L)$.
(ii) The *one-legged inclusion* is the functor $\iota\colon \mathcal{C} \to \mathsf{Fel}(\mathcal{C})$, $\iota(X) = \langle X\rangle$, $\iota(f) = (\mathrm{id}_1, f)$, where here $1 = \{*\}$ denotes the one-element index set.
(iii) The *legless cat* is $\mathbf{0} \coloneqq \langle\ \rangle_\varnothing = (0,1,\varnothing)$.
(iv) The *cephalic cat* is $\mathbf{h} \coloneqq \iota(0)$, the unipedal cat whose leg is the head.
(v) The *caudal cat* is $\mathbf{1} \coloneqq \iota(1)$, the unipedal cat whose leg is the tail.
(vi) The *leg functor* is $\pi\colon \mathsf{Fel}(\mathcal{C}) \to \mathbf{Set}$, $\pi(H,T,(L_\alpha)_{\alpha\in A}) = A$, $\pi(\varphi,F) = \varphi$.
(vii) For $\alpha \in A$, the *α-th leg inclusion* of a cat K is the morphism $j_\alpha = (\alpha, \mathrm{id}_{L_\alpha})\colon \iota(L_\alpha) \to K$, where α also denotes the function $1 \to A$ with value α.

Proposition 3.22. *The one-legged inclusion ι is fully faithful, and its essential image is the full subcategory of unipedal cats.*

Proof. By Corollary 3.11, $\mathsf{Fel}(\mathcal{C})(\iota X, \iota Y) \cong \mathcal{C}(X,Y)$ via $(\mathrm{id}_1, f) \mapsto f$, which is inverse to the action of ι. A unipedal cat $(H,T,(L_a)_{a\in\{a\}})$ is isomorphic to $\iota(L_a)$ by the morphism with gait the bijection $\{a\} \to 1$ and identity component, which is an isomorphism by Proposition 3.23 below; and by the same proposition every cat isomorphic to a unipedal cat is unipedal. □

Proposition 3.23. *A feline morphism $(\varphi,F)\colon K \to K'$ is an isomorphism if and only if φ is a bijection and every F_α is an isomorphism.*

Proof. If (ψ,G) is inverse to (φ,F), then $\psi\varphi = \mathrm{id}$ and $\varphi\psi = \mathrm{id}$, so φ is bijective, and $G_{\varphi(\alpha)}F_\alpha = \mathrm{id}$, $F_{\psi(\beta)}G_\beta = \mathrm{id}$; taking $\beta = \varphi(\alpha)$ shows that F_α is an isomorphism with inverse $G_{\varphi(\alpha)}$. Conversely, if φ is bijective and each F_α invertible, then $(\varphi^{-1}, (F^{-1}_{\varphi^{-1}(\beta)})_\beta)$ is an inverse. □

6. Examples

Proposition 3.24. *Let $\mathbb{1}$ be the terminal category, with a single object $*$ and only the identity morphism. Then $\mathsf{Fel}(\mathbb{1})$ is isomorphic to* **Set***, via the leg functor π.*

Proof. The object $*$ is both initial and terminal, and it is the only object. Hence a $\mathbb{1}$-cat is a tuple $(*, *, (*)_{\alpha\in A})$, determined by the small set A, and every small set arises. A morphism with gait φ has all components equal to id_*, so it is determined by φ, and every function arises. Thus π is bijective on objects and on morphisms. □

Example 3.25 (Sets)**.** A **Set**-cat is a family of small sets $(L_\alpha)_{\alpha\in A}$ equipped with the head $\varnothing$ and a chosen singleton as tail. A morphism consists of a function $\varphi\colon A \to B$ and functions $L_\alpha \to L'_{\varphi(\alpha)}$. We shall see in Corollary 7.22 that $\mathsf{Fel}(\mathbf{Set})$ is equivalent to the arrow category $\mathbf{Set}^{\to}$, via $K \mapsto (\coprod_\alpha L_\alpha \to A)$.

Example 3.26 (Spaces)**.** In **Top** the head is the empty space and the tail is a one-point space. A **Top**-cat is a family of spaces; note that the legs are not glued to one another, and in particular a cat is not a topological space. The topological space naturally associated with a cat, its disjoint union, is its body (Definition 7.10).

Example 3.27 (Pointed categories)**.** If $\mathcal{C}$ is pointed, for instance $\mathcal{C} = \mathbf{Ab}$, **Grp**, $\mathbf{Vect}_k$ or $\mathbf{Set}_*$, then the head and the tail of every cat are zero objects, hence isomorphic to each other, and every shoulder, hip and spine is a zero morphism. The distinction between head and tail is then a matter of labelling alone. In Proposition 7.5 and Remark 4.17 we shall see that pointedness nevertheless has consequences for the theory.

Example 3.28 (Rings)**.** In the category **Ring** of unital rings, the head is $\mathbb{Z}$ and the tail is the zero ring. A **Ring**-cat is therefore a family of rings (R_α), each equipped with its unique ring homomorphisms $\mathbb{Z} \to R_\alpha \to 0$.

Example 3.29 (Bounded posets)**.** Let P be a poset with least element $\bot$ and greatest element $\top$, regarded as a category. A P-cat is a family $(p_\alpha)_{\alpha\in A}$ of elements of P, and a morphism $\langle p_\alpha\rangle_A \to \langle q_\beta\rangle_B$ is a function $\varphi\colon A \to B$ with $p_\alpha \le q_{\varphi(\alpha)}$ for all α. Although P is a preorder, $\mathsf{Fel}(P)$ is not: the two functions $1 \to 2$ are the gaits of two distinct morphisms $\langle\bot\rangle \to \langle\bot,\bot\rangle$. When $P = \{\bot = \top\}$, $\mathsf{Fel}(P)$ is isomorphic to **Set** by Proposition 3.24.

Example 3.30 (Categories)**.** In the category **Cat** of small categories, the head is the empty category and the tail is the terminal category $\mathbb{1}$. A **Cat**-cat is a small family of small categories; it is a family of categories, not itself a category.

Example 3.31 (Opposite of sets)**.** In $\mathbf{Set}^{\mathrm{op}}$ the initial object is a singleton and the terminal object is $\varnothing$. A $\mathbf{Set}^{\mathrm{op}}$-cat has a singleton for a head and the empty set for a tail; a morphism $\langle X_\alpha\rangle_A \to \langle Y_\beta\rangle_B$ consists of $\varphi\colon A \to B$ and functions $Y_{\varphi(\alpha)} \to X_\alpha$.

7. Remarks on the definition

Remark 3.32 (Why the head is initial)**.** If, in Definition 3.2, H and T were allowed to be arbitrary objects equipped with morphisms $h_\alpha\colon H \to L_\alpha$ and $t_\alpha\colon L_\alpha \to T$, then the commutativity of (3.1) would become a genuine condition, the components F_H and F_T of a morphism would become genuine data, and the resulting structure would be a functor $\mathfrak{F}_A \to \mathcal{C}$, that is, a cone over the legs together with a cocone under them and a morphism $H \to T$ equal to every composite $t_\alpha h_\alpha$ (when $A = \varnothing$, simply a morphism $H \to T$). The theory of such diagrams is a special case of the theory of diagrams in $\mathcal{C}$ and is not considered here. The requirement that H be initial and T terminal is exactly what makes the head and tail inert.

Remark 3.33 (Standing assumption on $\mathcal{C}$)**.** We have required $\mathcal{C}$ to have an initial object, as part of the definition of the theory. Most results of later chapters also require a terminal object, since otherwise $\mathsf{Fel}(\mathcal{C})$ is empty (Proposition 3.16). Where the initial object plays an essential rôle in a proof, beyond the existence of cats, we shall say so and give a counterexample for headless cats.

CHAPTER 4

Anatomy: kittens, headless, tailless and legless cats

Standing Assumption 4.1. In Sections 1 to 5, $\mathcal{C}$ is a category with an initial object 0 and a terminal object 1. In Sections 6 to 8 the hypotheses are stated individually, since the point of those sections is to vary them.

This chapter studies the full subcategories of $\mathsf{Fel}(\mathcal{C})$ obtained by restricting the number of legs, and the variants of Definition 3.2 obtained by removing the head or the tail.

1. Legless cats

Definition 4.2. A $\mathcal{C}$-cat is *legless* if its leg set is empty. We write $\mathsf{Fel}_{\varnothing}(\mathcal{C})$ for the full subcategory of legless cats.

A legless cat is a pair (H, T) consisting of an initial and a terminal object, joined by its spine. The normalized legless cat is $\mathbf{0} = (0, 1, \varnothing)$.

Proposition 4.3. *Between any two legless cats there is exactly one morphism, and it is an isomorphism. Hence $\mathsf{Fel}_{\varnothing}(\mathcal{C})$ is a nonempty groupoid in which every hom-set is a singleton, and the functor $\mathsf{Fel}_{\varnothing}(\mathcal{C}) \to \mathbb{1}$ is an equivalence.*

Proof. By Corollary 3.11, $\mathsf{Fel}(\mathcal{C})(K, K')$ is in bijection with the set of functions $\varnothing \to \varnothing$, which is a singleton. The unique morphism is an isomorphism by Proposition 3.23, the empty function being bijective. □

Proposition 4.4. *A $\mathcal{C}$-cat is an initial object of $\mathsf{Fel}(\mathcal{C})$ if and only if it is legless. Moreover the initial object of $\mathsf{Fel}(\mathcal{C})$ is strict: every morphism $K \to \mathbf{0}$ is an isomorphism.*

Proof. If K is legless then $\mathsf{Fel}(\mathcal{C})(K, K') \cong \prod_{\alpha \in \varnothing}(\cdots)$ is a singleton for every K', by Corollary 3.11. Conversely, if K is initial then there is a morphism $K \to \mathbf{0}$, whose gait is a function $\pi(K) \to \varnothing$; hence $\pi(K) = \varnothing$. The same argument shows that any K admitting a morphism to $\mathbf{0}$ is legless, and then the morphism is an isomorphism by Proposition 4.3. □

Warning 4.5. The cephalic cat $\mathbf{h} = \iota(0)$, whose single leg is an initial object, is *not* initial in $\mathsf{Fel}(\mathcal{C})$. Indeed, by Corollary 3.11,

$$\mathsf{Fel}(\mathcal{C})(\mathbf{h}, K) \cong \coprod_{\alpha \in \pi(K)} \mathcal{C}(0, L_\alpha) \cong \pi(K),$$

which is a singleton only when K is unipedal. The functor ι therefore does not preserve initial objects. We return to this bijection in Lemma 6.3, where it is the fact that makes the leg functor representable.

2. Unipedal and n-legged cats

Definition 4.6. For a cardinal n (small), $\mathsf{Fel}_n(\mathcal{C})$ denotes the full subcategory of n-legged cats. A feline morphism is *isopodal* if its gait is a bijection and *vertical* if its gait is an identity.

By Proposition 3.22, $\iota\colon \mathcal{C} \to \mathsf{Fel}_1(\mathcal{C})$ is an equivalence. For $n \geq 2$ the category $\mathsf{Fel}_n(\mathcal{C})$ is not equivalent to $\mathcal{C}^n$ in general, since its morphisms may have arbitrary, not necessarily bijective, gaits. For example, for $\mathcal{C} = \mathbb{1}$ and $n = 2$, all objects of $\mathsf{Fel}_2(\mathbb{1})$ are isomorphic and each has an endomorphism monoid with four elements (the functions $2 \to 2$), whereas $\mathbb{1}^2$ is the terminal category.

Proposition 4.7. *For a small set A, let $\mathsf{Fel}(\mathcal{C})_A$ denote the subcategory of cats with leg set A and vertical morphisms. Then $\mathsf{Fel}(\mathcal{C})_A \simeq \mathcal{C}^A$, the A-fold power of $\mathcal{C}$, and this restricts to an isomorphism on normalized cats.*

Proof. A vertical morphism $K \to K'$ between cats with leg set A is, by Proposition 3.9, a family $(F_\alpha\colon L_\alpha \to L'_\alpha)_\alpha$, that is, a morphism in $\mathcal{C}^A$. The functor $K \mapsto (L_\alpha)_\alpha$ is therefore fully faithful; it is bijective on normalized cats, and every cat is vertically isomorphic to its normalization by Proposition 3.20. □

3. Kittens

Definition 4.8. A *$\mathcal{C}$-kitten* is a $\mathcal{C}$-cat whose leg set is finite. We write $\mathsf{Kit}(\mathcal{C})$ for the full subcategory of $\mathsf{Fel}(\mathcal{C})$ spanned by kittens.

Let $\mathsf{Fam}_{\mathrm{fin}}(\mathcal{C})$ denote the full subcategory of $\mathsf{Fam}(\mathcal{C})$ on families indexed by finite sets.

Proposition 4.9. *The equivalence of Theorem 3.13 restricts to an equivalence $\mathsf{Kit}(\mathcal{C}) \simeq \mathsf{Fam}_{\mathrm{fin}}(\mathcal{C})$. In particular $\mathsf{Kit}(\mathbb{1}) \simeq$ **FinSet**, the category of finite sets.*

Proof. U preserves leg sets. The second assertion follows from Proposition 3.24. □

Proposition 4.10. *The full subcategory $\mathsf{Kit}(\mathcal{C})$ contains the legless cat and the caudal cat and is closed under finite coproducts in $\mathsf{Fel}(\mathcal{C})$. A coproduct in $\mathsf{Fel}(\mathcal{C})$ of an infinite family of kittens is a kitten if and only if all but finitely many of the summands are legless.*

Proof. By Proposition 6.5 below, the leg set of a coproduct is the disjoint union of the leg sets. A finite disjoint union of finite sets is finite; an infinite disjoint union of finite sets is finite if and only if all but finitely many of them are empty. □

(The forward reference is harmless: Proposition 6.5 is proved directly from Corollary 3.11.)

Remark 4.11. Kittens reappear in three places: in Corollary 6.10 we show that $\mathsf{Kit}(\mathcal{C})$ is closed under the finite limits that exist in $\mathsf{Fel}(\mathcal{C})$; in Theorem 12.7 we show that every cat is canonically a filtered colimit of kittens; and in Section 6 we compute the rig of isomorphism classes of kittens.

4. Litters

Since $\mathsf{Kit}(\mathcal{C})$ has the initial object $\mathbf{0}$ and the terminal object $\mathbf{1}$ of $\mathsf{Fel}(\mathcal{C})$ (the latter by Proposition 6.2 below), the notion of a $\mathsf{Kit}(\mathcal{C})$-cat is defined.

Definition 4.12. A *litter* in $\mathcal{C}$ is a $\mathsf{Kit}(\mathcal{C})$-cat, that is, a small family of $\mathcal{C}$-kittens equipped with a legless head and a terminal tail (a unipedal kitten whose leg is terminal). The *flattening* of a litter, written in normalized form $\mathcal{L} = \langle K_a \rangle_{a \in A}$, with $K_a = \langle L_{a,\alpha} \rangle_{\alpha \in A_a}$, is the cat $\flat(\mathcal{L}) = \langle L_{a,\alpha} \rangle_{(a,\alpha) \in \coprod_a A_a}$; on morphisms, $\flat(\varphi, (\varphi_a, F_a)_a)$ is the morphism with gait $(a, \alpha) \mapsto (\varphi(a), \varphi_a(\alpha))$ and components $F_{a,\alpha}$.

Proposition 4.13. *Flattening is a functor* $\flat \colon \mathsf{Fel}(\mathsf{Kit}(\mathcal{C})) \to \mathsf{Fel}(\mathcal{C})$. *It is essentially surjective. It is not faithful, and it is not full, as soon as* $\mathcal{C}$ *is nonempty (which it is, having an initial object).*

Proof. Functoriality is the associativity of composition of gaits and of components. Every cat K is isomorphic to the flattening of the litter $\langle \iota(L_\alpha) \rangle_{\alpha \in A}$ of its unipedal kittens.

Not faithful: let $\mathcal{L} = \langle \mathbf{0} \rangle_1$ be the litter with a single, legless, kitten, and $\mathcal{L}' = \langle \mathbf{0}, \mathbf{0} \rangle_2$. There are two morphisms $\mathcal{L} \to \mathcal{L}'$ (with gaits the two functions $1 \to 2$), and both flatten to the unique morphism $\mathbf{0} \to \mathbf{0}$.

Not full: fix an object X of $\mathcal{C}$, and let $\mathcal{L} = \langle \langle X, X \rangle \rangle_1$ be the litter consisting of one two-legged kitten and $\mathcal{L}' = \langle \iota X, \iota X \rangle_2$ the litter of two unipedal kittens. Both flatten to two-legged cats with both legs equal to X, so there is an isomorphism $\flat(\mathcal{L}) \to \flat(\mathcal{L}')$ with bijective gait and identity components. It is not the flattening of a morphism $\mathcal{L} \to \mathcal{L}'$: such a morphism maps the unique kitten of $\mathcal{L}$ into a single unipedal kitten of $\mathcal{L}'$, so its flattening has a gait that is not injective. □

Remark 4.14. Flattening is a special case of the body functor of Definition 7.10, applied to the category $\mathsf{Fel}(\mathcal{C})$, which has small coproducts; see Proposition 11.2.

5. Chloe cats

The following class of cats is defined in terms of the connected objects of Definition 2.9. It plays a rôle in Chapter 13, and we record its elementary properties here.

Definition 4.15 (Chloe Condition)**.** A $\mathcal{C}$-cat K satisfies the *Chloe Condition* if every leg of K is a connected object of $\mathcal{C}$. A cat satisfying the Chloe Condition is called a *Chloe cat*, and $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$ denotes the full subcategory of

Chloe cats. By extension, an object X of $\mathcal{C}$ satisfies the Chloe Condition if the unipedal cat ιX does, that is, if X is connected.

Proposition 4.16. (i) *Every legless cat satisfies the Chloe Condition.*
(ii) *No leg of a Chloe cat is initial. In particular the cephalic cat* $\mathbf{h}$ *does not satisfy the Chloe Condition.*
(iii) $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$ *is closed in* $\mathsf{Fel}(\mathcal{C})$ *under isomorphisms, under the small coproducts of Proposition 6.5, and under reindexing: if* K' *is Chloe and* $\varphi\colon A \to \pi K'$*, then* $\langle L'_{\varphi(\alpha)}\rangle_{\alpha\in A}$ *is Chloe.*
(iv) *The equivalence* U *of Theorem 3.13 restricts to an equivalence* $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C}^\circ)$.

Proof. (i) is vacuous. (ii) is Remark 2.10. (iii) Connectedness is invariant under isomorphism, and the legs of $\coprod_i K_i$ and of $\varphi^* K'$ are legs of the K_i and of K', by Propositions 3.23 and 6.5. (iv) The legs of a Chloe cat form a family in $\mathcal{C}^\circ$. □

Remark 4.17. If $\mathcal{C}$ is pointed then $\mathcal{C}^\circ$ is empty (Remark 2.10), and the only Chloe cats are the legless ones. Thus in **Ab**, **Grp** or $\mathbf{Vect}_k$, every cat with at least one leg fails the Chloe Condition.

Example 4.18. (i) In **Set**, a cat satisfies the Chloe Condition if and only if each of its legs is a singleton (Example 2.11). The leg functor restricts to an equivalence $\mathsf{Fel}^{\mathrm{Ch}}(\mathbf{Set}) \simeq \mathbf{Set}$.
(ii) In **Top**, a cat satisfies the Chloe Condition if and only if each of its legs is a nonempty connected space.
(iii) Consider the category $\mathsf{Fel}(\mathcal{C})$ itself, which has the initial object $\mathbf{0}$ and the terminal object $\mathbf{1}$ (Propositions 4.4 and 6.2). We shall see in Proposition 7.15 that the connected objects of $\mathsf{Fel}(\mathcal{C})$ are the unipedal cats; hence a $\mathsf{Fel}(\mathcal{C})$-cat (a cat of cats) satisfies the Chloe Condition if and only if each of its legs is a unipedal cat.

6. Headless cats

In this section $\mathcal{C}$ is an arbitrary category.

Definition 4.19. A *headless* (or *acephalous*) $\mathcal{C}$-cat is a pair $(T, (L_\alpha)_{\alpha\in A})$ with T a terminal object, A a small set, and L_α objects of $\mathcal{C}$, equipped with the hips $t_\alpha\colon L_\alpha \to T$. A morphism $(T, L) \to (T', L')$ is a tuple $(\varphi, F_T, (F_\alpha))$ such that $t'_{\varphi(\alpha)} F_\alpha = F_T t_\alpha$ for all α. The resulting category is $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})$; it is empty if $\mathcal{C}$ has no terminal object.

Proposition 4.20. *Let* $\mathcal{C}$ *be any category.*
(i) *The conditions on a morphism of headless cats are automatic,* F_T *is uniquely determined and invertible, and the forgetful functor* $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \to \mathsf{Fam}(\mathcal{C})$ *is fully faithful.*
(ii) $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ *if and only if* $\mathcal{C}$ *has a terminal object.*

(iii) *If $\mathcal{C}$ has an initial object, the* decapitation *functor* $\mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})$, $(H, T, L) \mapsto (T, L)$, *is an equivalence.*

Proof. (i) is proved exactly as Proposition 3.9(ii),(iii). (ii) is proved as Theorem 3.13(ii). For (iii), the composite of decapitation with the forgetful functor to $\mathsf{Fam}(\mathcal{C})$ is the functor U of Theorem 3.13; if $\mathcal{C}$ has no terminal object both $\mathsf{Fel}(\mathcal{C})$ and $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})$ are empty, and otherwise U and $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \to \mathsf{Fam}(\mathcal{C})$ are equivalences, so decapitation is one by the two-out-of-three property of equivalences. □

Example 4.21. Let $\mathbf{Set}_{\neq\varnothing}$ be the category of nonempty small sets. It has terminal objects but no initial object, so it has no cats, while $\mathsf{Fel}^{\mathrm{hl}}(\mathbf{Set}_{\neq\varnothing}) \simeq \mathsf{Fam}(\mathbf{Set}_{\neq\varnothing})$. Under the equivalence of Corollary 7.22 below, the latter corresponds to the full subcategory of $\mathbf{Set}^{\to}$ on surjections.

7. Tailless cats

In this section $\mathcal{C}$ is an arbitrary category.

Definition 4.22. A *tailless* (or *acaudal*) $\mathcal{C}$-cat is a pair $(H, (L_\alpha)_{\alpha\in A})$ with H initial, equipped with its shoulders $h_\alpha \colon H \to L_\alpha$. Morphisms are tuples $(\varphi, F_H, (F_\alpha))$ with $F_\alpha h_\alpha = h'_{\varphi(\alpha)} F_H$. The resulting category is $\mathsf{Fel}^{\mathrm{tl}}(\mathcal{C})$.

Proposition 4.23. *Let $\mathcal{C}$ be any category.*

(i) *The forgetful functor $\mathsf{Fel}^{\mathrm{tl}}(\mathcal{C}) \to \mathsf{Fam}(\mathcal{C})$ is fully faithful, and it is an equivalence if and only if $\mathcal{C}$ has an initial object.*

(ii) *Suppose $\mathcal{C}$ has an initial object. The* caudectomy *functor $\mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}^{\mathrm{tl}}(\mathcal{C})$, $(H, T, L) \mapsto (H, L)$, is fully faithful; it is an equivalence if and only if $\mathcal{C}$ has a terminal object.*

Proof. (i) is proved as Proposition 4.20. (ii): full faithfulness follows from Proposition 3.9. If $\mathcal{C}$ has a terminal object then caudectomy is an equivalence by two-out-of-three as before. If $\mathcal{C}$ has no terminal object then $\mathsf{Fel}(\mathcal{C})$ is empty but $\mathsf{Fel}^{\mathrm{tl}}(\mathcal{C})$ contains $(0, \varnothing)$. □

Example 4.24. For the poset $(\mathbb{N}, \leq)$ we have $\mathsf{Fel}(\mathbb{N}) = \varnothing$ but $\mathsf{Fel}^{\mathrm{tl}}(\mathbb{N}) \simeq \mathsf{Fam}(\mathbb{N})$, whose objects are small families of natural numbers $(n_\alpha)_{\alpha\in A}$ and whose morphisms $(n_\alpha)_A \to (m_\beta)_B$ are functions $\varphi \colon A \to B$ with $n_\alpha \leq m_{\varphi(\alpha)}$.

8. Summary of the anatomical variants

Let $\mathsf{Fel}^{\mathrm{hl,tl}}(\mathcal{C})$ denote the category of cats with neither head nor tail; by definition, it is $\mathsf{Fam}(\mathcal{C})$. The preceding results combine as follows.

Theorem 4.25. *For an arbitrary category $\mathcal{C}$ (with $\mathsf{Fel}(\mathcal{C})$ interpreted as empty when $\mathcal{C}$ lacks an initial or a terminal object):*

(i) $\mathsf{Fel}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ *if and only if $\mathcal{C}$ has an initial and a terminal object;*

(ii) $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ *if and only if $\mathcal{C}$ has a terminal object;*

(iii) $\mathsf{Fel}^{\mathrm{tl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ *if and only if $\mathcal{C}$ has an initial object;*

(iv) $\mathsf{Fel}^{\mathrm{hl,tl}}(\mathcal{C}) = \mathsf{Fam}(\mathcal{C})$.

In each case the equivalence is the functor forgetting the head and the tail.

PROOF. In each case the forgetful functor is fully faithful by the Parallel Morphism Lemma, and it is essentially surjective exactly when the required extremal objects exist, because $\mathsf{Fam}(\mathcal{C})$ is never empty. □

Remark 4.26. Theorem 4.25 shows that the head and the tail contribute nothing to the structure of the category of cats, but the existence of an initial and a terminal object in $\mathcal{C}$ determines whether cats exist at all. The existence of an initial object also bears on the relation between $\mathcal{C}$ and $\mathsf{Fam}(\mathcal{C})$, as shown in Chapter 6: when $\mathcal{C}$ has an initial object, $\mathsf{Fam}(\mathcal{C})$ has limits of a given shape exactly when $\mathcal{C}$ does (Theorem 6.8), whereas without one $\mathsf{Fam}(\mathcal{C})$ may have limits that $\mathcal{C}$ lacks (Counterexample 6.13).

CHAPTER 5

Feline functors and natural transformations

Standing Assumption 5.1. Throughout this chapter, $\mathcal{C}$, $\mathcal{D}$ and $\mathcal{E}$ are categories each possessing an initial and a terminal object, chosen as in Standing Assumption 3.18. Functors between them are *not* assumed to preserve initial or terminal objects.

1. Felinization

Definition 5.2. Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor. Its *felinization* is the functor

$$\begin{aligned} \mathsf{Fel}(F)\colon \mathsf{Fel}(\mathcal{C}) &\longrightarrow \mathsf{Fel}(\mathcal{D}), \\ (H, T, (L_\alpha)_{\alpha\in A}) &\longmapsto \langle FL_\alpha\rangle_{\alpha\in A}, \\ (\varphi, (f_\alpha)) &\longmapsto (\varphi, (Ff_\alpha)). \end{aligned}$$

For a natural transformation $\tau\colon F \Rightarrow G$ between functors $\mathcal{C} \to \mathcal{D}$, its felinization $\mathsf{Fel}(\tau)\colon \mathsf{Fel}(F) \Rightarrow \mathsf{Fel}(G)$ has components $\mathsf{Fel}(\tau)_K = (\mathrm{id}_A, (\tau_{L_\alpha})_{\alpha\in A})$.

The head and tail of $\mathsf{Fel}(F)(K)$ are the chosen objects $0_\mathcal{D}$ and $1_\mathcal{D}$, not $F(H)$ and $F(T)$; this is why no preservation hypothesis on F is needed.

Proposition 5.3. (i) *$\mathsf{Fel}(F)$ is a functor and $\mathsf{Fel}(\tau)$ is a natural transformation.*

(ii) *$\mathsf{Fel}(G \circ F) = \mathsf{Fel}(G) \circ \mathsf{Fel}(F)$ for $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$, and $\mathsf{Fel}(\mathrm{id}_\mathcal{C}) = N$ is the normalization functor of Proposition 3.20, which is naturally isomorphic to $\mathrm{id}_{\mathsf{Fel}(\mathcal{C})}$.*

(iii) *Fel preserves vertical composition and identities of natural transformations, and whiskering: $\mathsf{Fel}(W\tau) = \mathsf{Fel}(W)\mathsf{Fel}(\tau)$ and $\mathsf{Fel}(\tau F') = \mathsf{Fel}(\tau)\mathsf{Fel}(F')$.*

(iv) *$\pi_\mathcal{D} \circ \mathsf{Fel}(F) = \pi_\mathcal{C}$ and $\mathsf{Fel}(F) \circ \iota_\mathcal{C} = \iota_\mathcal{D} \circ F$.*

In particular the restriction of Fel to normalized cats is a strict 2-functor $\mathsf{Fel}^{\mathrm{nrm}}$ from the 2-category of locally small categories with chosen initial and terminal objects, arbitrary functors, and arbitrary natural transformations, to the 2-category of (large) locally small categories.

Proof. (i) $\mathsf{Fel}(F)$ preserves composition because

$$(\psi, (Fg_\beta)) \circ (\varphi, (Ff_\alpha)) = (\psi\varphi, (Fg_{\varphi(\alpha)}Ff_\alpha)) = (\psi\varphi, (F(g_{\varphi(\alpha)}f_\alpha))),$$

and preserves identities since F does. Naturality of $\mathsf{Fel}(\tau)$ at $(\varphi, f)\colon K \to K'$ requires

$$(\varphi, (Gf_\alpha)) \circ (\mathrm{id}_A, (\tau_{L_\alpha})) = (\mathrm{id}_B, (\tau_{L'_\beta})) \circ (\varphi, (Ff_\alpha)),$$

that is, $Gf_\alpha \circ \tau_{L_\alpha} = \tau_{L'_{\varphi(\alpha)}} \circ Ff_\alpha$, which is naturality of τ. (ii) Both sides send K to $\langle GFL_\alpha \rangle$ and (φ, f) to (φ, GFf). The statement about $\mathrm{id}_\mathcal{C}$ is the definition of N. (iii) and (iv) are verified componentwise in the same manner. The final assertion follows since on normalized cats N is the identity. □

Example 5.4. Let $\mathbb{1}$ be the terminal category, and recall $\mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$ (Proposition 3.24). The functors $\mathbb{1} \to \mathcal{C}$ correspond to objects of $\mathcal{C}$; we write $\ulcorner X \urcorner \colon \mathbb{1} \to \mathcal{C}$ for the functor with value X. Then, under $\mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$:

(i) $\mathsf{Fel}(\ulcorner X \urcorner) \colon \mathbf{Set} \to \mathsf{Fel}(\mathcal{C})$ sends a set S to the cat $\langle X \rangle_{s \in S}$ with S legs, each equal to X;
(ii) $\mathsf{Fel}(!_\mathcal{C}) \colon \mathsf{Fel}(\mathcal{C}) \to \mathbf{Set}$, for the unique functor $!_\mathcal{C} \colon \mathcal{C} \to \mathbb{1}$, is the leg functor π.

We write $\Delta_X \coloneqq \mathsf{Fel}(\ulcorner X \urcorner)$. The functors Δ_0 and Δ_1 will play a central rôle in Chapter 7.

Proposition 5.5. *Let $F \colon \mathcal{C} \to \mathcal{D}$ be a functor and let P be one of the properties* faithful, full, fully faithful, essentially surjective, equivalence. *Then F has P if and only if $\mathsf{Fel}(F)$ has P.*

Proof. By Corollary 3.11 the action of $\mathsf{Fel}(F)$ on hom-sets is the map

$$\coprod_\varphi \prod_\alpha \mathcal{C}(L_\alpha, L'_{\varphi(\alpha)}) \longrightarrow \coprod_\varphi \prod_\alpha \mathcal{D}(FL_\alpha, FL'_{\varphi(\alpha)})$$

given by F in each factor. A coproduct of products of injections is injective, and a coproduct of products of surjections is surjective (the latter using the axiom of choice for infinite products). Hence F faithful (resp. full) implies $\mathsf{Fel}(F)$ faithful (resp. full). If F is essentially surjective, choose for each leg Y_β of a $\mathcal{D}$-cat an object X_β with an isomorphism $FX_\beta \cong Y_\beta$; then $\mathsf{Fel}(F)\langle X_\beta \rangle \cong \langle Y_\beta \rangle$ by Proposition 3.23. Conversely, since $\mathsf{Fel}(F)\iota_\mathcal{C} = \iota_\mathcal{D} F$ and $\iota_\mathcal{C}, \iota_\mathcal{D}$ are fully faithful (Proposition 3.22), the action of F on $\mathcal{C}(X, Y)$ is identified with the action of $\mathsf{Fel}(F)$ on $\mathsf{Fel}(\mathcal{C})(\iota X, \iota Y)$; so $\mathsf{Fel}(F)$ faithful or full implies the same for F. If $\mathsf{Fel}(F)$ is essentially surjective then for each Y there is K with $\mathsf{Fel}(F)K \cong \iota Y$; then K is unipedal by Proposition 3.23, say $K \cong \iota X$, and $FX \cong Y$ by full faithfulness of $\iota_\mathcal{D}$. An equivalence is a fully faithful essentially surjective functor. □

2. Feline functors

Definition 5.6. A feline morphism (φ, F) is *reindexing* if each of its leg components F_α is an isomorphism.

For example, the leg inclusions $j_\alpha \colon \iota(L_\alpha) \to K$ are reindexing. In Theorem 10.3 we show that the reindexing morphisms are exactly the cartesian morphisms for the leg functor.

Definition 5.7. A functor $\Phi \colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}(\mathcal{D})$ is *feline* if $\pi_\mathcal{D} \circ \Phi = \pi_\mathcal{C}$. A feline functor is *fibred* if it maps reindexing morphisms to reindexing morphisms. A *feline natural transformation* $\tau \colon \Phi \Rightarrow \Psi$ between feline

functors is a natural transformation each of whose components is vertical. We write $\mathsf{FelFun}(\mathcal{C},\mathcal{D})$ for the category of fibred feline functors and feline natural transformations.

When $\mathcal{C}$ is not small, the categories $\mathsf{Fun}(\mathcal{C},\mathcal{D})$ and $\mathsf{FelFun}(\mathcal{C},\mathcal{D})$ may have large hom-sets; this is an exception to Convention 2.1, which is harmless since we make no use of the size of these hom-sets.

By Proposition 5.3, felinization defines a functor

$$\mathsf{Fel}_{\mathcal{C},\mathcal{D}}\colon \mathsf{Fun}(\mathcal{C},\mathcal{D}) \longrightarrow \mathsf{FelFun}(\mathcal{C},\mathcal{D}),$$

since $\mathsf{Fel}(F)$ is feline and preserves isomorphisms of legs, and $\mathsf{Fel}(\tau)$ is vertical.

Definition 5.8. For a feline functor $\Phi\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}(\mathcal{D})$, its *unipedal restriction* $\Phi^\natural\colon \mathcal{C} \to \mathcal{D}$ is defined as follows. For an object X, the cat $\Phi(\iota X)$ has leg set $\pi(\iota X) = 1$; let $\Phi^\natural(X)$ be its unique leg. For $g\colon X \to Y$, the morphism $\Phi(\iota g)$ has gait $\pi(\iota g) = \mathrm{id}_1$; let $\Phi^\natural(g)$ be its unique leg component. For a feline natural transformation $\tau\colon \Phi \Rightarrow \Psi$ let $\tau^\natural_X$ be the unique leg component of $\tau_{\iota X}$.

$\Phi^\natural$ is a functor because Φ and ι are, and $\tau^\natural$ is natural because τ is.

Theorem 5.9 (Classification of feline functors). *The felinization functor $\mathsf{Fel}_{\mathcal{C},\mathcal{D}}$ from $\mathsf{Fun}(\mathcal{C},\mathcal{D})$ to $\mathsf{FelFun}(\mathcal{C},\mathcal{D})$ is an equivalence of categories. A quasi-inverse is given by unipedal restriction $\Phi \mapsto \Phi^\natural$, and $\mathsf{Fel}(F)^\natural = F$ for every functor F.*

Proof. The identity $\mathsf{Fel}(F)^\natural = F$ holds by construction, as does $\mathsf{Fel}(\tau)^\natural = \tau$. It therefore suffices to show (a) that $\mathsf{Fel}_{\mathcal{C},\mathcal{D}}$ is full, and (b) that every fibred feline functor Φ is isomorphic in $\mathsf{FelFun}(\mathcal{C},\mathcal{D})$ to $\mathsf{Fel}(\Phi^\natural)$. Faithfulness is immediate from $\mathsf{Fel}(\tau)^\natural = \tau$. Once $\mathsf{Fel}_{\mathcal{C},\mathcal{D}}$ is known to be an equivalence, unipedal restriction $R\colon \Phi \mapsto \Phi^\natural$, which is a functor $\mathsf{FelFun}(\mathcal{C},\mathcal{D}) \to \mathsf{Fun}(\mathcal{C},\mathcal{D})$ with $R \circ \mathsf{Fel}_{\mathcal{C},\mathcal{D}} = \mathrm{id}$, is a quasi-inverse: if E is a quasi-inverse of $\mathsf{Fel}_{\mathcal{C},\mathcal{D}}$, then $R \cong R \circ \mathsf{Fel}_{\mathcal{C},\mathcal{D}} \circ E = E$.

We first establish an identity in $\mathsf{Fel}(\mathcal{C})$. Let $(\psi, G)\colon K \to K'$ be a morphism, with $K = \langle L_\alpha\rangle_{\alpha\in A}$ up to normalization, K' with leg set B, and $\alpha \in A$. Then

$$(\psi, G) \circ j_\alpha = j_{\psi(\alpha)} \circ \iota(G_\alpha)\colon \iota(L_\alpha) \longrightarrow K', \tag{5.1}$$

since both sides have gait $1 \to B$ with value $\psi(\alpha)$ and leg component G_α.

(b) Let Φ be a fibred feline functor and K a cat with leg set A. For $\alpha \in A$, the morphism $\Phi(j_\alpha)\colon \Phi(\iota L_\alpha) \to \Phi(K)$ has gait $\pi(j_\alpha) = \alpha$ and is reindexing, since j_α is; so its unique leg component is an isomorphism

$$c_{K,\alpha}\colon \Phi^\natural(L_\alpha) \xrightarrow{\cong} \Phi(K)_\alpha,$$

where $\Phi(K)_\alpha$ denotes the α-th leg of $\Phi(K)$. Define $\theta_K\colon \Phi(K) \to \mathsf{Fel}(\Phi^\natural)(K)$ to be the vertical morphism with components $c^{-1}_{K,\alpha}$. It is an isomorphism by

Proposition 3.23. For naturality, let $(\psi, G)\colon K \to K'$. Applying Φ to (5.1) and comparing the leg components of the two sides gives

$$\Phi(\psi, G)_\alpha \circ c_{K,\alpha} = c_{K',\psi(\alpha)} \circ \Phi^\natural(G_\alpha),$$

where $\Phi(\psi, G)_\alpha\colon \Phi(K)_\alpha \to \Phi(K')_{\psi(\alpha)}$ is the α-th leg component of $\Phi(\psi, G)$, whose gait is ψ because Φ is feline. Rearranged, this says $c^{-1}_{K',\psi(\alpha)} \circ \Phi(\psi, G)_\alpha = \Phi^\natural(G_\alpha) \circ c^{-1}_{K,\alpha}$ for all α, which is the equation $\theta_{K'} \circ \Phi(\psi, G) = \mathsf{Fel}(\Phi^\natural)(\psi, G) \circ \theta_K$.

(a) Let $\sigma\colon \mathsf{Fel}(F) \Rightarrow \mathsf{Fel}(G)$ be a feline natural transformation, and set $\tau = \sigma^\natural\colon F \Rightarrow G$. For a cat K and $\alpha \in \pi(K)$, naturality of σ with respect to j_α gives $\sigma_K \circ \mathsf{Fel}(F)(j_\alpha) = \mathsf{Fel}(G)(j_\alpha) \circ \sigma_{\iota L_\alpha}$. Both $\mathsf{Fel}(F)(j_\alpha)$ and $\mathsf{Fel}(G)(j_\alpha)$ have identity leg components, so comparing leg components yields $(\sigma_K)_\alpha = \tau_{L_\alpha}$. Hence $\sigma = \mathsf{Fel}(\tau)$. □

Remark 5.10. Theorem 5.9 is the feline form of the familiar fact that fibred functors between family fibrations $\mathsf{Fam}(\mathcal{C}) \to \mathbf{Set}$ and $\mathsf{Fam}(\mathcal{D}) \to \mathbf{Set}$ correspond, up to isomorphism, to functors $\mathcal{C} \to \mathcal{D}$; see [4] for the general theory of fibred functors.

Counterexample 5.11 (A feline functor that is not fibred)**.** Suppose $\mathcal{C}$ has small coproducts. Define $\Phi_\Sigma\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}(\mathcal{C})$ by

$$\Phi_\Sigma(K) = \Big\langle \coprod_{\gamma\in A} L_\gamma \Big\rangle_{\alpha\in A}, \qquad \Phi_\Sigma(\varphi, F) = \Big(\varphi, \big(\textstyle\coprod_\gamma F_\gamma\big)_{\alpha\in A}\Big),$$

where $\coprod_\gamma F_\gamma\colon \coprod_{\gamma\in A} L_\gamma \to \coprod_{\delta\in B} L'_\delta$ is the morphism whose restriction to the γ-th summand is $\mathrm{in}_{\varphi(\gamma)} \circ F_\gamma$. Thus every leg of $\Phi_\Sigma(K)$ is a copy of the coproduct of all the legs of K. This is a feline functor. It is not fibred when $\mathcal{C} = \mathbf{Set}$: for the two-legged cat $K = \langle\{*\}, \{*\}\rangle_{\{a,b\}}$, the leg inclusion $j_a\colon \iota\{*\} \to K$ is reindexing, but $\Phi_\Sigma(j_a)$ has leg component the coprojection $1 \to 1 \sqcup 1$, which is not an isomorphism. Consequently Φ_Σ is not isomorphic, via a feline natural isomorphism, to $\mathsf{Fel}(F)$ for any functor F, since fibredness is invariant under vertical natural isomorphism. The hypothesis of fibredness in Theorem 5.9 is therefore necessary.

3. Feline adjunctions and equivalences

Proposition 5.12. *Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{C}$ be functors. Then $F \dashv G$ if and only if $\mathsf{Fel}(F) \dashv \mathsf{Fel}(G)$. More precisely, every adjunction $F \dashv G$ induces an adjunction $\mathsf{Fel}(F) \dashv \mathsf{Fel}(G)$, and every adjunction $\mathsf{Fel}(F) \dashv \mathsf{Fel}(G)$ restricts along ι to an adjunction $F \dashv G$.*

Proof. Suppose given natural bijections $\vartheta_{X,Y}\colon \mathcal{D}(FX, Y) \cong \mathcal{C}(X, GY)$. For $K = \langle L_\alpha\rangle_A$ and $K' = \langle L'_\beta\rangle_B$, Corollary 3.11 gives bijections

$$\begin{aligned}\mathsf{Fel}(\mathcal{D})(\mathsf{Fel}(F)K, K') &\cong \prod_\alpha \coprod_\beta \mathcal{D}(FL_\alpha, L'_\beta) \xrightarrow{\prod\coprod \vartheta} \prod_\alpha \coprod_\beta \mathcal{C}(L_\alpha, GL'_\beta)\\ &\cong \mathsf{Fel}(\mathcal{C})(K, \mathsf{Fel}(G)K'),\end{aligned}$$

which preserve gaits. Naturality in K and K' reduces, leg by leg, to naturality of ϑ. (For non-normalized K we compose with the normalization isomorphism.) Conversely, an adjunction $\mathsf{Fel}(F) \dashv \mathsf{Fel}(G)$ gives natural bijections

$$\mathcal{D}(FX, Y) \cong \mathsf{Fel}(\mathcal{D})(\mathsf{Fel}(F)\iota X, \iota Y) \cong \mathsf{Fel}(\mathcal{C})(\iota X, \mathsf{Fel}(G)\iota Y) \cong \mathcal{C}(X, GY),$$

using $\mathsf{Fel}(F)\iota = \iota F$, $\mathsf{Fel}(G)\iota = \iota G$ and the full faithfulness of ι. □

Corollary 5.13. *There are adjunctions* $\Delta_0 \dashv \pi \dashv \Delta_1$ *between* **Set** $\cong \mathsf{Fel}(\mathbb{1})$ *and* $\mathsf{Fel}(\mathcal{C})$.

Proof. An object X of $\mathcal{C}$ is initial if and only if $\ulcorner X \urcorner \colon \mathbb{1} \to \mathcal{C}$ is left adjoint to $!_{\mathcal{C}} \colon \mathcal{C} \to \mathbb{1}$, and terminal if and only if $\ulcorner X \urcorner$ is right adjoint to $!_{\mathcal{C}}$. Hence $\ulcorner 0 \urcorner \dashv !_{\mathcal{C}} \dashv \ulcorner 1 \urcorner$. Apply Proposition 5.12 and Example 5.4. □

Corollary 5.13 will be extended in Theorem 7.3. We note that the adjunctions $\Delta_0 \dashv \pi \dashv \Delta_1$ are merely the felinizations of the definitions of initial and terminal object.

Corollary 5.14. $\mathcal{C} \simeq \mathcal{D}$ *implies* $\mathsf{Fel}(\mathcal{C}) \simeq \mathsf{Fel}(\mathcal{D})$*, by a feline equivalence.*

Proof. Apply Proposition 5.5 to an equivalence $F \colon \mathcal{C} \to \mathcal{D}$. □

The converse of Corollary 5.14 holds as well, for arbitrary (not necessarily feline) equivalences; see Theorem 7.17.

4. Functors that are not feline

Many natural endofunctors of $\mathsf{Fel}(\mathcal{C})$ change the number of legs. For a fixed cat M, the functor $K \mapsto K \sqcup M$ (with $\sqcup$ the coproduct of Proposition 6.5) has leg sets $\pi(K) \sqcup \pi(M)$, and, if $\mathcal{C}$ has binary products, the functor $K \mapsto K \times M$ of Theorem 6.8 has leg sets $\pi(K) \times \pi(M)$. If M is not legless, respectively not unipedal, these functors are not isomorphic to any feline functor, since they change the cardinality of the leg set of some cat (for instance of $\iota(0)$), and a natural isomorphism preserves leg sets up to bijection.

By contrast, the functors $\Delta_1 \circ \pi$, which replaces every leg by the tail, and $\Delta_0 \circ \pi$, which replaces every leg by the head, are feline, and in fact fibred, since every morphism in their image has identity leg components; in accordance with Theorem 5.9 they are the felinizations of the constant functors $\mathcal{C} \to \mathcal{C}$ with values 1 and 0. They are also the monad $\Delta_1\pi$ of the adjunction $\pi \dashv \Delta_1$ and the comonad $\Delta_0\pi$ of the adjunction $\Delta_0 \dashv \pi$.

CHAPTER 6

Limits and colimits

Standing Assumption 6.1. Throughout this chapter $\mathcal{C}$ has an initial object 0 and a terminal object 1. In Counterexamples 6.13 and 6.19 we consider headless cats in categories without initial objects; this is stated explicitly there.

1. Extremal objects

We have already shown (Proposition 4.4) that the initial objects of $\mathsf{Fel}(\mathcal{C})$ are precisely the legless cats, and that the initial object is strict.

Proposition 6.2. *The caudal cat $\mathbf{1} = \iota(1)$ is a terminal object of $\mathsf{Fel}(\mathcal{C})$. A cat is terminal if and only if it is unipedal and its leg is a terminal object of $\mathcal{C}$.*

Proof. For every cat K', $\mathsf{Fel}(\mathcal{C})(K', \mathbf{1}) \cong \prod_\alpha \mathcal{C}(L'_\alpha, 1)$ by Corollary 3.11, which is a singleton. Terminal objects are isomorphic to $\mathbf{1}$, and by Proposition 3.23 the cats isomorphic to $\mathbf{1}$ are precisely the unipedal cats whose leg is isomorphic to 1. □

Lemma 6.3. *The leg functor is representable by the cephalic cat: there is a natural isomorphism $\pi \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}, -)$, whose component at K sends $\alpha \in \pi(K)$ to the morphism $(\alpha, !_{L_\alpha}) \colon \mathbf{h} \to K$.*

Proof. By Corollary 3.11, $\mathsf{Fel}(\mathcal{C})(\mathbf{h}, K) \cong \coprod_\alpha \mathcal{C}(0, L_\alpha)$, and each $\mathcal{C}(0, L_\alpha)$ is a singleton. Naturality: for $(\varphi, F) \colon K \to K'$, $(\varphi, F) \circ (\alpha, !) = (\varphi(\alpha), F_\alpha \circ !) = (\varphi(\alpha), !)$. □

Remark 6.4. The head of any particular cat carries no information. What Lemma 6.3 uses is the existence of an initial object in $\mathcal{C}$, which makes the set of legs of a cat *representable*, so that the leg functor preserves all limits. In Proposition 8.14 we show that, for headless cats in a category $\mathcal{C}$ with a terminal object, the leg functor is representable if and only if $\mathcal{C}$ has an initial object.

2. Coproducts and extensivity

Proposition 6.5. *Let $(K_i)_{i \in I}$ be a small family of cats, where K_i has legs $(L_{i,\alpha})_{\alpha \in A_i}$. The cat*

$$\coprod_{i \in I} K_i \coloneqq \langle L_{i,\alpha} \rangle_{(i,\alpha) \in \coprod_i A_i},$$

with coprojections $\mathrm{in}_i = (\alpha \mapsto (i,\alpha), \mathrm{id})\colon K_i \to \coprod_i K_i$*, is a coproduct of the family. In particular* $\mathsf{Fel}(\mathcal{C})$ *has all small coproducts, and* π *preserves them.*

PROOF. By Corollary 3.11, for any cat M with legs $(M_\beta)_{\beta\in B}$,

$$\mathsf{Fel}(\mathcal{C})\Big(\coprod_i K_i, M\Big) \cong \prod_{(i,\alpha)} \coprod_\beta \mathcal{C}(L_{i,\alpha}, M_\beta) = \prod_i \prod_{\alpha\in A_i} \coprod_\beta \mathcal{C}(L_{i,\alpha}, M_\beta)$$
$$\cong \prod_i \mathsf{Fel}(\mathcal{C})(K_i, M),$$

and the composite bijection is precomposition with the coprojections. □

Corollary 6.6. *Every cat* K *is the coproduct of its unipedal subcats: the leg inclusions* $j_\alpha\colon \iota(L_\alpha) \to K$ *exhibit* K *as* $\coprod_\alpha \iota(L_\alpha)$*.*

PROOF. Apply Proposition 6.5 to the family $(\iota L_\alpha)_\alpha$; the resulting coproduct is isomorphic to K by an isomorphism with bijective gait $(\alpha, *) \mapsto \alpha$ and identity components, under which the coprojections correspond to the j_α. □

Recall that a category $\mathcal{E}$ with small coproducts is (infinitary) *extensive* if, for every small family $(X_i)_{i\in I}$ of objects, the functor

$$\coprod\colon \prod_{i\in I} \mathcal{E}/X_i \longrightarrow \mathcal{E}\Big/\coprod_i X_i, \qquad (Y_i \to X_i)_i \longmapsto \Big(\coprod_i Y_i \to \coprod_i X_i\Big) \tag{6.1}$$

is an equivalence of categories. For the finitary notion and its theory see [**2**].

Proposition 6.7. $\mathsf{Fel}(\mathcal{C})$ *is extensive. Moreover coproducts in* $\mathsf{Fel}(\mathcal{C})$ *are disjoint: for* $i \neq j$ *the pullback of* in_i *and* in_j *exists and is the legless cat.*

PROOF. Let $(K_i)_{i\in I}$ be a family with leg sets A_i, and let $(\gamma, G)\colon M \to \coprod_i K_i$ be an object of the slice over the coproduct, M having leg set B. For each i let $B_i = \gamma^{-1}(\{i\} \times A_i)$ and let M_i be the cat $\langle M_\beta\rangle_{\beta\in B_i}$ with the morphism $G_i\colon M_i \to K_i$ given by $\beta \mapsto \gamma(\beta)$ (read in A_i) and the components G_β. Since $B = \coprod_i B_i$, there is an isomorphism over $\coprod_i K_i$ between $\coprod_i (M_i \to K_i)$ and $(M \to \coprod_i K_i)$ with bijective gait and identity components; so (6.1) is essentially surjective. A morphism over $\coprod_i K_i$ between $\coprod_i M_i$ and $\coprod_i M_i'$ has a gait compatible with the maps to $\coprod_i A_i$, hence maps B_i into B_i' for each i; it is therefore the coproduct of a unique family of morphisms $M_i \to M_i'$ over K_i. So (6.1) is fully faithful. For disjointness, a cone $(P \to K_i, P \to K_j)$ over the cospan $(\mathrm{in}_i, \mathrm{in}_j)$ must have a gait sending each leg of P into both $\{i\} \times A_i$ and $\{j\} \times A_j$, which are disjoint; hence P is legless, and the legless cat, being initial, is the pullback. □

3. Limits

THEOREM 6.8. *Let* $\mathcal{J}$ *be a small category. Then* $\mathsf{Fel}(\mathcal{C})$ *has all limits of shape* $\mathcal{J}$ *if and only if* $\mathcal{C}$ *does. In that case, the limit of a diagram*

$D\colon \mathcal{J} \to \mathsf{Fel}(\mathcal{C})$, $j \mapsto K_j = \langle L_{j,a}\rangle_{a\in A_j}$, $(u\colon j \to j') \mapsto (\varphi_u, F_u)$, *is*

$$\lim_j K_j = \Big\langle \lim_{j\in\mathcal{J}} L_{j,a_j} \Big\rangle_{(a_j)_j \in \lim_j A_j},$$

where $\lim_j A_j$ *is computed in* **Set** *and, for* $a = (a_j)_j \in \lim_j A_j$*, the limit* $\lim_j L_{j,a_j}$ *is that of the diagram* $D_a\colon \mathcal{J} \to \mathcal{C}$ *given by* $j \mapsto L_{j,a_j}$ *and* $u \mapsto (F_u)_{a_j}$*. The leg functor preserves all limits that exist.*

Proof. The leg functor preserves all limits because it is representable (Lemma 6.3).

Sufficiency. Suppose $\mathcal{C}$ has limits of shape $\mathcal{J}$. For $a = (a_j) \in \lim_j A_j$ we have $\varphi_u(a_j) = a_{j'}$ for every $u\colon j \to j'$, so $(F_u)_{a_j}\colon L_{j,a_j} \to L_{j',a_{j'}}$ is well typed, and functoriality of D (that is, $(\varphi_{vu}, F_{vu}) = (\varphi_v, F_v)(\varphi_u, F_u)$, whose components give $(F_{vu})_{a_j} = (F_v)_{a_{j'}}(F_u)_{a_j}$) shows that D_a is a functor. Choose a limit $(\Lambda_a, (p_{a,j})_j)$ of D_a and set $\Lambda = \langle \Lambda_a\rangle_{a\in\lim A_j}$. Define $P_j\colon \Lambda \to K_j$ to have gait the projection $\mathrm{pr}_j\colon \lim_j A_j \to A_j$ and components $p_{a,j}$. These form a cone: $(\varphi_u, F_u)P_j$ has gait $\varphi_u \mathrm{pr}_j = \mathrm{pr}_{j'}$ and components $(F_u)_{a_j} p_{a,j} = p_{a,j'}$. Let $(Q_j = (\psi_j, G_j)\colon M \to K_j)_j$ be another cone, $M = \langle M_b\rangle_{b\in B}$. The gaits ψ_j form a cone in **Set**, so there is a unique $\psi\colon B \to \lim_j A_j$ with $\mathrm{pr}_j\psi = \psi_j$. For each b, the morphisms $(G_j)_b\colon M_b \to L_{j,\psi_j(b)}$ form a cone over $D_{\psi(b)}$, by the cone condition on the Q_j; hence there is a unique $g_b\colon M_b \to \Lambda_{\psi(b)}$ with $p_{\psi(b),j} g_b = (G_j)_b$. The morphism $(\psi, (g_b))\colon M \to \Lambda$ satisfies $P_j(\psi, g) = Q_j$, and it is the unique such morphism because its gait and components are forced.

Necessity. Suppose $\mathsf{Fel}(\mathcal{C})$ has limits of shape $\mathcal{J}$ and let $E\colon \mathcal{J} \to \mathcal{C}$ be a diagram. Let Λ be a limit of ιE. Since π preserves limits, $\pi(\Lambda) \cong \lim_j \pi(\iota E_j) = \lim_j 1$, which is a singleton. Hence Λ is unipedal, so $\Lambda \cong \iota(X)$ for some X (Proposition 3.22), and the limit cone lies, up to isomorphism, in the image of the fully faithful functor ι. Fully faithful functors reflect limits, so X is a limit of E. □

Corollary 6.9. *Suppose the relevant limits exist in* $\mathcal{C}$*.*

(i) *The product of cats* $K = \langle L_\alpha\rangle_A$ *and* $K' = \langle L'_\beta\rangle_B$ *is* $K \times K' = \langle L_\alpha \times L'_\beta\rangle_{(\alpha,\beta)\in A\times B}$*.*
(ii) *The equalizer of* $(\varphi, F), (\psi, G)\colon K \to K'$ *is* $\langle \mathrm{Eq}(F_\alpha, G_\alpha)\rangle_{\alpha\in A_0}$ *where* $A_0 = \{\alpha : \varphi(\alpha) = \psi(\alpha)\}$*.*
(iii) *The pullback of* $K \to M \leftarrow K'$ *with gaits* $\varphi\colon A \to \pi M$ *and* $\psi\colon B \to \pi M$ *has legs* $L_\alpha \times_{M_{\varphi(\alpha)}} L'_\beta$ *indexed by* $\{(\alpha, \beta) \in A \times B : \varphi(\alpha) = \psi(\beta)\}$*.*

Proof. These are the cases of Theorem 6.8 for the discrete category with two objects, the parallel pair, and the cospan. □

Corollary 6.10. *The full subcategory* $\mathsf{Kit}(\mathcal{C})$ *of kittens is closed under all finite limits that exist in* $\mathsf{Fel}(\mathcal{C})$*.*

PROOF. By Theorem 6.8, the leg set of a limit of a finite diagram of kittens is the limit of a finite diagram of finite sets, which is a subset of a finite product of finite sets. □

Remark 6.11. The full subcategory of cats satisfying the Chloe Condition (Definition 4.15) is closed under coproducts (Proposition 4.16) but not, in general, under products. In the category **Gph** of Example 2.11, which has an initial object, a terminal object (a vertex with a loop) and finite products, the cat $\iota(K_2)$, whose leg is the graph with two vertices joined by an edge, is Chloe, while $\iota(K_2) \times \iota(K_2) \cong \iota(K_2 \times K_2)$ is not, because $K_2 \times K_2$ is disconnected (Example 13.15). In **Set** and **Top**, where products of connected objects are connected and the terminal object is connected, Chloe cats are closed under finite products.

Remark 6.12. The sufficiency half of Theorem 6.8 is the elementary computation of limits in $\mathsf{Fam}(\mathcal{C})$. The necessity half depends on the initial object of $\mathcal{C}$, through Lemma 6.3. The following counterexample shows that it fails for headless cats.

Counterexample 6.13. Let $\mathcal{C}$ be the poset $\{x, y, \top\}$ with $x < \top$ and $y < \top$ and x, y incomparable. It has a terminal object $\top$ and no initial object, and the product $x \times y$ (that is, the meet $x \wedge y$) does not exist, since x and y have no common lower bound. Nevertheless the category of headless cats $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ has all binary products. For families $(X_a)_{a\in A}$ and $(Y_b)_{b\in B}$, let $P = \{(a,b) \in A \times B : X_a \wedge Y_b \text{ exists}\}$, which excludes exactly the pairs with $\{X_a, Y_b\} = \{x, y\}$, and set $Z_{(a,b)} = X_a \wedge Y_b$, with projections $(P, Z) \to (A, X)$ and $(P, Z) \to (B, Y)$ given by $(a, b) \mapsto a$ and $(a, b) \mapsto b$ and the inequalities $Z_{(a,b)} \le X_a$, $Z_{(a,b)} \le Y_b$. For an object W of $\mathcal{C}$,

$$\begin{aligned}\mathsf{Fam}(\mathcal{C})\big((1, W), (P, Z)\big) &= \{(a,b) \in P : W \le X_a \wedge Y_b\} \\ &= \{(a,b) \in A \times B : W \le X_a,\ W \le Y_b\},\end{aligned}$$

because $W \le X_a$ and $W \le Y_b$ force $X_a \wedge Y_b$ to exist. The right-hand side is $\mathsf{Fam}(\mathcal{C})((1, W), (A, X)) \times \mathsf{Fam}(\mathcal{C})((1, W), (B, Y))$, naturally in W, and the bijection is composition with the two projections. Since every family is a coproduct of one-element families and $\mathsf{Fam}(\mathcal{C})(\coprod_i (1, W_i), -) \cong \prod_i \mathsf{Fam}(\mathcal{C})((1, W_i), -)$, it follows that (P, Z) is a product. In particular the product of $(1, x)$ and $(1, y)$ is the empty family.

4. Colimits

Recall that a category is *connected* if it is nonempty and any two objects are joined by a zigzag, and that a *connected colimit* is a colimit of a diagram indexed by a small connected category.

THEOREM 6.14. *The category* $\mathsf{Fel}(\mathcal{C})$ *has all small colimits if and only if* $\mathcal{C}$ *has all small connected colimits. In that case, the colimit of a diagram* $D\colon \mathcal{J} \to \mathsf{Fel}(\mathcal{C})$*, with notation as in Theorem 6.8, is computed as follows. Let* $\mathcal{G} = \int(\pi D)$ *be the category of elements of the functor* $\pi D\colon \mathcal{J} \to$ **Set**,

with objects (j, a), $a \in A_j$, *and morphisms* $u\colon (j,a) \to (j', \varphi_u(a))$. *For each connected component* c *of* $\mathcal{G}$ *let* $D_c\colon c \to \mathcal{C}$ *be the functor* $(j,a) \mapsto L_{j,a}$, $u \mapsto (F_u)_a$. *Then*

$$\operatorname*{colim}_j K_j = \left\langle \operatorname{colim} D_c \right\rangle_{c \in \pi_0(\mathcal{G})},$$

and π *preserves all colimits that exist.*

Proof. The leg functor preserves all colimits because it is a left adjoint (Corollary 5.13).

Sufficiency. Suppose $\mathcal{C}$ has small connected colimits. Each component c of $\mathcal{G}$ is a small connected category, so $\operatorname{colim} D_c$ exists; let $q_{(j,a)}\colon L_{j,a} \to \operatorname{colim} D_c$ be its coprojections, for $(j,a) \in c$. By Lemma 2.17, $\operatorname{colim}_j A_j \cong \pi_0(\mathcal{G})$ with coprojections $a \mapsto [(j,a)]$. Let $\Xi = \langle \operatorname{colim} D_c \rangle_{c \in \pi_0(\mathcal{G})}$ and define $Q_j\colon K_j \to \Xi$ with gait $a \mapsto [(j,a)]$ and components $q_{(j,a)}$. These form a cocone, because $q_{(j',\varphi_u(a))} \circ (F_u)_a = q_{(j,a)}$ is the cocone condition for D_c. Given another cocone $(R_j = (\rho_j, G_j)\colon K_j \to M)_j$ with $M = \langle M_b \rangle_{b \in B}$, the gaits form a cocone in **Set** and induce a unique $\rho\colon \pi_0(\mathcal{G}) \to B$ with $\rho([(j,a)]) = \rho_j(a)$. For each component c, the morphisms $(G_j)_a\colon L_{j,a} \to M_{\rho(c)}$, for $(j,a) \in c$, form a cocone on D_c by the cocone condition on the R_j, and so induce a unique $g_c\colon \operatorname{colim} D_c \to M_{\rho(c)}$. The morphism $(\rho, (g_c))\colon \Xi \to M$ is the unique morphism compatible with the cocones.

Necessity. Suppose $\mathsf{Fel}(\mathcal{C})$ has small colimits, and let $E\colon \mathcal{J} \to \mathcal{C}$ with $\mathcal{J}$ small and connected. Let Ξ be a colimit of ιE. Since π preserves colimits, $\pi(\Xi) \cong \operatorname{colim}_{\mathcal{J}} 1 \cong \pi_0(\mathcal{J})$, a singleton by Lemma 2.17. Hence $\Xi \cong \iota X$ for some X, and X is a colimit of E because ι is fully faithful. □

Remark 6.15. The same proof shows that $\mathsf{Fel}(\mathcal{C})$ has all small connected colimits if and only if $\mathcal{C}$ does; since $\mathsf{Fel}(\mathcal{C})$ always has small coproducts and every colimit is a coproduct of connected colimits (over the components of the indexing category), this is equivalent to Theorem 6.14. The statement for colimits of a *fixed* connected shape is more delicate, because the components of $\mathcal{G}$ need not have the same shape as $\mathcal{J}$, nor be finite when $\mathcal{J}$ is. The following counterexample shows that finite cocompleteness of $\mathcal{C}$ does not imply that $\mathsf{Fel}(\mathcal{C})$ has coequalizers.

Counterexample 6.16. Let $\mathcal{C}$ be the totally ordered set $\mathbb{Q} \cap [0,1]$, regarded as a category. It has an initial object 0, a terminal object 1, and all finite colimits (finite joins are maxima, and coequalizers are trivial in a poset). Let $(q_n)_{n \in \mathbb{N}}$ be a strictly increasing sequence in $\mathcal{C}$ converging to the irrational number $r = 1/\sqrt{2}$, and let $K = \langle q_n \rangle_{n \in \mathbb{N}}$. The morphisms

$$f = (\mathrm{id}_{\mathbb{N}}, (q_n \le q_n)_n), \qquad g = (n \mapsto n+1, (q_n \le q_{n+1})_n)\colon K \longrightarrow K$$

have no coequalizer in $\mathsf{Fel}(\mathcal{C})$. Indeed, suppose $e\colon K \to Q$ were a coequalizer. Then $\pi(Q)$ is the coequalizer in **Set** of id and the successor map on $\mathbb{N}$, which is a singleton; so $Q \cong \iota(z)$ for some $z \in \mathcal{C}$, and e amounts to the inequalities $q_n \le z$. For every $w \in \mathcal{C}$, the morphisms $K \to \iota(w)$ coequalizing f and g are all morphisms $K \to \iota(w)$, which exist if and only if $q_n \le w$ for all n, that is,

if and only if $w \geq r$. The universal property therefore requires $z \leq w$ for all rational $w \geq r$ in $\mathcal{C}$ and $z \geq q_n$ for all n, so $z = r$, which is not in $\mathcal{C}$.

Proposition 6.17. *The category $\mathsf{Kit}(\mathcal{C})$ of kittens has all finite colimits if and only if $\mathcal{C}$ has all finite connected colimits; in that case they are computed as in Theorem 6.14 and agree with the colimits in $\mathsf{Fel}(\mathcal{C})$.*

Proof. If $\mathcal{J}$ is finite and each K_j is a kitten, the category $\mathcal{G}$ of Theorem 6.14 is finite, so its components are finite connected categories, and the construction in the proof of Theorem 6.14 uses only finite connected colimits in $\mathcal{C}$. The result is a kitten since $\pi_0(\mathcal{G})$ is finite, and it is a colimit in $\mathsf{Fel}(\mathcal{C})$, hence in the full subcategory $\mathsf{Kit}(\mathcal{C})$. Conversely, the adjunction $\pi \dashv \Delta_1$ restricts to finite sets and kittens, so $\pi\colon \mathsf{Kit}(\mathcal{C}) \to \mathbf{FinSet}$ preserves finite colimits, and the necessity argument of Theorem 6.14 applies verbatim to finite connected $\mathcal{J}$. □

5. Monomorphisms and epimorphisms

Proposition 6.18. *A feline morphism $(\varphi, F)\colon K \to K'$ is a monomorphism if and only if φ is injective and each F_α is a monomorphism in $\mathcal{C}$.*

Proof. Sufficiency: if $(\varphi, F)(\chi, G) = (\varphi, F)(\chi', G')$ then $\varphi\chi = \varphi\chi'$, so $\chi = \chi'$, and $F_{\chi(b)}G_b = F_{\chi(b)}G'_b$, so $G_b = G'_b$. Necessity: $\pi \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}, -)$ is representable, hence preserves monomorphisms, so φ is injective. If $g, g'\colon X \to L_\alpha$ satisfy $F_\alpha g = F_\alpha g'$, then the morphisms $j_\alpha\iota(g), j_\alpha\iota(g')\colon \iota X \to K$ have equal composites with (φ, F), namely $(\varphi(\alpha), F_\alpha g)$; hence they are equal, and $g = g'$. □

Counterexample 6.19. The necessity of injectivity in Proposition 6.18 depends on the head. Let $\mathcal{C}$ be the poset $\{x_1, x_2, z\}$ with $x_1 < z$, $x_2 < z$ and x_1, x_2 incomparable; it has a terminal object z but no initial object. In $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$, consider the morphism $m\colon (2, (x_1, x_2)) \to (1, z)$ with the unique gait $2 \to 1$. For any family (B, W), a morphism $(B, W) \to (2, (x_1, x_2))$ must send b to the index i with $W_b = x_i$, and exists only if no W_b equals z; so there is at most one morphism from any object to $(2, (x_1, x_2))$, and m is a monomorphism. Its gait is not injective.

Proposition 6.20. *A feline morphism $(\varphi, F)\colon K \to K'$ is an epimorphism if and only if φ is surjective and, for each $\beta \in B$, the family $(F_\alpha\colon L_\alpha \to L'_\beta)_{\alpha \in \varphi^{-1}(\beta)}$ is jointly epimorphic in $\mathcal{C}$ (that is, $gF_\alpha = g'F_\alpha$ for all $\alpha \in \varphi^{-1}(\beta)$ implies $g = g'$).*

Proof. Sufficiency: if $(\chi, G)(\varphi, F) = (\chi', G')(\varphi, F)$ then $\chi\varphi = \chi'\varphi$, so $\chi = \chi'$ as φ is surjective; and $G_\beta F_\alpha = G'_\beta F_\alpha$ for all $\alpha \in \varphi^{-1}(\beta)$, so $G_\beta = G'_\beta$. Necessity: suppose $\beta \notin \operatorname{im}\varphi$. Let K'' be $K' \sqcup \iota(L'_\beta)$, and consider the coprojection $\mathrm{in}_1\colon K' \to K''$ and the morphism $\mathrm{in}'\colon K' \to K''$ which agrees with in_1 except that it sends the leg β to the new leg, with identity component. Then $\mathrm{in}_1 \neq \mathrm{in}'$ but $\mathrm{in}_1(\varphi, F) = \mathrm{in}'(\varphi, F)$. Hence φ is surjective. Now fix β and $g, g'\colon L'_\beta \to Y$ with $gF_\alpha = g'F_\alpha$ for all $\alpha \in \varphi^{-1}(\beta)$. Let K'' be

obtained from K' by replacing the leg β by Y, and let $G, G' \colon K' \to K''$ have identity gait, components g resp. g' at β, and identity components elsewhere. Then $G(\varphi, F) = G'(\varphi, F)$, so $G = G'$ and $g = g'$. □

Remark 6.21. The proofs of Propositions 6.18 and 6.20 are not dual to one another. The characterization of monomorphisms uses the head, through the representability of π, and fails without it (Counterexample 6.19). The characterization of epimorphisms uses only test morphisms built from coproducts and from the replacement of one leg by another object, and in particular does not use the tail. This asymmetry reflects the fact that $\mathsf{Fel}(\mathcal{C})$ is a free completion under coproducts, and not under products.

6. Preservation properties

Proposition 6.22. (i) *The leg functor π preserves all limits and colimits.*

(ii) *The one-legged inclusion ι preserves all limits that exist in $\mathcal{C}$, and all connected colimits that exist in $\mathcal{C}$; it preserves neither the initial object nor binary coproducts.*

(iii) *Δ_0 preserves all colimits, and Δ_1 preserves all limits and all colimits.*

Proof. (i) was shown in Theorems 6.8 and 6.14. (ii) If X is a limit of $E \colon \mathcal{J} \to \mathcal{C}$, the cone $\iota X \to \iota E$ is a limit cone. Indeed, the sufficiency argument in the proof of Theorem 6.8 uses only the limits of the diagrams D_a for $a \in \lim_j A_j$; for $D = \iota E$ we have $\lim_j 1 = 1$ and $D_* = E$, whose limit exists. Explicitly, a cone $M \to \iota E$ consists of the unique gait $\pi M \to 1$ and, for each leg of M, a cone over E, which factors uniquely through X. Similarly, the sufficiency argument of Theorem 6.14 with $\int(\pi\iota E) \cong \mathcal{J}$ connected uses only the colimit of E, so ι preserves connected colimits that exist. The initial object 0 is sent to $\mathbf{h}$, which is not initial (Warning 4.5); and $\iota(X \sqcup Y)$ is unipedal while $\iota X \sqcup \iota Y$ is two-legged. (iii) Δ_0 is a left adjoint; Δ_1 is a right adjoint (Corollary 5.13) and also a left adjoint (Theorem 7.3). □

CHAPTER 7

Adjunctions and equivalences

Standing Assumption 7.1. Throughout this chapter $\mathcal{C}$ and $\mathcal{D}$ have initial and terminal objects, chosen as in Standing Assumption 3.18.

1. The feline adjoint string

Definition 7.2. The *points functor* $\Gamma\colon \mathsf{Fel}(\mathcal{C}) \to \mathbf{Set}$ is defined by

$$\Gamma(K) = \coprod_{\alpha\in\pi(K)} \mathcal{C}(1, L_\alpha), \qquad \Gamma(\varphi, F)(\alpha, x) = (\varphi(\alpha), F_\alpha x).$$

An element of $\Gamma(K)$ is a *point* of K: a leg together with a global element of that leg. The natural transformation $p\colon \Gamma \Rightarrow \pi$, $p_K(\alpha, x) = \alpha$, is called the *points-to-legs* transformation.

Recall from Example 5.4 the functors $\Delta_X\colon \mathbf{Set} \to \mathsf{Fel}(\mathcal{C})$, $\Delta_X(S) = \langle X\rangle_{s\in S}$. Thus $\Delta_0(S)$ is the cat with S legs each of which is a head, and $\Delta_1(S)$ is the cat with S legs each of which is a tail.

Theorem 7.3 (The feline adjoint string). *There are adjunctions*

$$\Delta_0 \;\dashv\; \pi \;\dashv\; \Delta_1 \;\dashv\; \Gamma.$$

The functors Δ_0 and Δ_1 are fully faithful, and $\pi\Delta_0 = \pi\Delta_1 = \mathrm{id}_{\mathbf{Set}}$. Moreover $\pi \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}, -)$ and $\Gamma \cong \mathsf{Fel}(\mathcal{C})(\mathbf{1}, -)$, and under these identifications the points-to-legs transformation p is precomposition with the unique morphism $\iota(!_1)\colon \mathbf{h} \to \mathbf{1}$.

Proof. The first two adjunctions are Corollary 5.13. Explicitly, by Corollary 3.11,

$$\mathsf{Fel}(\mathcal{C})(\Delta_0 S, K) \cong \prod_{s\in S} \coprod_{\alpha} \mathcal{C}(0, L_\alpha) \cong \mathbf{Set}(S, \pi K),$$
$$\mathsf{Fel}(\mathcal{C})(K, \Delta_1 S) \cong \prod_{\alpha} \coprod_{s\in S} \mathcal{C}(L_\alpha, 1) \cong \mathbf{Set}(\pi K, S).$$

For the third,

$$\mathsf{Fel}(\mathcal{C})(\Delta_1 S, K) \cong \prod_{s\in S} \coprod_{\alpha} \mathcal{C}(1, L_\alpha) = \mathbf{Set}(S, \Gamma K),$$

naturally in S and K: a morphism $\Delta_1 S \to K$ with gait φ and components $x_s\colon 1 \to L_{\varphi(s)}$ corresponds to the function $s \mapsto (\varphi(s), x_s)$. The functors Δ_0, Δ_1 are fully faithful since $\mathsf{Fel}(\mathcal{C})(\Delta_X S, \Delta_X S') \cong \prod_s \coprod_{s'} \mathcal{C}(X, X) \cong \mathbf{Set}(S, S')$ for $X \in \{0, 1\}$. The equalities $\pi\Delta_X = \mathrm{id}$ hold by definition. The

representability of π is Lemma 6.3, and $\mathsf{Fel}(\mathcal{C})(\mathbf{1}, K) \cong \coprod_\alpha \mathcal{C}(1, L_\alpha) = \Gamma(K)$ by Corollary 3.11. Finally, precomposition with $\iota(!_1)$ sends $(\alpha, x)\colon \mathbf{1} \to K$ to $(\alpha, x \circ !_1) = (\alpha, !)$, which corresponds to α. □

Remark 7.4. In terms of families, π is the projection $\mathsf{Fam}(\mathcal{C}) \to \mathbf{Set}$ of the family fibration, Δ_1 and Δ_0 are its sections by fibrewise terminal and initial objects, and $\Gamma = \mathsf{Fam}(\mathcal{C})(\mathbf{1}, -)$ is the global-sections functor of $\mathsf{Fam}(\mathcal{C})$. That a fibration with fibrewise terminal objects stable under reindexing has its terminal-object section as a right adjoint is a general fact [4]. The adjunction $\Delta_1 \dashv \Gamma$ is an instance of the general adjunction between the copower functor $S \mapsto \coprod_S \mathbf{1}$ and global sections, valid in any category with coproducts and a terminal object.

Proposition 7.5. *The functor Δ_0 has a left adjoint if and only if $\mathcal{C}$ is pointed. In that case $\Delta_0 \cong \Delta_1$ and $\pi \dashv \Delta_0 \dashv \pi$, so that π is both left and right adjoint to Δ_0.*

Proof. Suppose $\Lambda \dashv \Delta_0$. Then $\mathsf{Fel}(\mathcal{C})(\iota X, \Delta_0 1) \cong \mathbf{Set}(\Lambda \iota X, 1)$ is a singleton for every X. But $\mathsf{Fel}(\mathcal{C})(\iota X, \Delta_0 1) = \mathsf{Fel}(\mathcal{C})(\iota X, \iota 0) \cong \mathcal{C}(X, 0)$. Thus 0 is terminal and $\mathcal{C}$ is pointed. Conversely, if $\mathcal{C}$ is pointed then $0 \cong 1$, so $\Delta_0 \cong \Delta_1$ (via Fel applied to the natural isomorphism $\ulcorner 0 \urcorner \cong \ulcorner 1 \urcorner$), and $\pi \dashv \Delta_1 \cong \Delta_0$. □

Proposition 7.6. *The functor Γ has a right adjoint if and only if either*

(i) *$\mathcal{C}$ is pointed, in which case $\Gamma \cong \pi$ and $\Gamma \dashv \Delta_1$; or*
(ii) *the global-sections functor $\mathcal{C}(1, -)\colon \mathcal{C} \to \mathbf{Set}$ has a right adjoint ∇, in which case $\Gamma \dashv \iota\nabla$.*

Proof. If $\mathcal{C}$ is pointed then $\mathcal{C}(1, L_\alpha) \cong \mathcal{C}(0, L_\alpha)$ is a singleton, so $p\colon \Gamma \Rightarrow \pi$ is an isomorphism, and $\pi \dashv \Delta_1$. If $\mathcal{C}(1, -) \dashv \nabla$, then naturally in K and S,

$$\mathsf{Fel}(\mathcal{C})(K, \iota\nabla S) \cong \prod_\alpha \mathcal{C}(L_\alpha, \nabla S) \cong \prod_\alpha \mathbf{Set}(\mathcal{C}(1, L_\alpha), S) \cong \mathbf{Set}(\Gamma K, S).$$

Conversely, suppose $\Gamma \dashv R$ and that $\mathcal{C}$ is not pointed. By Lemma 2.8, $\mathcal{C}(1, 0) = \varnothing$, so $\Gamma(\mathbf{h}) = \varnothing$ and $\pi(RS) \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}, RS) \cong \mathbf{Set}(\Gamma\mathbf{h}, S) = \mathbf{Set}(\varnothing, S)$ is a singleton for every S. Hence $RS \cong \iota(\nabla S)$ for an object ∇S, unique up to isomorphism, and

$$\mathcal{C}(X, \nabla S) \cong \mathsf{Fel}(\mathcal{C})(\iota X, RS) \cong \mathbf{Set}(\Gamma \iota X, S) = \mathbf{Set}(\mathcal{C}(1, X), S)$$

naturally in X. By the parametrized representability theorem, ∇ extends uniquely to a functor right adjoint to $\mathcal{C}(1, -)$. □

Proposition 7.7. *The one-legged inclusion $\iota\colon \mathcal{C} \to \mathsf{Fel}(\mathcal{C})$ has no right adjoint. More generally, if $\mathcal{D}$ is a category with an initial object, no functor $G\colon \mathcal{D} \to \mathsf{Fel}(\mathcal{C})$ whose values are unipedal cats has a right adjoint.*

Proof. A left adjoint preserves initial objects. The initial objects of $\mathsf{Fel}(\mathcal{C})$ are the legless cats (Proposition 4.4), and $G(0_\mathcal{D})$ is unipedal. For ι one may also argue directly, without using the initial object of $\mathcal{C}$: if $\iota \dashv R$ then $\mathcal{C}(R\mathbf{0}, R\mathbf{0}) \cong \mathsf{Fel}(\mathcal{C})(\iota R\mathbf{0}, \mathbf{0}) = \varnothing$, which is absurd. □

Remark 7.8. The second argument in the proof of Proposition 7.7 applies equally to headless cats in a category without an initial object: the obstruction to a right adjoint of ι is the legless cat, not the head.

Proposition 7.9. *If $\mathcal{C}$ is pointed, then, writing $\Delta = \Delta_0 \cong \Delta_1$, there is an infinite string of adjunctions $\cdots \dashv \Delta \dashv \pi \dashv \Delta \dashv \pi \dashv \cdots$. If $\mathcal{C}$ is not pointed, the string $\Delta_0 \dashv \pi \dashv \Delta_1 \dashv \Gamma$ cannot be extended to the left.*

Proof. Immediate from Proposition 7.5. □

2. The body

Definition 7.10. Suppose $\mathcal{C}$ has small coproducts, chosen once and for all. The *body* of a cat $K = (H, T, (L_\alpha)_{\alpha\in A})$ is the object $\Sigma K \coloneqq \coprod_{\alpha\in A} L_\alpha$ of $\mathcal{C}$. For a morphism $(\varphi, F)\colon K \to K'$, $\Sigma(\varphi, F)\colon \Sigma K \to \Sigma K'$ is the morphism whose restriction to the α-th summand is $\mathrm{in}_{\varphi(\alpha)} \circ F_\alpha$.

Proposition 7.11. *The one-legged inclusion ι has a left adjoint if and only if $\mathcal{C}$ has small coproducts; in that case the body functor is left adjoint to ι, and $\Sigma\iota \cong \mathrm{id}_{\mathcal{C}}$.*

Proof. For a cat K and an object Y, $\mathsf{Fel}(\mathcal{C})(K, \iota Y) \cong \prod_\alpha \mathcal{C}(L_\alpha, Y)$ by Corollary 3.11. A left adjoint Λ to ι therefore satisfies $\mathcal{C}(\Lambda K, Y) \cong \prod_\alpha \mathcal{C}(L_\alpha, Y)$ naturally in Y, which says precisely that ΛK is a coproduct of the legs of K. Every small family of objects is the family of legs of some cat, so ι has a left adjoint if and only if every small family has a coproduct. The counit $\Sigma\iota Y = \coprod_1 Y \to Y$ is an isomorphism since ι is fully faithful. □

Proposition 7.12. *Suppose $\mathcal{C}$ has small coproducts. Then:*

(i) *Σ preserves all colimits;*
(ii) *$\Sigma(\mathbf{0}) \cong 0$, $\Sigma(\mathbf{h}) \cong 0$, and $\Sigma(\Delta_X S) \cong S \cdot X$, the S-fold copower of X;*
(iii) *Σ is neither faithful nor conservative.*

Proof. (i) Σ is a left adjoint. (ii) is immediate from the definition, the coproduct of the empty family and of a family of initial objects being initial. (iii) The two morphisms $\mathbf{h} \to \Delta_0(2)$ (with gaits the two functions $1 \to 2$) have the same body, namely the unique morphism $0 \to 0 \sqcup 0$; and the unique morphism $\mathbf{0} \to \mathbf{h}$ is not an isomorphism (its gait is not bijective), although its body $0 \to 0$ is. □

Proposition 7.13. *Let $\mathcal{C}$ be one of* **Set**, **Top** *or* **Cat**, *and let $\nabla\colon$ **Set** $\to \mathcal{C}$ be, respectively, the identity, the indiscrete-topology functor, or the functor sending a set to the indiscrete (chaotic) category on it. Then there is a string of five adjoint functors*

$$\Delta_0 \ \dashv\ \pi \ \dashv\ \Delta_1 \ \dashv\ \Gamma \ \dashv\ \iota\nabla,$$

*which cannot be extended in either direction. For $\mathcal{C} =$ **Set** the points functor is isomorphic to the body functor, so the string reads $\Delta_\varnothing \dashv \pi \dashv \Delta_1 \dashv \Sigma \dashv \iota$.*

PROOF. In each case ∇ is right adjoint to $\mathcal{C}(1,-)$, which is respectively the identity, the underlying-set functor, and the object-set functor; so Theorem 7.3 and Proposition 7.6 give the string. It cannot be extended to the left because $\mathcal{C}$ is not pointed (Proposition 7.5), and it cannot be extended to the right by Proposition 7.7, since $\iota\nabla$ is a functor from **Set**, which has an initial object, with unipedal values. For $\mathcal{C} = \mathbf{Set}$, $\mathbf{Set}(1, L_\alpha) \cong L_\alpha$ naturally, so $\Gamma K \cong \coprod_\alpha L_\alpha = \Sigma K$. □

Remark 7.14. In the situation of Proposition 7.13, the last four functors $\pi \dashv \Delta_1 \dashv \Gamma \dashv \iota\nabla$ form an adjoint quadruple in which the second and fourth functors are fully faithful (for $\iota\nabla$ because ι is fully faithful by Proposition 3.22 and each of the three functors ∇ is fully faithful), and π preserves finite products by Theorem 6.8. This is the shape of the adjoint quadruple "pieces $\dashv$ discrete $\dashv$ points $\dashv$ codiscrete" in Lawvere's axiomatic cohesion [8]. We record the resemblance and nothing more: we have not verified any of the further axioms of that theory, and we draw no conclusions from the coincidence of shapes.

3. Connected objects and reconstruction

Proposition 7.15. *A cat is a connected object of* $\mathsf{Fel}(\mathcal{C})$*, in the sense of Definition 2.9, if and only if it is unipedal.*

PROOF. Let X be an object of $\mathcal{C}$ and $(K_i)_{i\in I}$ a family of cats. By Corollary 3.11 and Proposition 6.5,

$$\mathsf{Fel}(\mathcal{C})\Big(\iota X, \coprod_i K_i\Big) \cong \coprod_{(i,\alpha)} \mathcal{C}(X, L_{i,\alpha}) \cong \coprod_i \mathsf{Fel}(\mathcal{C})(\iota X, K_i),$$

and the composite is inverse to the canonical map (2.1); so unipedal cats are connected, and so are cats isomorphic to them. Conversely, let K be connected with leg set A. Applying (2.1) to the empty family gives $\mathsf{Fel}(\mathcal{C})(K, \mathbf{0}) = \varnothing$, so $A \neq \varnothing$. Suppose $A = A_1 \sqcup A_2$ with both parts nonempty, and let K_1, K_2 be the corresponding subcats, so that $K \cong K_1 \sqcup K_2$. The identity of K corresponds to a morphism $K \to K_1 \sqcup K_2$ whose gait is a bijection $A \to A_1 \sqcup A_2$, and this morphism does not factor through either coprojection, whose image misses A_2 or A_1. This contradicts connectedness. Hence $|A| = 1$. □

Remark 7.16. In the terminology of Section 5, the objects of $\mathsf{Fel}(\mathcal{C})$ satisfying the Chloe Condition are the unipedal cats, and a cat of cats satisfies the Chloe Condition if and only if its legs are unipedal (Example 4.18(iii)). By Corollary 6.6, every cat is a coproduct of unipedal cats; in the terminology of Definition 12.12, every object of $\mathsf{Fel}(\mathcal{C})$ admits a Gideon Decomposition (Example 12.13(v)).

THEOREM 7.17 (Reconstruction). *Let $E\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}(\mathcal{D})$ be an equivalence of categories. Then E restricts to an equivalence between the full subcategories of unipedal cats, and consequently there is an equivalence $F\colon \mathcal{C} \to \mathcal{D}$ with $E \circ \iota_{\mathcal{C}} \cong \iota_{\mathcal{D}} \circ F$. In particular, $\mathsf{Fel}(\mathcal{C}) \simeq \mathsf{Fel}(\mathcal{D})$ implies $\mathcal{C} \simeq \mathcal{D}$.*

PROOF. An equivalence preserves and reflects connected objects, since the definition of a connected object is invariant under equivalence. By Proposition 7.15, E therefore restricts to an equivalence $E_1\colon \mathsf{Fel}_1(\mathcal{C}) \to \mathsf{Fel}_1(\mathcal{D})$. Composing with the equivalences $\iota_{\mathcal{C}}\colon \mathcal{C} \to \mathsf{Fel}_1(\mathcal{C})$ and a quasi-inverse of $\iota_{\mathcal{D}}$ gives F. □

Remark 7.18. Theorem 7.17 is the feline form of the standard fact that a category can be recovered, up to equivalence, from its free coproduct completion as the full subcategory of connected objects. The equivalence E in Theorem 7.17 is not assumed to commute with the leg functors. The next proposition shows that it nevertheless does so up to isomorphism.

Proposition 7.19. *For every equivalence $E\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}(\mathcal{D})$ there is a natural isomorphism $\pi_{\mathcal{D}} \circ E \cong \pi_{\mathcal{C}}$.*

PROOF. The cephalic cat $\mathbf{h}_{\mathcal{C}} = \iota(0)$ is an initial object of the full subcategory $\mathsf{Fel}_1(\mathcal{C})$ of unipedal cats, since $\iota\colon \mathcal{C} \to \mathsf{Fel}_1(\mathcal{C})$ is an equivalence (Proposition 3.22) and 0 is initial in $\mathcal{C}$. By the proof of Theorem 7.17, E restricts to an equivalence $\mathsf{Fel}_1(\mathcal{C}) \to \mathsf{Fel}_1(\mathcal{D})$, which preserves initial objects; hence $E(\mathbf{h}_{\mathcal{C}}) \cong \mathbf{h}_{\mathcal{D}}$. By Lemma 6.3,

$$\pi_{\mathcal{D}}(EK) \cong \mathsf{Fel}(\mathcal{D})(\mathbf{h}_{\mathcal{D}}, EK) \cong \mathsf{Fel}(\mathcal{D})(E\mathbf{h}_{\mathcal{C}}, EK) \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}_{\mathcal{C}}, K) \cong \pi_{\mathcal{C}}(K),$$

naturally in K. □

Corollary 7.20. *The construction $\mathcal{C} \mapsto \mathsf{Fel}(\mathcal{C})$ is not idempotent up to equivalence: $\mathsf{Fel}(\mathsf{Fel}(\mathbb{1})) \not\simeq \mathsf{Fel}(\mathbb{1})$.*

PROOF. By Proposition 3.24, $\mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$, which has an initial and a terminal object, so $\mathsf{Fel}(\mathsf{Fel}(\mathbb{1}))$ is defined. If it were equivalent to $\mathsf{Fel}(\mathbb{1})$, then Theorem 7.17 would give $\mathsf{Fel}(\mathbb{1}) \simeq \mathbb{1}$, that is, $\mathbf{Set} \simeq \mathbb{1}$, which is false. □

4. Computations

The following theorem computes the category of cats in a functor category $\mathbf{Set}^{\mathcal{J}}$. It is the main computational result of this monograph.

THEOREM 7.21. *Let $\mathcal{J}$ be a small category, and let $\mathcal{J}^{\triangleright}$ be $\mathcal{J}$ with a terminal object $\top$ adjoined (Definition 2.15). Then there is an equivalence of categories*

$$\Theta\colon \mathsf{Fel}(\mathbf{Set}^{\mathcal{J}}) \longrightarrow \mathbf{Set}^{\mathcal{J}^{\triangleright}}$$

with $\mathrm{ev}_{\top} \circ \Theta = \pi$. Explicitly, for a cat K with legs $L_\alpha\colon \mathcal{J} \to \mathbf{Set}$,

$$\Theta(K)|_{\mathcal{J}} = \coprod_{\alpha \in A} L_\alpha, \qquad \Theta(K)(\top) = A,$$

and $\Theta(K)(j \to \top)\colon \coprod_\alpha L_\alpha(j) \to A$ is the map $(\alpha, x) \mapsto \alpha$.

PROOF. Note that $\mathbf{Set}^{\mathcal{J}}$ has an initial object (the constant functor at $\varnothing$) and a terminal object (the constant functor at a singleton), so $\mathsf{Fel}(\mathbf{Set}^{\mathcal{J}})$ is defined. A functor $G\colon \mathcal{J}^{\triangleright} \to \mathbf{Set}$ is the same as a functor $G|_{\mathcal{J}}\colon \mathcal{J} \to \mathbf{Set}$, a set $G(\top)$, and a cocone $(G(j \to \top))_j$ from $G|_{\mathcal{J}}$ to $G(\top)$; the cocone condition is the functoriality of G on composites $j \to j' \to \top$. The data $\Theta(K)$ satisfy this condition, since each $L_\alpha(u)$ preserves the summand α. For a morphism $(\varphi, F)\colon K \to K'$, let $\Theta(\varphi, F)$ be the natural transformation with component φ at $\top$ and component $(\alpha, x) \mapsto (\varphi(\alpha), F_{\alpha,j}(x))$ at $j \in \mathcal{J}$; it is natural at $j \to \top$ by construction and at $u\colon j \to j'$ by naturality of F_α. Functoriality of Θ is clear, and $\mathrm{ev}_\top \Theta = \pi$.

Θ *is fully faithful.* Let $\theta\colon \Theta(K) \to \Theta(K')$ be a natural transformation and $\varphi = \theta_\top\colon A \to B$. Naturality at $j \to \top$ says that θ_j maps the summand $L_\alpha(j)$ into the summand $L'_{\varphi(\alpha)}(j)$, defining functions $F_{\alpha,j}\colon L_\alpha(j) \to L'_{\varphi(\alpha)}(j)$; naturality at $u\colon j \to j'$ says that $F_\alpha = (F_{\alpha,j})_j$ is a natural transformation. Then $\theta = \Theta(\varphi, F)$, and (φ, F) is uniquely determined by θ.

Θ *is essentially surjective.* Given $G\colon \mathcal{J}^{\triangleright} \to \mathbf{Set}$, let $A = G(\top)$ and, for $\alpha \in A$, let $L_\alpha(j) = G(j \to \top)^{-1}(\alpha) \subseteq G(j)$. For $u\colon j \to j'$, the map $G(u)$ sends $L_\alpha(j)$ into $L_\alpha(j')$ by the cocone condition, so L_α is a functor $\mathcal{J} \to \mathbf{Set}$. The evident bijections $\coprod_\alpha L_\alpha(j) \cong G(j)$ and id_A form a natural isomorphism $\Theta\langle L_\alpha\rangle_A \cong G$. □

Corollary 7.22. *There is an equivalence* $\mathsf{Fel}(\mathbf{Set}) \simeq \mathbf{Set}^{\to}$ *between the category of* **Set***-cats and the arrow category of* **Set***, sending a cat* K *to the function* $\Sigma K = \coprod_\alpha L_\alpha \to \pi K$*. Under this equivalence the functors of Proposition 7.13 correspond to*

$$(S \mapsto (\varnothing \to S)) \;\dashv\; \mathrm{cod} \;\dashv\; (S \mapsto \mathrm{id}_S) \;\dashv\; \mathrm{dom} \;\dashv\; (S \mapsto (S \to 1)).$$

PROOF. Take $\mathcal{J} = \mathbb{1}$ in Theorem 7.21; then $\mathbf{Set}^{\mathbb{1}} = \mathbf{Set}$ and $\mathbb{1}^{\triangleright} = [1]$, and $\mathbf{Set}^{[1]} = \mathbf{Set}^{\to}$. The images of $\Delta_\varnothing S$, $\Delta_1 S$ and ιS are $\varnothing \to S$, $S \times 1 \to S$ and $S \to 1$; π corresponds to the codomain and $\Sigma = \Gamma$ to the domain. □

Corollary 7.23. *For every small category* $\mathcal{D}$ *there is an equivalence*

$$\mathsf{Fel}(\mathsf{PSh}(\mathcal{D})) \simeq \mathsf{PSh}(\mathcal{D}^{\triangleleft}),$$

under which the leg functor corresponds to evaluation at the adjoined initial object $\bot$*.*

PROOF. Apply Theorem 7.21 with $\mathcal{J} = \mathcal{D}^{\mathrm{op}}$, and use $(\mathcal{D}^{\mathrm{op}})^{\triangleright} = (\mathcal{D}^{\triangleleft})^{\mathrm{op}}$, under which $\top$ corresponds to $\bot$. □

Remark 7.24. In Corollary 7.23, the adjoined object $\bot$ is an initial object of $\mathcal{D}^{\triangleleft}$ and the leg functor is evaluation of presheaves at it. A presheaf P on $\mathcal{D}^{\triangleleft}$ assigns to $\bot$ the set of legs, and to each $d \in \mathcal{D}$ the set of elements of all legs at d, each element being sent by the restriction map $P(d) \to P(\bot)$ to the leg containing it. We shall see in Theorem 8.10 that, quite generally, the legs of a cat are the elements of its Yoneda presheaf at the head.

5. Failure of duality

Since an initial object of $\mathcal{C}$ is a terminal object of $\mathcal{C}^{\mathrm{op}}$ and vice versa, *head–tail reversal* $(H, T, L) \mapsto (T, H, L)$ is a bijection between the classes of $\mathcal{C}$-cats and of $\mathcal{C}^{\mathrm{op}}$-cats. In general it does not extend to an equivalence of categories in either variance.

Proposition 7.25. *In general* $\mathsf{Fel}(\mathcal{C}^{\mathrm{op}}) \not\simeq \mathsf{Fel}(\mathcal{C})$ *and* $\mathsf{Fel}(\mathcal{C}^{\mathrm{op}}) \not\simeq \mathsf{Fel}(\mathcal{C})^{\mathrm{op}}$.

Proof. Take $\mathcal{C} = \mathbb{1}$. Then $\mathsf{Fel}(\mathbb{1}^{\mathrm{op}}) = \mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$, so an equivalence $\mathsf{Fel}(\mathbb{1}^{\mathrm{op}}) \simeq \mathsf{Fel}(\mathbb{1})^{\mathrm{op}}$ would give $\mathbf{Set} \simeq \mathbf{Set}^{\mathrm{op}}$. This is false: the initial object $\varnothing$ of $\mathbf{Set}$ is strict, whereas the initial object of $\mathbf{Set}^{\mathrm{op}}$, a singleton, is not strict in $\mathbf{Set}^{\mathrm{op}}$ (there are morphisms $2 \to 1$ in $\mathbf{Set}^{\mathrm{op}}$, namely functions $1 \to 2$, that are not isomorphisms). For the first assertion take $\mathcal{C} = \mathbf{Set}$. By Theorem 7.17, $\mathsf{Fel}(\mathbf{Set}^{\mathrm{op}}) \simeq \mathsf{Fel}(\mathbf{Set})$ would imply $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{Set}$. □

Remark 7.26. Unwinding the definitions, $\mathsf{Fel}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}$ is equivalent to the category whose objects are small families in $\mathcal{C}$ and whose morphisms $(A, X) \to (B, Y)$ are pairs consisting of a function $\psi\colon B \to A$ and morphisms $X_{\psi(b)} \to Y_b$ of $\mathcal{C}$. This is the dual of the family construction, namely the free completion of $\mathcal{C}$ under small products. One could call its objects *retrograde cats*; they are not studied here.

CHAPTER 8

Feline Yoneda theory

Standing Assumption 8.1. Throughout this chapter $\mathcal{C}$ is a locally small category with an initial object 0 and a terminal object 1. Presheaf categories are as in Convention 2.2.

1. The feline Yoneda functor

Definition 8.2. The *feline Yoneda functor* $\mathbb{Y}\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{PSh}(\mathcal{C})$ sends a cat K with legs $(L_\alpha)_{\alpha\in A}$ to the presheaf

$$\mathbb{Y}K \coloneqq \coprod_{\alpha\in A} y(L_\alpha), \qquad (\mathbb{Y}K)(X) = \coprod_{\alpha\in A} \mathcal{C}(X, L_\alpha),$$

and a morphism $(\varphi, F)\colon K \to K'$ to the natural transformation given by $(\mathbb{Y}(\varphi,F))_X(\alpha, g) = (\varphi(\alpha), F_\alpha g)$.

Remark 8.3. By Corollary 3.11, $(\mathbb{Y}K)(X) \cong \mathsf{Fel}(\mathcal{C})(\iota X, K)$, naturally in X and K. Thus $\mathbb{Y} \cong \iota^* \circ y_{\mathsf{Fel}(\mathcal{C})}$ is, up to natural isomorphism, the composite of the Yoneda embedding of $\mathsf{Fel}(\mathcal{C})$ with restriction along ι. This description will be exploited in Section 3.

Lemma 8.4 (Feline Yoneda Lemma)**.** *For a cat K and a presheaf P on $\mathcal{C}$, the map*

$$\mathsf{PSh}(\mathcal{C})(\mathbb{Y}K, P) \longrightarrow \prod_{\alpha\in\pi(K)} P(L_\alpha), \qquad \theta \longmapsto \big(\theta_{L_\alpha}(\alpha, \mathrm{id}_{L_\alpha})\big)_\alpha,$$

is a bijection, natural in K and P. In particular the left-hand side is a small set.

Proof. Since $\mathbb{Y}K$ is the coproduct of the $y(L_\alpha)$, with coprojections sending $g \in \mathcal{C}(X, L_\alpha)$ to (α, g), a natural transformation $\mathbb{Y}K \to P$ is the same as a family of natural transformations $y(L_\alpha) \to P$, which by the Yoneda lemma correspond to elements of $P(L_\alpha)$ via evaluation at id_{L_α}. Naturality in P is immediate; naturality in K means that for $(\varphi, F)\colon K \to K'$ the induced map $\prod_\beta P(L'_\beta) \to \prod_\alpha P(L_\alpha)$ is $(x_\beta) \mapsto (P(F_\alpha)x_{\varphi(\alpha)})_\alpha$, which follows from $(\mathbb{Y}(\varphi,F))_{L_\alpha}(\alpha, \mathrm{id}) = (\varphi(\alpha), F_\alpha)$ and the naturality of θ. □

Theorem 8.5. *The feline Yoneda functor $\mathbb{Y}$ is fully faithful.*

Proof. Take $P = \mathbb{Y}K'$ in Lemma 8.4:

$$\mathsf{PSh}(\mathcal{C})(\mathbb{Y}K, \mathbb{Y}K') \cong \prod_\alpha (\mathbb{Y}K')(L_\alpha) = \prod_\alpha \coprod_\beta \mathcal{C}(L_\alpha, L'_\beta) \cong \mathsf{Fel}(\mathcal{C})(K, K').$$

The composite sends $\mathbb{Y}(\varphi, F)$ to $\big((\mathbb{Y}(\varphi,F))_{L_\alpha}(\alpha,\mathrm{id})\big)_\alpha = ((\varphi(\alpha), F_\alpha))_\alpha$, which corresponds to (φ, F). Hence the action of $\mathbb{Y}$ on hom-sets is the inverse of this bijection. □

THEOREM 8.6. *For a presheaf P on $\mathcal{C}$ the following are equivalent.*

(i) *$P \cong \mathbb{Y}K$ for some cat K.*
(ii) *P is a small coproduct of representable presheaves.*
(iii) *The category of elements $\int P$ is a coproduct $\coprod_{i\in I} \mathcal{E}_i$ of categories indexed by a small set I, each of which has a terminal object.*

In this case the leg set of K is in bijection with I, and the legs of K are, up to isomorphism, the first coordinates of the terminal objects of the $\mathcal{E}_i$.

PROOF. (i)⇔(ii) is the definition of $\mathbb{Y}$, every small family of objects being the family of legs of a cat.

(ii)⇒(iii). The category of elements of a coproduct of presheaves is the coproduct of their categories of elements, and $\int y(X) \cong \mathcal{C}/X$, the slice category, which has the terminal object (X, id_X).

(iii)⇒(ii). Let (X_i, x_i) be a terminal object of $\mathcal{E}_i$. For an object Y of $\mathcal{C}$, define

$$\coprod_{i\in I} \mathcal{C}(Y, X_i) \longrightarrow P(Y), \qquad (i, f) \longmapsto P(f)(x_i).$$

It is natural in Y. It is surjective: an element $y \in P(Y)$ is an object (Y, y) of some $\mathcal{E}_i$, and the unique morphism $(Y, y) \to (X_i, x_i)$ is an $f\colon Y \to X_i$ with $P(f)(x_i) = y$. It is injective: if $P(f)(x_i) = P(f')(x_{i'}) = y$, then f and f' are morphisms in $\int P$ from (Y, y) to (X_i, x_i) and to $(X_{i'}, x_{i'})$, so $i = i'$ because the $\mathcal{E}_i$ are distinct components, and $f = f'$ by terminality. Hence $P \cong \coprod_i y(X_i)$.

For the final assertion: each $\mathcal{E}_i$ has a terminal object and is therefore connected, so the $\mathcal{E}_i$ are exactly the connected components of $\int P$. If $P \cong \mathbb{Y}K$, then $\int P \cong \coprod_\alpha \mathcal{C}/L_\alpha$, whose components are indexed by $\pi(K)$ and have terminal objects (L_α, id); terminal objects are unique up to isomorphism. □

Remark 8.7. Theorems 8.5 and 8.6 are the feline forms of the standard facts that the free coproduct completion of $\mathcal{C}$ embeds fully faithfully in $\mathsf{PSh}(\mathcal{C})$ as the coproducts of representables, and that a presheaf is a coproduct of representables if and only if each connected component of its category of elements has a terminal object and the set of components is small. (For large $\mathcal{C}$ the smallness condition is not automatic: the constant presheaf at 1 on a large discrete category has a proper class of components. For categories with an initial object it is automatic, by Proposition 8.8 below.)

2. Evaluation at the head and at the tail

Proposition 8.8. *For every presheaf P on $\mathcal{C}$, the map*

$$\pi_0(\textstyle\int P) \longrightarrow P(0), \qquad [(X, x)] \longmapsto P(!_X)(x),$$

is a well-defined bijection, where $!_X\colon 0 \to X$.

Proof. If $f\colon (X,x)\to(X',x')$ is a morphism of $\int P$, so that $P(f)(x')=x$, then $P(!_X)(x)=P(!_X)P(f)(x')=P(f\circ !_X)(x')=P(!_{X'})(x')$ because $f\circ !_X=!_{X'}$; hence the map is constant on components. It is surjective because $(0,z)\mapsto P(\mathrm{id}_0)(z)=z$. It is injective because every object (X,x) is connected to $(0,P(!_X)(x))$ by the morphism $!_X\colon (0,P(!_X)x)\to(X,x)$; so two objects with the same image are connected to the same object $(0,z)$. □

Remark 8.9. Proposition 8.8 is the special case of Lemma 2.18 asserting that the colimit of a functor $\mathcal{C}^{\mathrm{op}}\to\mathbf{Set}$ is its value at a terminal object of $\mathcal{C}^{\mathrm{op}}$, combined with Lemma 2.17; the latter requires $\mathcal{C}$ to be small, but the direct argument above does not.

Theorem 8.10 (Head–Tail Evaluation). *There are natural isomorphisms*

$$\mathrm{ev}_0\circ\mathbb{Y}\cong\pi,\qquad \mathrm{ev}_1\circ\mathbb{Y}\cong\Gamma,$$

and, more generally, $\mathrm{ev}_X\circ\mathbb{Y}\cong\mathsf{Fel}(\mathcal{C})(\iota X,-)$ *for every object* X*. The points-to-legs transformation* $p\colon\Gamma\Rightarrow\pi$ *corresponds to* $\mathbb{Y}(-)(!_1)\colon \mathrm{ev}_1\circ\mathbb{Y}\Rightarrow\mathrm{ev}_0\circ\mathbb{Y}$*, where* $!_1\colon 0\to 1$*. Moreover the connected components of* $\int\mathbb{Y}K$ *are in canonical bijection with the legs of* K*.*

Proof. $(\mathbb{Y}K)(0)=\coprod_\alpha\mathcal{C}(0,L_\alpha)\cong\pi(K)$ and $(\mathbb{Y}K)(1)=\coprod_\alpha\mathcal{C}(1,L_\alpha)=\Gamma(K)$, naturally in K. The general statement is Remark 8.3. The map $(\mathbb{Y}K)(!_1)$ sends (α,x) to $(\alpha,x\circ!_1)$, which corresponds to α. The final assertion follows from Proposition 8.8. □

Thus the legs of a cat are the elements of its Yoneda presheaf at the head, and its points are the elements at the tail. We stress that the content of Theorem 8.10 is the Yoneda lemma together with the definition of an initial object.

3. Feline presheaves

Definition 8.11. A presheaf $Q\colon\mathsf{Fel}(\mathcal{C})^{\mathrm{op}}\to\mathbf{Set}$ is *feline* if it sends small coproducts in $\mathsf{Fel}(\mathcal{C})$ to products: for every small family $(K_i)_{i\in I}$ the map $Q(\coprod_i K_i)\to\prod_i Q(K_i)$ induced by the coprojections is a bijection (for $I=\varnothing$ this says that $Q(\mathbf{0})$ is a singleton). We write $\mathsf{PSh}_{\sqcup}(\mathsf{Fel}(\mathcal{C}))$ for the full subcategory of feline presheaves.

Theorem 8.12. *Restriction along* ι *is an equivalence of categories*

$$\iota^*\colon\mathsf{PSh}_{\sqcup}(\mathsf{Fel}(\mathcal{C}))\longrightarrow\mathsf{PSh}(\mathcal{C}),$$

with quasi-inverse $P\mapsto\widehat{P}:=\mathsf{PSh}(\mathcal{C})(\mathbb{Y}(-),P)$*; explicitly* $\widehat{P}(K)\cong\prod_\alpha P(L_\alpha)$*. Every representable presheaf* $\mathsf{Fel}(\mathcal{C})(-,K)$ *is feline, and* $\iota^*\mathsf{Fel}(\mathcal{C})(-,K)\cong\mathbb{Y}K$*.*

Proof. By Lemma 8.4, $\widehat{P}(K)\cong\prod_\alpha P(L_\alpha)$ naturally. Since the leg set of $\coprod_i K_i$ is $\coprod_i A_i$ (Proposition 6.5), $\widehat{P}$ is feline, and $\widehat{P}(\iota X)\cong P(X)$, so $\iota^*\widehat{P}\cong P$. Conversely, if Q is feline, then using Corollary 6.6,

$$Q(K)\cong Q\Big(\coprod_\alpha\iota L_\alpha\Big)\cong\prod_\alpha Q(\iota L_\alpha)=\prod_\alpha(\iota^*Q)(L_\alpha)\cong\widehat{\iota^*Q}(K),$$

naturally in K; so $Q \cong \widehat{\iota^* Q}$. Finally, a natural transformation between feline presheaves is determined by its components at unipedal cats, by the same decomposition, so ι^* is faithful, and the two isomorphisms above show that it is full and essentially surjective. Representables send colimits to limits, so they are feline, and $\iota^* \mathsf{Fel}(\mathcal{C})(-, K) \cong \mathbb{Y}K$ is Remark 8.3. □

Remark 8.13. Theorem 8.12 is the universal property of the free coproduct completion applied to the cocomplete category $\mathbf{Set}^{\mathrm{op}}$; compare Theorem 12.4. Combined with the ordinary Yoneda embedding of $\mathsf{Fel}(\mathcal{C})$, it yields a second proof of Theorem 8.5.

Proposition 8.14. *Let $\mathcal{C}$ be a category with a terminal object, and let $\pi \colon \mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \to \mathbf{Set}$ be the leg functor on headless cats. Then π is representable if and only if $\mathcal{C}$ has an initial object, in which case it is represented by the headless cat $(1, (0))$, the decapitated cephalic cat. By contrast, Γ is always represented by $(1, (1))$.*

Proof. If $\mathcal{C}$ has an initial object, argue as in Lemma 6.3. Conversely, suppose $\pi \cong \mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})(R, -)$. The leg functor preserves coproducts (including the empty coproduct), so R is a connected object of $\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$; the proof of Proposition 7.15, which does not use the head, shows that R is unipedal, say $R \cong (1, (Y))$. Then $\mathcal{C}(Y, X) \cong \mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})(R, (1, (X))) \cong \pi(1, (X)) = 1$ for all X, so Y is initial. The statement about Γ is proved as in Theorem 7.3. □

4. Natural transformations between the basic functors

The ordinary Yoneda lemma, applied to the representable functors $\pi \cong \mathsf{Fel}(\mathcal{C})(\mathbf{h}, -)$ and $\Gamma \cong \mathsf{Fel}(\mathcal{C})(\mathbf{1}, -)$, computes all natural transformations between them. Nothing feline happens in the following proposition beyond the identification of representing objects.

Proposition 8.15.
(i) *The only natural transformation $\pi \Rightarrow \pi$ is the identity.*
(ii) *The only natural transformation $\Gamma \Rightarrow \Gamma$ is the identity.*
(iii) *The points-to-legs transformation p is the only natural transformation $\Gamma \Rightarrow \pi$.*
(iv) *There is a natural transformation $\pi \Rightarrow \Gamma$ if and only if $\mathcal{C}$ is pointed, and it is then unique and invertible.*

Proof. By the Yoneda lemma,

$$\mathrm{Nat}\big(\mathsf{Fel}(\mathcal{C})(M, -), \mathsf{Fel}(\mathcal{C})(M', -)\big) \cong \mathsf{Fel}(\mathcal{C})(M', M).$$

(i) $\mathsf{Fel}(\mathcal{C})(\mathbf{h}, \mathbf{h}) \cong \mathcal{C}(0, 0)$ is a singleton. (ii) $\mathsf{Fel}(\mathcal{C})(\mathbf{1}, \mathbf{1}) \cong \mathcal{C}(1, 1)$ is a singleton. (iii) $\mathsf{Fel}(\mathcal{C})(\mathbf{h}, \mathbf{1}) \cong \mathcal{C}(0, 1)$ is a singleton, and p is a natural transformation. (iv) $\mathsf{Fel}(\mathcal{C})(\mathbf{1}, \mathbf{h}) \cong \mathcal{C}(1, 0)$, which is nonempty if and only if $\mathcal{C}$ is pointed (Lemma 2.8); in that case the unique element is an isomorphism. □

5. Restricted Yoneda on connected objects

For use in Chapter 13 we record a variant of the feline Yoneda functor in which only connected objects are used as probes. Let $\mathcal{C}^\circ$ be the full subcategory of connected objects (Definition 2.9), and let $y^\circ\colon \mathcal{C} \to \mathsf{PSh}(\mathcal{C}^\circ)$, $y^\circ(X) = \mathcal{C}(-, X)|_{\mathcal{C}^\circ}$, be the restricted Yoneda functor.

Definition 8.16. The *restricted feline Yoneda functor* is $\mathbb{Y}^\circ\colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{PSh}(\mathcal{C}^\circ)$, $\mathbb{Y}^\circ K = \coprod_\alpha y^\circ(L_\alpha)$, defined on morphisms as in Definition 8.2. Note that $\mathbb{Y}^\circ(\iota X) \cong y^\circ(X)$ canonically; we identify the two.

Proposition 8.17. *Let K be a cat all of whose legs are connected. Then for every cat K' the map $\mathbb{Y}^\circ\colon \mathsf{Fel}(\mathcal{C})(K, K') \to \mathsf{PSh}(\mathcal{C}^\circ)(\mathbb{Y}^\circ K, \mathbb{Y}^\circ K')$ is a bijection. In particular, $\mathbb{Y}^\circ$ is fully faithful on the full subcategory of cats with connected legs.*

Proof. The proof of Lemma 8.4 applies verbatim with $\mathcal{C}^\circ$ in place of $\mathcal{C}$, because each L_α is an object of $\mathcal{C}^\circ$, and gives $\mathsf{PSh}(\mathcal{C}^\circ)(\mathbb{Y}^\circ K, P) \cong \prod_\alpha P(L_\alpha)$. With $P = \mathbb{Y}^\circ K'$ the right-hand side is $\prod_\alpha \coprod_\beta \mathcal{C}(L_\alpha, L'_\beta) \cong \mathsf{Fel}(\mathcal{C})(K, K')$, and as in Theorem 8.5 the composite is inverse to the action of $\mathbb{Y}^\circ$. □

Counterexample 8.18. The hypothesis on the legs of K in Proposition 8.17 cannot be dropped. In **Set**, the connected objects are the singletons, so $\mathsf{PSh}(\mathbf{Set}^\circ) \simeq \mathbf{Set}$ and $\mathbb{Y}^\circ K \cong \Sigma K$ is the disjoint union of the legs. The two-legged cat $\langle 1, 1\rangle$ and the unipedal cat $\iota(2)$ have isomorphic images, but they are not isomorphic in $\mathsf{Fel}(\mathbf{Set})$, their leg sets having different cardinalities.

6. Margaret Compatibility

The restricted feline Yoneda functor leads to the following notion, which compares a cat with an object of $\mathcal{C}$ as seen from connected objects.

Definition 8.19 (Margaret Compatibility)**.** Let K be a $\mathcal{C}$-cat and X an object of $\mathcal{C}$. A *cocone* from K to X is a family $\beta = (\beta_\alpha\colon L_\alpha \to X)_{\alpha \in \pi K}$; by Corollary 3.11 it is the same as a morphism $\beta\colon K \to \iota X$. The cocone β is *Margaret compatible* if, for every connected object Y of $\mathcal{C}$, the map

$$m^\beta_Y\colon \coprod_{\alpha \in \pi K} \mathcal{C}(Y, L_\alpha) \longrightarrow \mathcal{C}(Y, X), \qquad (\alpha, g) \longmapsto \beta_\alpha \circ g,$$

is a bijection. In words: every morphism from a connected object into X factors through exactly one leg, and through that leg in exactly one way.

Proposition 8.20. *For a cocone $\beta\colon K \to \iota X$ the following are equivalent:*

(i) *β is Margaret compatible;*
(ii) *for every connected Y, postcomposition with β is a bijection from $\mathsf{Fel}(\mathcal{C})(\iota Y, K)$ to $\mathsf{Fel}(\mathcal{C})(\iota Y, \iota X)$;*
(iii) *$\mathbb{Y}^\circ(\beta)\colon \mathbb{Y}^\circ K \to \mathbb{Y}^\circ(\iota X) \cong y^\circ X$ is an isomorphism of presheaves on $\mathcal{C}^\circ$ (Definition 8.16).*

PROOF. Under the bijections $\mathsf{Fel}(\mathcal{C})(\iota Y, K) \cong \coprod_\alpha \mathcal{C}(Y, L_\alpha) = (\mathbb{Y}^\circ K)(Y)$ and $\mathsf{Fel}(\mathcal{C})(\iota Y, \iota X) \cong \mathcal{C}(Y, X) = (y^\circ X)(Y)$, both postcomposition with β and the component of $\mathbb{Y}^\circ(\beta)$ at Y become m_Y^β. A morphism of presheaves is an isomorphism if and only if each component is a bijection. □

Example 8.21. (i) In **Set**, the connected objects are the singletons, and m_1^β is identified with the map $\coprod_\alpha L_\alpha \to X$ induced by β. So a cocone in **Set** is Margaret compatible if and only if it is a coproduct cocone.

(ii) In a pointed category every cocone is Margaret compatible, vacuously.

(iii) In **Top**, let $X = \mathbb{Q}$ with its usual topology, and let $K = \langle \{q\} \rangle_{q \in \mathbb{Q}}$ with β_q the inclusion of the point q. Every continuous map from a connected space Y to $\mathbb{Q}$ is constant, since connected subsets of $\mathbb{Q}$ have at most one point, and so factors uniquely through exactly one β_q. Hence β is Margaret compatible. It is not a coproduct cocone, since the coproduct of the points of $\mathbb{Q}$ is $\mathbb{Q}$ with the discrete topology.

Proposition 8.22. (i) *If $\beta\colon K \to \iota X$ is Margaret compatible and $\theta\colon K' \to K$ is an isomorphism of cats, then $\beta\theta$ is Margaret compatible; if $f\colon X \to X'$ is an isomorphism, then $\iota(f)\beta$ is Margaret compatible.*

(ii) *If $\beta^i\colon K_i \to \iota X_i$ are Margaret compatible for $i \in I$ and $\coprod_i X_i$ exists in $\mathcal{C}$, then the cocone $\coprod_i K_i \to \iota(\coprod_i X_i)$ with components $\mathrm{in}_i \circ \beta^i_\alpha$ is Margaret compatible.*

PROOF. (i) Postcomposition with an isomorphism is bijective. (ii) For connected Y,

$$\coprod_{(i,\alpha)} \mathcal{C}(Y, L_{i,\alpha}) = \coprod_i \coprod_\alpha \mathcal{C}(Y, L_{i,\alpha}) \xrightarrow{\coprod_i m_Y^{\beta^i}} \coprod_i \mathcal{C}(Y, X_i) \xrightarrow{\kappa} \mathcal{C}\Big(Y, \coprod_i X_i\Big)$$

is a composite of bijections, κ being bijective by connectedness of Y, and it equals m_Y for the coproduct cocone. □

CHAPTER 9

Monoidal and enriched cats

Standing Assumption 9.1. In Sections 1 to 3, $\mathcal{C}$ is a category with initial and terminal objects 0 and 1. In Sections 1 and 2 it carries a monoidal structure $(\otimes, I, a, l, r)$, with associator $a_{X,Y,Z} \colon (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ and unitors $l_X \colon I \otimes X \to X$, $r_X \colon X \otimes I \to X$. The hypotheses of Section 4 are stated there.

1. The feline tensor product

Definition 9.2. The *feline tensor product* of cats $K = \langle L_\alpha \rangle_{\alpha \in A}$ and $K' = \langle L'_\beta \rangle_{\beta \in B}$ (normalized without loss of generality) is

$$K \otimes K' \coloneqq \langle L_\alpha \otimes L'_\beta \rangle_{(\alpha,\beta) \in A \times B},$$

and on morphisms $(\varphi, F) \otimes (\psi, G) \coloneqq (\varphi \times \psi, (F_\alpha \otimes G_\beta)_{(\alpha,\beta)})$. The *feline unit* is $\mathbf{I} \coloneqq \iota(I)$. The associator $\mathbf{a}_{K,K',K''}$ has gait $((\alpha,\beta),\gamma) \mapsto (\alpha,(\beta,\gamma))$ and components $a_{L_\alpha, L'_\beta, L''_\gamma}$; the unitors $\mathbf{l}_K$ and $\mathbf{r}_K$ have gaits $(*,\alpha) \mapsto \alpha$ and $(\alpha,*) \mapsto \alpha$ and components l_{L_α} and r_{L_α}.

THEOREM 9.3. *$(\mathsf{Fel}(\mathcal{C}), \otimes, \mathbf{I}, \mathbf{a}, \mathbf{l}, \mathbf{r})$ is a monoidal category. If $\mathcal{C}$ is braided (resp. symmetric) with braiding c, then $\mathsf{Fel}(\mathcal{C})$ is braided (resp. symmetric) with braiding of gait $(\alpha,\beta) \mapsto (\beta,\alpha)$ and components c_{L_α, L'_β}. The leg functor is a strict monoidal functor $(\mathsf{Fel}(\mathcal{C}), \otimes, \mathbf{I}) \to (\mathbf{Set}, \times, 1)$, and ι is a strong monoidal functor.*

PROOF. Functoriality of $\otimes$ follows from that of $\times$ on **Set** and of $\otimes$ on $\mathcal{C}$. The associator and unitors are isomorphisms by Proposition 3.23, and natural because their gaits are natural in **Set** and their components are natural in $\mathcal{C}$. Every coherence axiom (the pentagon and triangle axioms, and in the braided case the hexagon axioms and, in the symmetric case, $c_{Y,X} c_{X,Y} = \mathrm{id}$) is an equation between morphisms of cats; by Proposition 3.9 such an equation holds if and only if it holds on gaits and on each leg component. On gaits it is the corresponding axiom for the cartesian monoidal structure of **Set**, and on leg components it is the corresponding axiom in $\mathcal{C}$. The leg functor sends $K \otimes K'$ to $A \times B$, $\mathbf{I}$ to 1, and the structural isomorphisms to those of **Set**, so it is strict monoidal. Finally $\iota X \otimes \iota Y = \langle X \otimes Y \rangle_{1 \times 1}$, which is isomorphic to $\iota(X \otimes Y)$ by the isomorphism with gait $1 \times 1 \to 1$ and identity component; these isomorphisms, together with $\mathbf{I} = \iota I$, make ι strong monoidal, the coherence conditions again reducing to identities on gaits and components. □

Proposition 9.4. *For every cat K, the functors $K \otimes -$ and $- \otimes K$ preserve small coproducts. In particular $K \otimes \mathbf{0} \cong \mathbf{0} \cong \mathbf{0} \otimes K$. This holds whether or not $\otimes$ preserves coproducts in $\mathcal{C}$.*

PROOF. For a family (K_i) with leg sets B_i, the canonical morphism $\coprod_i (K \otimes K_i) \to K \otimes \coprod_i K_i$ has gait the canonical bijection $\coprod_i (A \times B_i) \to A \times \coprod_i B_i$ and identity components, so it is an isomorphism by Proposition 3.23. □

Remark 9.5. If the tensor product of two connected objects of $\mathcal{C}$ is connected, then the feline tensor product of two cats satisfying the Chloe Condition satisfies it, since the legs of $K \otimes K'$ are the objects $L_\alpha \otimes L'_\beta$. This holds for $(\mathbf{Set}, \times)$ and $(\mathbf{Top}, \times)$ but not for $(\mathbf{Gph}, \times)$; the consequences for Gideon Decompositions are drawn in Section 3.

Proposition 9.6. *Suppose $\mathcal{C}$ has small coproducts. For cats K and K' let $\mu_{K,K'} \colon \Sigma(K \otimes K') \to \Sigma K \otimes \Sigma K'$ be the morphism whose restriction to the summand $L_\alpha \otimes L'_\beta$ is $\mathrm{in}_\alpha \otimes \mathrm{in}_\beta$. Then $\mu_{K,K'}$ is an isomorphism for all K, K' if and only if $\otimes$ preserves small coproducts in each variable separately. In that case Σ, with the inverses of the $\mu_{K,K'}$ and the canonical isomorphism $\Sigma\mathbf{I} = \coprod_1 I \cong I$, is a strong monoidal functor.*

PROOF. If $\otimes$ preserves coproducts in each variable, $\mu_{K,K'}$ is the composite of the canonical isomorphisms $\coprod_{\alpha,\beta} L_\alpha \otimes L'_\beta \cong \coprod_\alpha (L_\alpha \otimes \coprod_\beta L'_\beta) \cong (\coprod_\alpha L_\alpha) \otimes \coprod_\beta L'_\beta$. Conversely, taking $K = \iota X$ shows that the canonical map $\coprod_\beta X \otimes L'_\beta \to X \otimes \coprod_\beta L'_\beta$ is invertible for every family (L'_β), and similarly in the other variable. The coherence conditions for a strong monoidal functor hold because both sides are induced by the coproduct injections. □

Remark 9.7. One might attempt to define the tensor product of cats componentwise, as $(H \otimes H', T \otimes T', (L_\alpha \otimes L'_\beta))$. This is a cat for all K, K' if and only if $0 \otimes 0$ is initial and $1 \otimes 1$ is terminal, since any initial object is isomorphic to 0 and any terminal object to 1. Neither condition holds in general. In $\mathbf{Set}^{\mathrm{op}}$ with its cartesian monoidal structure, the product of two copies of the initial object (a singleton) is a two-element set, which is not initial in $\mathbf{Set}^{\mathrm{op}}$. In $\mathbf{Set}$ with the cocartesian monoidal structure $\sqcup$, the tensor $1 \sqcup 1$ of two copies of the terminal object is not terminal. This is why Definition 9.2 renormalizes the head and the tail.

2. Monoidal cats and graded monoids

Definition 9.8. A *monoidal $\mathcal{C}$-cat* is a monoid in the monoidal category $(\mathsf{Fel}(\mathcal{C}), \otimes, \mathbf{I})$: a cat K with morphisms $\mu \colon K \otimes K \to K$ and $\eta \colon \mathbf{I} \to K$ satisfying the associativity and unit axioms. A morphism of monoidal cats is a monoid morphism. The category of monoidal $\mathcal{C}$-cats is denoted $\mathbf{Mon}(\mathsf{Fel}(\mathcal{C}))$.

Definition 9.9. Let M be a monoid in **Set** with unit e. An *M-graded monoid* in $\mathcal{C}$ is a family $(X_m)_{m\in M}$ of objects with morphisms $\mu_{m,n}\colon X_m \otimes X_n \to X_{mn}$ and $\eta\colon I \to X_e$ such that for all $m, n, p \in M$

$$\mu_{mn,p} \circ (\mu_{m,n} \otimes \mathrm{id}) = \mu_{m,np} \circ (\mathrm{id} \otimes \mu_{n,p}) \circ a,$$
$$\mu_{e,m} \circ (\eta \otimes \mathrm{id}) = l, \qquad \mu_{m,e} \circ (\mathrm{id} \otimes \eta) = r.$$

A morphism $(M, X) \to (N, Y)$ of graded monoids is a monoid homomorphism $f\colon M \to N$ together with morphisms $g_m\colon X_m \to Y_{f(m)}$ commuting with the multiplications and units. We write $\mathsf{GrMon}(\mathcal{C})$ for the resulting category.

THEOREM 9.10. *There is an equivalence* $\mathbf{Mon}(\mathsf{Fel}(\mathcal{C})) \simeq \mathsf{GrMon}(\mathcal{C})$*, which is an isomorphism on normalized cats. Under it, the underlying set of legs of a monoidal cat, with the multiplication given by the gait of μ, is the grading monoid.*

PROOF. Let (K, μ, η) be a monoidal cat with $K = \langle L_\alpha \rangle_A$. The morphism μ has a gait $m\colon A \times A \to A$ and components $\mu_{\alpha,\beta}\colon L_\alpha \otimes L_\beta \to L_{m(\alpha,\beta)}$; the morphism η has a gait $1 \to A$, picking out $e \in A$, and a component $\eta\colon I \to L_e$. By Proposition 3.9, the associativity and unit axioms of the monoid K hold if and only if they hold on gaits and on components. Since π is strict monoidal (Theorem 9.3), the axioms on gaits say that (A, m, e) is a monoid in **Set**, and the axioms on components are exactly those of Definition 9.9. Similarly, a morphism of monoidal cats is a morphism of cats whose gait is a monoid homomorphism and whose components satisfy the compatibilities of Definition 9.9. □

Example 9.11.
(i) For $(\mathcal{C}, \otimes) = (\mathbf{Ab}, \otimes_{\mathbb{Z}})$, an M-graded monoid is a family of abelian groups (X_m) with bilinear associative unital multiplications $X_m \times X_n \to X_{mn}$; taking the direct sum, this is the same as an M-graded ring. Thus monoidal **Ab**-cats with leg monoid M are M-graded rings, and monoidal **Ab**-cats with leg monoid $\mathbb{N}$ are $\mathbb{N}$-graded rings.
(ii) For $(\mathcal{C}, \otimes) = (\mathbf{Set}, \times)$, an M-graded monoid in **Set** is the same as a monoid $N = \coprod_m X_m$ together with a monoid homomorphism $N \to M$ (sending X_m to m). Thus $\mathbf{Mon}(\mathsf{Fel}(\mathbf{Set}))$ is equivalent to the arrow category of the category of monoids, in accordance with $\mathsf{Fel}(\mathbf{Set}) \simeq \mathbf{Set}^{\to}$ (Corollary 7.22).
(iii) For $\mathcal{C} = \mathbb{1}$, a monoidal $\mathbb{1}$-cat is a monoid in **Set**.

Remark 9.12. Theorem 9.10 identifies monoidal cats with graded monoids, a classical notion. When $\mathcal{C}$ is symmetric, commutative monoids in $\mathsf{Fel}(\mathcal{C})$ correspond to graded monoids with commutative grading monoid and commutative multiplication in the sense determined by the symmetry of $\mathcal{C}$. Any signs in this commutativity come from the symmetry of $\mathcal{C}$ (for instance, the Koszul signs of chain complexes involve the homological degree of the legs), not from the grading monoid: a commutative monoid in $\mathsf{Fel}(\mathbf{Ab})$ with leg monoid $\mathbb{N}$ is a commutative, not a graded-commutative, $\mathbb{N}$-graded ring.

3. Self-enrichment

The category $\mathsf{Fel}(\mathbf{Set})$ has finite products (Theorem 6.8), and so is a cartesian monoidal category. The following theorem shows that every category of cats is enriched in **Set**-cats.

THEOREM 9.13. *For $\mathcal{C}$-cats $K = \langle L_\alpha \rangle_A$ and $K' = \langle L'_\beta \rangle_B$, let*

$$[K, K'] \coloneqq \Big\langle \prod_{\alpha \in A} \mathcal{C}(L_\alpha, L'_{\varphi(\alpha)}) \Big\rangle_{\varphi \in \mathbf{Set}(A,B)} \in \mathsf{Fel}(\mathbf{Set}).$$

Let $\mathrm{c} \colon [K', K''] \times [K, K'] \to [K, K'']$ *be the morphism of* **Set***-cats with gait* $(\psi, \varphi) \mapsto \psi\varphi$ *and components* $(G, F) \mapsto (G_{\varphi(\alpha)} F_\alpha)_\alpha$*, and let* $\mathrm{u} \colon \mathbf{1} \to [K, K]$ *have gait* $* \mapsto \mathrm{id}_A$ *and component* $* \mapsto (\mathrm{id}_{L_\alpha})_\alpha$*. These data make* $\mathsf{Fel}(\mathcal{C})$ *a category enriched in* $(\mathsf{Fel}(\mathbf{Set}), \times, \mathbf{1})$*, whose underlying ordinary category, computed with* $\Gamma = \mathsf{Fel}(\mathbf{Set})(\mathbf{1}, -)$*, is* $\mathsf{Fel}(\mathcal{C})$*.*

PROOF. By Corollary 6.9, $[K', K''] \times [K, K']$ has legs indexed by pairs (ψ, φ), with leg the product of the corresponding legs, so c is well typed. The associativity and unit axioms of an enriched category are equations between morphisms of **Set**-cats, and by Proposition 3.9 they reduce to equations on gaits, which are associativity and unitality of composition of functions, and on components, which are associativity and unitality of composition in $\mathcal{C}$. Finally, $\Gamma[K, K'] = \coprod_\varphi \prod_\alpha \mathcal{C}(L_\alpha, L'_{\varphi(\alpha)}) \cong \mathsf{Fel}(\mathcal{C})(K, K')$ by Corollary 3.11, and the composition induced by c on points is that of $\mathsf{Fel}(\mathcal{C})$. □

Remark 9.14. Under $\mathsf{Fel}(\mathbf{Set}) \simeq \mathbf{Set}^{\to}$, a category enriched in $(\mathbf{Set}^{\to}, \times)$ amounts to a pair of ordinary categories with the same objects and an identity-on-objects functor between them. The enrichment of Theorem 9.13 corresponds to the identity-on-objects functor from $\mathsf{Fel}(\mathcal{C})$ to the category whose objects are cats and whose morphisms are functions between leg sets, that is, to the leg functor. No information beyond π is contained in the self-enrichment.

4. Additive enrichment

In this section $\mathcal{C}$ is a *preadditive* category, that is, a category enriched in abelian groups, possessing an initial object.

Lemma 9.15. *In a preadditive category every initial object is a zero object. In particular head and tail of every $\mathcal{C}$-cat are zero objects, and $\mathcal{C}$-cats exist.*

PROOF. Let 0 be initial. Then $\mathcal{C}(0, 0)$ is the trivial group, so id_0 is the zero morphism. For any $f \colon X \to 0$, bilinearity of composition gives $f = \mathrm{id}_0 \circ f = 0 \circ f = 0$. Hence $\mathcal{C}(X, 0)$ is a singleton. □

In particular $\mathcal{C}$ is pointed, so by Remark 4.17 the only $\mathcal{C}$-cats satisfying the Chloe Condition are the legless ones, and by Example 12.13(iv) only the zero objects of $\mathcal{C}$ admit Gideon Decompositions.

Definition 9.16. The *additively enriched category of cats* $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ has as objects the $\mathcal{C}$-cats and as hom-groups

$$\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(K, K') \coloneqq \prod_{\alpha\in A} \bigoplus_{\beta\in B} \mathcal{C}(L_\alpha, L'_\beta),$$

whose elements we write as matrices $(f_{\beta\alpha})$ with $f_{\beta\alpha}\colon L_\alpha \to L'_\beta$ such that for each α only finitely many $f_{\beta\alpha}$ are nonzero (*column-finite matrices*). Composition is matrix multiplication, $(gf)_{\gamma\alpha} = \sum_\beta g_{\gamma\beta} f_{\beta\alpha}$, and identities are identity matrices.

Lemma 9.17. $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ *is a preadditive category.*

Proof. The sum defining $(gf)_{\gamma\alpha}$ is finite because f is column-finite. For fixed α, $(gf)_{\gamma\alpha} \neq 0$ only if $g_{\gamma\beta} \neq 0$ for one of the finitely many β with $f_{\beta\alpha} \neq 0$, and each of these columns of g is finite; hence gf is column-finite. Associativity, unitality and bilinearity follow from the corresponding properties in $\mathcal{C}$ and the finiteness of all sums involved. □

Theorem 9.18. *Let $\mathcal{C}$ be preadditive with an initial object* 0.

(i) *In $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$, every cat K is the coproduct of the unipedal cats $\iota(L_\alpha)$, the α-th coprojection being the matrix whose only possibly nonzero entry is id_{L_α}, in row α; if K is a kitten, this is a biproduct.*

(ii) *In $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ the cephalic cat $\mathbf{h} = \iota(0)$ is isomorphic to the legless cat $\mathbf{0}$.*

(iii) *If $\mathcal{C}$ has binary biproducts, then $\iota(X \oplus Y) \cong \langle X, Y\rangle$ in $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$.*

(iv) *Consequently, the number of legs is not an isomorphism invariant in $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$, and there is no functor P from the underlying category of $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ to* **Set** *with $P(K)$ in bijection with $\pi(K)$ for all K.*

(v) *The underlying ordinary category of $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ is not equivalent to $\mathsf{Fel}(\mathcal{C}_0)$, where $\mathcal{C}_0$ is the underlying ordinary category of $\mathcal{C}$.*

Proof. (i) By definition,

$$\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(K, M) = \prod_\alpha \bigoplus_\beta \mathcal{C}(L_\alpha, M_\beta) = \prod_\alpha \mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(\iota L_\alpha, M),$$

and the bijection is precomposition with the stated coprojections. For finite A a coproduct in a preadditive category is a biproduct.

(ii) $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(\mathbf{h}, \mathbf{0}) = \bigoplus_\varnothing = 0$ and $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(\mathbf{0}, \mathbf{h}) = \prod_\varnothing = 0$, and $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})(\mathbf{h}, \mathbf{h}) = \mathcal{C}(0, 0) = 0$; so the zero morphisms are mutually inverse.

(iii) The matrices $\binom{p_X}{p_Y}\colon \iota(X \oplus Y) \to \langle X, Y\rangle$ and $(i_X \;\; i_Y)\colon \langle X, Y\rangle \to \iota(X \oplus Y)$, formed from the biproduct projections and injections, are mutually inverse by the biproduct identities.

(iv) By (ii), $\mathbf{h} \cong \mathbf{0}$ although $|\pi\mathbf{h}| = 1 \neq 0 = |\pi\mathbf{0}|$. A functor preserves isomorphisms, so $P(\mathbf{h}) \cong P(\mathbf{0})$, which is incompatible with $P(\mathbf{h}) \cong 1$ and $P(\mathbf{0}) \cong \varnothing$.

(v) The underlying category of $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ is preadditive and has the initial object $\mathbf{0}$ (by (i), as the empty coproduct), which is therefore a zero object by

Lemma 9.15. In $\mathsf{Fel}(\mathcal{C}_0)$ the initial object $\mathbf{0}$ is strict (Proposition 4.4); if it were also terminal, then every cat would admit a morphism to $\mathbf{0}$ and hence be legless, which fails for $\iota(0)$. So $\mathsf{Fel}(\mathcal{C}_0)$ is not pointed, and an equivalence would preserve pointedness. (For instance, if $\mathcal{C}$ is the trivial preadditive category, $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ is equivalent to the terminal category, every cat being a coproduct of zero objects, while $\mathsf{Fel}(\mathcal{C}_0) \cong \mathbf{Set}$.) □

Example 9.19. Let $\mathcal{C}$ be the full subcategory of $\mathbf{Ab}$ on the objects 0 and $\mathbb{Z}$. The functor sending a cat K to $\bigoplus_\alpha L_\alpha$ is an equivalence between $\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$ and the category of free abelian groups. Indeed, it is fully faithful because a homomorphism $\bigoplus_\alpha L_\alpha \to \bigoplus_\beta L'_\beta$ is determined by its restrictions to the summands, each of which lands in a finite sub-sum; and it is essentially surjective since every free abelian group is of the form $\bigoplus_A \mathbb{Z}$. The legs equal to 0 are invisible, in accordance with Theorem 9.18(ii).

Remark 9.20. Theorem 9.18 shows that the naive enrichment of cat theory over abelian groups does not merely enrich the theory of Chapter 3 but changes it: legs may be merged and split, and a leg equal to the head may be amputated without changing the isomorphism class of the cat. The felinization of an enriched category is therefore not the enrichment of the felinization. We do not develop enrichment over a general base $\mathcal{V}$. One would expect the formula $\prod_\alpha \coprod_\beta \mathcal{C}(L_\alpha, L'_\beta)$ for hom-objects to define a $\mathcal{V}$-category when $\mathcal{V}$ has small products and small coproducts and $\otimes$ preserves coproducts in each variable, but we have not verified the axioms and make no claim to this effect.

CHAPTER 10

Feline fibrations and derived constructions

Standing Assumption 10.1. Throughout this chapter $\mathcal{C}$ has an initial object 0 and a terminal object 1. Additional hypotheses are stated in each section.

1. The leg fibration

Definition 10.2. For a function $\varphi\colon A \to B$ and a cat K' with leg set B, the *reindexing* of K' along φ is the cat $\varphi^* K' := \langle L'_{\varphi(\alpha)} \rangle_{\alpha \in A}$, and the *canonical lift* of φ is the morphism $\bar{\varphi} = (\varphi, (\mathrm{id}_{L'_{\varphi(\alpha)}})_\alpha)\colon \varphi^* K' \to K'$.

THEOREM 10.3. *The leg functor $\pi\colon \mathsf{Fel}(\mathcal{C}) \to \mathbf{Set}$ is a Grothendieck fibration. A feline morphism is cartesian if and only if it is reindexing in the sense of Definition 5.6, that is, all of its leg components are isomorphisms. The canonical lifts form a cleavage which, on normalized cats, is split: $(\psi\varphi)^* = \varphi^*\psi^*$ and $\mathrm{id}^* = \mathrm{id}$. The fibre over a set A is equivalent to $\mathcal{C}^A$.*

PROOF. Let $(\varphi, F)\colon K \to K'$ be reindexing. Given $(\psi, G)\colon M \to K'$ and $\chi\colon \pi M \to A$ with $\varphi\chi = \psi$, a morphism $(\chi, H)\colon M \to K$ satisfies $(\varphi, F)(\chi, H) = (\psi, G)$ if and only if $F_{\chi(c)} H_c = G_c$ for all c, that is, $H_c = F^{-1}_{\chi(c)} G_c$. So (φ, F) is cartesian. In particular canonical lifts are cartesian, and π is a fibration. Conversely, suppose $(\varphi, F)\colon K \to K'$ is cartesian. Since $\bar{\varphi}\colon \varphi^* K' \to K'$ is also cartesian over φ, the universal properties provide vertical morphisms $(\mathrm{id}, V)\colon K \to \varphi^* K'$ and $(\mathrm{id}, W)\colon \varphi^* K' \to K$ with $\bar{\varphi}(\mathrm{id}, V) = (\varphi, F)$ and $(\varphi, F)(\mathrm{id}, W) = \bar{\varphi}$; uniqueness in the universal properties shows that they are mutually inverse. Comparing components, $V_\alpha = F_\alpha$, and F_α is an isomorphism. The splitting identities hold on normalized cats by inspection, and the statement about fibres is Proposition 4.7. □

Remark 10.4. By Proposition 4.16(iii), cats satisfying the Chloe Condition are closed under reindexing, so the canonical lifts of Theorem 10.3 remain in the full subcategory $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$, and the restriction of π to $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$ is again a fibration. The opcartesian lifts of Theorem 10.6, when they exist, need not remain in $\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$: in **Set**, the opcartesian lift of $2 \to 1$ at the Chloe cat $\langle 1, 1 \rangle$ has the two-element set as its leg, which is not connected.

Proposition 10.5. *Let $\mathcal{C}^{(-)}\colon \mathbf{Set}^{\mathrm{op}} \to \mathbf{CAT}$ be the functor $A \mapsto \mathcal{C}^A$, $\varphi \mapsto \varphi^*$ (precomposition). Then $\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C})$ is isomorphic, over **Set**, to the Grothendieck construction $\int \mathcal{C}^{(-)}$ of Definition 2.20. Hence $\mathsf{Fel}(\mathcal{C})$ is equivalent over **Set** to $\int \mathcal{C}^{(-)}$.*

PROOF. An object of $\int \mathcal{C}^{(-)}$ is a pair (A, L) with $L \in \mathcal{C}^A$, and a morphism $(A, L) \to (B, L')$ is a pair (φ, F) with $F\colon L \to \varphi^* L'$ in $\mathcal{C}^A$, that is, a family $F_\alpha\colon L_\alpha \to L'_{\varphi(\alpha)}$. The composite $(\psi, G) \circ (\varphi, F) = (\psi\varphi, \varphi^*(G) \circ F)$ has components $G_{\varphi(\alpha)} F_\alpha$. This is exactly $\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C})$ under Proposition 3.9; the second assertion follows from Proposition 3.20. □

THEOREM 10.6. *The leg functor is an opfibration, and hence a bifibration, if and only if $\mathcal{C}$ has small coproducts. In that case a morphism $(\varphi, F)\colon K \to K'$ is opcartesian if and only if, for each $\beta \in \pi K'$, the family $(F_\alpha\colon L_\alpha \to L'_\beta)_{\alpha \in \varphi^{-1}(\beta)}$ is a coproduct cocone. In particular the opcartesian lift of $\varphi\colon A \to B$ at K is $(\varphi, \mathrm{in})\colon K \to \varphi_! K \coloneqq \langle \coprod_{\varphi(\alpha)=\beta} L_\alpha \rangle_{\beta \in B}$.*

PROOF. Suppose (φ, F) has the stated property. Given $(\psi, G)\colon K \to M$ and $\chi\colon B \to \pi M$ with $\chi\varphi = \psi$, a morphism $(\chi, H)\colon K' \to M$ with $(\chi, H)(\varphi, F) = (\psi, G)$ must satisfy $H_\beta F_\alpha = G_\alpha$ for all $\alpha \in \varphi^{-1}(\beta)$; since $(F_\alpha)_{\alpha \in \varphi^{-1}(\beta)}$ is a coproduct cocone and $G_\alpha\colon L_\alpha \to M_{\psi(\alpha)} = M_{\chi(\beta)}$, there is exactly one such H_β. So (φ, F) is opcartesian, and if $\mathcal{C}$ has small coproducts the morphisms (φ, in) are opcartesian lifts.

Conversely, let $(\varphi, F)\colon K \to K'$ be opcartesian and $\beta \in B$. Given a cocone $(g_\alpha\colon L_\alpha \to Z)_{\alpha \in \varphi^{-1}(\beta)}$, let M be obtained from K' by replacing the leg β by Z, and let $(\varphi, G)\colon K \to M$ have components g_α for $\alpha \in \varphi^{-1}(\beta)$ and F_α otherwise. Taking $\chi = \mathrm{id}_B$, opcartesianness yields a unique vertical morphism $(\mathrm{id}, H)\colon K' \to M$ with $(\mathrm{id}, H)(\varphi, F) = (\varphi, G)$; its component $h_0 = H_\beta$ satisfies $h_0 F_\alpha = g_\alpha$ for all $\alpha \in \varphi^{-1}(\beta)$. If $h\colon L'_\beta \to Z$ is any morphism with the same property, let (id, H^h) be the vertical morphism with component h at β and component $H_{\beta'}$ at every $\beta' \neq \beta$. Then $(\mathrm{id}, H^h)(\varphi, F) = (\varphi, G)$, so $(\mathrm{id}, H^h) = (\mathrm{id}, H)$ and $h = h_0$. Hence $(F_\alpha)_{\alpha \in \varphi^{-1}(\beta)}$ is a coproduct cocone. Finally, if π is an opfibration, let $(X_s)_{s \in S}$ be a small family in $\mathcal{C}$ and take an opcartesian lift of $S \to 1$ at $\langle X_s \rangle_S$; by what has been shown, its unique leg is a coproduct of the X_s. □

Proposition 10.7 (Factorizations)**.** *Every feline morphism $(\varphi, F)\colon K \to K'$ factors as a vertical morphism followed by a cartesian one,*

$$(\varphi, F) = \bar{\varphi} \circ (\mathrm{id}_A, F)\colon K \longrightarrow \varphi^* K' \longrightarrow K',$$

and this factorization is unique up to a unique vertical isomorphism. If $\mathcal{C}$ has small coproducts it also factors as an opcartesian morphism followed by a vertical one, $(\varphi, F) = (\mathrm{id}_B, \tilde{F}) \circ (\varphi, \mathrm{in})$, where $\tilde{F}_\beta\colon \coprod_{\varphi(\alpha)=\beta} L_\alpha \to L'_\beta$ is induced by the F_α.

PROOF. Both identities are verified on gaits and components. Uniqueness up to unique vertical isomorphism is the general uniqueness of cartesian (resp. opcartesian) lifts. □

2. Tail-absorbing morphisms

Remark 3.12 described feline morphisms as pairs (u, θ) with $u\colon \mathfrak{F}_A \to \mathfrak{F}_B$ leg-preserving. If u is allowed to send legs to the tail, we obtain a larger class of morphisms.

Definition 10.8. Let K and K' be cats with leg sets A and B, and write $B_{\mathsf{t}} = B \sqcup \{\mathsf{t}\}$. A *tail-absorbing morphism* $K \rightsquigarrow K'$ is a pair (φ, F) where $\varphi\colon A \to B_{\mathsf{t}}$ and $F = (F_\alpha)_{\alpha \in \varphi^{-1}(B)}$ with $F_\alpha\colon L_\alpha \to L'_{\varphi(\alpha)}$. The composite of $(\varphi, F)\colon K \rightsquigarrow K'$ and $(\psi, G)\colon K' \rightsquigarrow K''$ has gait $\alpha \mapsto \mathsf{t}$ if $\varphi(\alpha) = \mathsf{t}$ and $\alpha \mapsto \psi(\varphi(\alpha))$ otherwise, and components $G_{\varphi(\alpha)} F_\alpha$ for those α with $\psi(\varphi(\alpha)) \neq \mathsf{t}$. The resulting category is denoted $\mathsf{Fel}^{\mathrm{ta}}(\mathcal{C})$.

A leg sent to t is *absorbed into the tail*; its component would be the unique morphism $L_\alpha \to T'$, and is omitted.

Proposition 10.9. *Tail-absorbing morphisms $K \rightsquigarrow K'$ correspond bijectively to pairs (u, θ) where $u\colon \mathfrak{F}_A \to \mathfrak{F}_B$ is a functor fixing h and t and sending each ℓ_α to some ℓ_β or to t, and $\theta\colon K \Rightarrow K' \circ u$ is a natural transformation; composition corresponds to $(v, \zeta) \circ (u, \theta) = (vu, \zeta u \circ \theta)$.*

Proof. As in Remark 3.12: such u correspond to functions $\varphi\colon A \to B_{\mathsf{t}}$, and the components of θ at h, at t and at legs sent to t are forced, all naturality squares involving them commuting by Lemma 2.5. □

Proposition 10.10. *Let $\mathbb{T}$ be the monad $K \mapsto K \sqcup \mathbf{1}$ on $\mathsf{Fel}(\mathcal{C})$, with unit the coprojection $K \to K \sqcup \mathbf{1}$ and multiplication $K \sqcup \mathbf{1} \sqcup \mathbf{1} \to K \sqcup \mathbf{1}$ the codiagonal on the two copies of $\mathbf{1}$. Then $\mathsf{Fel}^{\mathrm{ta}}(\mathcal{C})$ is isomorphic to the Kleisli category $\mathsf{Kl}(\mathbb{T})$, by an isomorphism that is the identity on objects.*

Proof. That $\mathbb{T}$ is a monad follows from the universal property of coproducts: both sides of each monad law are morphisms out of a coproduct, and they agree on each summand. Using the coproduct of Proposition 6.5, the leg set of $K' \sqcup \mathbf{1}$ is $B \sqcup \{*\}$, which we identify with B_{t}, and the leg at t is 1. A Kleisli morphism $K \to K' \sqcup \mathbf{1}$ therefore consists of a function $\varphi\colon A \to B_{\mathsf{t}}$, components $L_\alpha \to L'_{\varphi(\alpha)}$ for $\varphi(\alpha) \in B$, and the unique components $L_\alpha \to 1$ otherwise; this is a tail-absorbing morphism. Kleisli composition $\mu \circ \mathbb{T}(g) \circ f$ sends a leg absorbed by f to the tail (via the codiagonal), and otherwise applies g; this is the composition of Definition 10.8. □

Example 10.11. For $\mathcal{C} = \mathbb{1}$, $\mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$, and $\mathsf{Fel}^{\mathrm{ta}}(\mathbb{1})$ is the category of small sets and partial functions, a leg absorbed into the tail being an element at which the function is undefined. This category is well known to be equivalent to the category $\mathbf{Set}_*$ of pointed sets.

Remark 10.12. One may also let u send legs to the head. A component $L_\alpha \to H'$ is then required, and such a component exists only if L_α admits a morphism to an initial object; if the initial object of $\mathcal{C}$ is strict, this forces L_α to be initial. In $\mathbf{Set}$, for example, only empty legs may be absorbed into the head. We do not pursue *head-absorbing* morphisms.

3. Kan extensions along the one-legged inclusion

Proposition 10.13. *Let $G\colon \mathcal{C} \to \mathcal{E}$ be a functor.*

(i) *If $\mathcal{E}$ has small coproducts, the functor $\widehat{G}\colon \mathsf{Fel}(\mathcal{C}) \to \mathcal{E}$, $\widehat{G}(K) = \coprod_\alpha G(L_\alpha)$, with $\widehat{G}(\psi, F)$ restricting on the α-th summand to $\mathrm{in}_{\psi(\alpha)} \circ G(F_\alpha)$, is a left Kan extension of G along ι, and $\widehat{G} \circ \iota \cong G$.*

(ii) *If $\mathcal{C}$ has small coproducts, $G \circ \Sigma$ is a right Kan extension of G along ι.*

Proof. (i) We identify $\widehat{G}\iota X = \coprod_1 G(X)$ with $G(X)$. Let $H\colon \mathsf{Fel}(\mathcal{C}) \to \mathcal{E}$ be a functor. Given $\sigma\colon G \Rightarrow H\iota$, let $\widehat{\sigma}_K\colon \widehat{G}(K) \to H(K)$ be the morphism whose restriction to the α-th summand is $H(j_\alpha) \circ \sigma_{L_\alpha}$. For $(\psi, F)\colon K \to K'$, (5.1) and naturality of σ give

$$H(\psi, F)H(j_\alpha)\sigma_{L_\alpha} = H(j_{\psi(\alpha)})H(\iota F_\alpha)\sigma_{L_\alpha} = H(j_{\psi(\alpha)})\sigma_{L'_{\psi(\alpha)}} G(F_\alpha),$$

which is the restriction of $\widehat{\sigma}_{K'} \circ \widehat{G}(\psi, F)$ to the α-th summand; so $\widehat{\sigma}$ is natural. Conversely, a natural transformation $\tau\colon \widehat{G} \Rightarrow H$ restricts to $\tau\iota\colon G \Rightarrow H\iota$. These constructions are mutually inverse: $\widehat{\sigma\iota} = \sigma$ because $j_* = \mathrm{id}$ for a unipedal cat; and the restriction of $\widehat{\tau\iota}_K$ to the α-th summand is $H(j_\alpha)\tau_{\iota L_\alpha} = \tau_K \circ \widehat{G}(j_\alpha)$, by naturality of τ, which is the restriction of τ_K, since $\widehat{G}(j_\alpha)$ is the α-th coprojection. The resulting bijection $\mathrm{Nat}(\widehat{G}, H) \cong \mathrm{Nat}(G, H\iota)$ is natural in H, which is the universal property of a left Kan extension.

(ii) Let $\eta\colon \mathrm{id} \Rightarrow \iota\Sigma$ and $\varepsilon\colon \Sigma\iota \Rightarrow \mathrm{id}$ be the unit and counit of $\Sigma \dashv \iota$. For $H\colon \mathsf{Fel}(\mathcal{C}) \to \mathcal{E}$, the maps

$$\begin{aligned} \mathrm{Nat}(H, G\Sigma) &\longrightarrow \mathrm{Nat}(H\iota, G), \qquad \tau \mapsto G\varepsilon \circ \tau\iota, \\ \mathrm{Nat}(H\iota, G) &\longrightarrow \mathrm{Nat}(H, G\Sigma), \qquad \rho \mapsto \rho\Sigma \circ H\eta, \end{aligned}$$

are mutually inverse. Indeed, by naturality of ρ and a triangle identity,

$$G\varepsilon \circ \rho\Sigma\iota \circ H\eta\iota = \rho \circ H(\iota\varepsilon \circ \eta\iota) = \rho,$$

and by naturality of τ and the other triangle identity,

$$G\varepsilon\Sigma \circ \tau\iota\Sigma \circ H\eta = G(\varepsilon\Sigma \circ \Sigma\eta) \circ \tau = \tau.$$

□

Remark 10.14. The comma category $\iota \downarrow K$ is isomorphic to $\coprod_\alpha \mathcal{C}/L_\alpha$, each $\mathcal{C}/L_\alpha$ having the terminal object (L_α, id); by Lemma 2.18, $\widehat{G}(K)$ is therefore also the colimit of G over $\iota \downarrow K$, so the left Kan extension of (i) is pointwise. Dually, $K \downarrow \iota$ is the category of cocones on the legs of K, whose initial object is the coproduct cocone, so the right Kan extension of (ii) is pointwise as well.

Remark 10.15. Part (i) is the coproduct-preserving extension of Theorem 12.4; part (ii) is an instance of the general fact that right Kan extension along a right adjoint is precomposition with its left adjoint.

4. The classifying space

Since $\mathsf{Fel}(\mathcal{C})$ is not small, we fix a second universe $\mathbb{U}'$ with $\mathbb{U} \in \mathbb{U}'$ and regard $\mathsf{Fel}(\mathcal{C})$ as a $\mathbb{U}'$-small category when $\mathcal{C}$ is $\mathbb{U}$-small. The nerve and classifying space are then defined as usual, relative to $\mathbb{U}'$.

THEOREM 10.16. *Let $\mathcal{C}$ be small. The classifying space of $\mathsf{Fel}(\mathcal{C})$ is contractible. More generally, every full subcategory of $\mathsf{Fel}(\mathcal{C})$ that contains a legless cat or a terminal cat has a contractible classifying space; this applies to $\mathsf{Kit}(\mathcal{C})$, to $\mathsf{Fel}_1(\mathcal{C})$, and to the full subcategory of cats with at least one leg.*

PROOF. A category $\mathcal{D}$ with an initial object i has a natural transformation from the constant functor at i to the identity functor; the induced homotopy between the identity of $B\mathcal{D}$ and a constant map shows that $B\mathcal{D}$ is contractible [**13, 11**]. The dual argument applies to a terminal object. A legless cat is initial, and the caudal cat is terminal, in any full subcategory containing them. □

Remark 10.17. Theorem 10.16 says that the homotopy type of the category of cats carries no information. This is a general property of categories with an initial or terminal object and has nothing to do with cats. The groupoid of cats and isomorphisms, considered next, is more informative.

5. The census of cats

Definition 10.18. Let $\mathsf{Iso}(\mathcal{C})$ be the class of isomorphism classes of objects of $\mathcal{C}$. The *multiplicity function* of a cat K is the function $m_K \colon \mathsf{Iso}(\mathcal{C}) \to \mathrm{Card}$, $m_K([X]) = \big|\{\alpha \in \pi K : L_\alpha \cong X\}\big|$.

The function m_K has small support, and its values are small cardinals.

THEOREM 10.19 (Census). *Two cats K and K' are isomorphic if and only if $m_K = m_{K'}$. Every function $m \colon \mathsf{Iso}(\mathcal{C}) \to \mathrm{Card}$ with small support and small values is the multiplicity function of some cat. Hence the isomorphism classes of $\mathcal{C}$-cats are in bijection with such functions.*

PROOF. If $(\varphi, F) \colon K \to K'$ is an isomorphism then φ is a bijection with $L_\alpha \cong L'_{\varphi(\alpha)}$ (Proposition 3.23), so φ restricts to bijections between the sets of legs in each isomorphism class, and $m_K = m_{K'}$. Conversely, if $m_K = m_{K'}$, choose for each class $[X]$ a bijection between the legs of K in the class $[X]$ and those of K', and for each leg an isomorphism with the corresponding leg (axiom of choice); this defines an isomorphism $K \to K'$ by Proposition 3.23. Given m, choose a representative X of each class in the support and take the cat with $m([X])$ legs equal to X for each such class. □

THEOREM 10.20. *Let K be a cat, and for each class $[X]$ in the support of m_K fix a representative X and let $S_{[X]}$ be the set of legs of K in the class $[X]$. Then*

$$\mathrm{Aut}(K) \cong \prod_{[X]} \mathrm{Aut}(X) \wr \mathrm{Sym}(S_{[X]}),$$

where for a group G and a set S, $G \wr \mathrm{Sym}(S)$ denotes the set $G^S \times \mathrm{Sym}(S)$ with multiplication $(f,\sigma)(g,\tau) = ((f \circ \tau)\cdot g, \sigma\tau)$, the product $\cdot$ being pointwise.

PROOF. Choosing isomorphisms $L_\alpha \cong X$ for each leg in the class $[X]$, we may assume, by transport of structure, that every leg of K in the class $[X]$ equals X. An automorphism (φ, F) of K has bijective gait preserving isomorphism classes of legs, so φ is a family $(\varphi_{[X]}) \in \prod_{[X]} \mathrm{Sym}(S_{[X]})$, and its components form a family $(F_\alpha) \in \prod_{[X]} \mathrm{Aut}(X)^{S_{[X]}}$. Composition is $(\varphi, F)(\psi, G) = (\varphi\psi, (F_{\psi(\alpha)} G_\alpha)_\alpha)$, which, on each factor, is the multiplication of the wreath product with $f = F$, $g = G$, $\sigma = \varphi$ and $\tau = \psi$. □

Example 10.21. (i) In $\mathsf{Fel}(\mathbb{1}) \cong \mathbf{Set}$, $\mathrm{Aut}(A) = \mathrm{Sym}(A)$.

(ii) In $\mathsf{Fel}(\mathbf{Set})$, a cat with n legs each of which is a singleton has automorphism group $\mathrm{Sym}(n)$, and a cat with n legs each of which has two elements has automorphism group $(\mathbb{Z}/2) \wr \mathrm{Sym}(n)$, the hyperoctahedral group of order $2^n n!$.

(iii) A quadrupedal **Set**-cat whose legs have 1, 1, 2 and 3 elements has automorphism group $\mathrm{Sym}(2) \times (\mathbb{Z}/2) \times \mathrm{Sym}(3)$, of order 24.

6. The kitten rig

In this section $\mathcal{C}$ is essentially small and symmetric monoidal. Let $\mathsf{Iso}(\mathcal{C})$ be the set of isomorphism classes of objects, which is a commutative monoid under $[X][Y] = [X \otimes Y]$ with unit $[I]$. For a commutative monoid M we write $\mathbb{N}[M]$ for the monoid semiring: finitely supported functions $M \to \mathbb{N}$ with pointwise addition and convolution product.

THEOREM 10.22. *The set $\mathcal{R}(\mathcal{C})$ of isomorphism classes of $\mathcal{C}$-kittens is a commutative semiring under $[K] + [K'] = [K \sqcup K']$ and $[K][K'] = [K \otimes K']$, with zero $[\mathbf{0}]$ and unit $[\mathbf{I}]$. The map $[K] \mapsto m_K$ is an isomorphism of semirings $\mathcal{R}(\mathcal{C}) \cong \mathbb{N}[\mathsf{Iso}(\mathcal{C})]$, and the leg count $[K] \mapsto |\pi K|$ corresponds to the augmentation $\mathbb{N}[\mathsf{Iso}(\mathcal{C})] \to \mathbb{N}$, $m \mapsto \sum_x m(x)$.*

PROOF. The operations are well defined on isomorphism classes since $\sqcup$ and $\otimes$ are functors. By Theorem 10.19, $[K] \mapsto m_K$ is a bijection onto finitely supported functions $\mathsf{Iso}(\mathcal{C}) \to \mathbb{N}$. It is additive since $m_{K \sqcup K'} = m_K + m_{K'}$ (Proposition 6.5), and multiplicative since the legs of $K \otimes K'$ are the $L_\alpha \otimes L'_\beta$, so that $m_{K \otimes K'}(z) = \sum_{xy=z} m_K(x) m_{K'}(y)$. The semiring axioms for $\mathcal{R}(\mathcal{C})$ follow, or may be read off from Theorem 9.3 and Proposition 9.4. The leg count is $\sum_x m_K(x)$. □

Example 10.23. (i) $\mathcal{R}(\mathbb{1}) \cong \mathbb{N}$.

(ii) Let $\mathcal{C}$ be the category of finite sets with the cocartesian monoidal structure $\sqcup$. Then $\mathsf{Iso}(\mathcal{C}) \cong (\mathbb{N}, +)$ and $\mathcal{R}(\mathcal{C}) \cong \mathbb{N}[x]$, the semiring of polynomials with natural-number coefficients: a kitten whose legs have $n_1, \ldots, n_k$ elements corresponds to $x^{n_1} + \cdots + x^{n_k}$. The leg count is the value at $x = 1$, and the cardinality of the body is the value at $x = 1$ of the formal derivative.

(iii) For finite sets with the cartesian structure, $\mathsf{Iso}(\mathcal{C}) \cong (\mathbb{N}, \cdot)$ and $\mathcal{R}(\mathcal{C}) \cong \mathbb{N}[(\mathbb{N}, \cdot)]$.

7. Feline toposes

Standing Assumption 10.24. In this section $\mathcal{E}$ is an elementary topos which has all small products. For example, $\mathcal{E}$ may be any Grothendieck topos.

Recall that an elementary topos has finite limits and finite colimits (in particular an initial object), is cartesian closed, and has a subobject classifier $\top\colon 1 \to \Omega$. We write Y^X for exponentials in $\mathcal{E}$.

Proposition 10.25. *$\mathsf{Fel}(\mathcal{E})$ is cartesian closed. For cats $K = \langle X_a \rangle_{a\in A}$ and $K' = \langle Y_b \rangle_{b\in B}$ the exponential is*

$$K'^K = \Big\langle \prod_{a\in A} Y_{\varphi(a)}^{X_a} \Big\rangle_{\varphi \in B^A},$$

with evaluation $K'^K \times K \to K'$ of gait $(\varphi, a) \mapsto \varphi(a)$ and components the composites $\prod_{a'} Y_{\varphi(a')}^{X_{a'}} \times X_a \to Y_{\varphi(a)}^{X_a} \times X_a \to Y_{\varphi(a)}$.

Proof. $\mathsf{Fel}(\mathcal{E})$ has finite products by Theorem 6.8. For a cat $M = \langle Z_q \rangle_{q\in Q}$, using Corollaries 3.11 and 6.9,

$$\begin{aligned}
\mathsf{Fel}(\mathcal{E})(M \times K, K') &\cong \prod_{q\in Q} \prod_{a\in A} \coprod_{b\in B} \mathcal{E}(Z_q \times X_a, Y_b) \cong \prod_q \prod_a \coprod_b \mathcal{E}(Z_q, Y_b^{X_a}) \\
&\cong \prod_q \coprod_{\varphi\in B^A} \prod_a \mathcal{E}(Z_q, Y_{\varphi(a)}^{X_a}) \cong \prod_q \coprod_\varphi \mathcal{E}\Big(Z_q, \prod_a Y_{\varphi(a)}^{X_a}\Big) \\
&\cong \mathsf{Fel}(\mathcal{E})(M, K'^K),
\end{aligned}$$

naturally in M, where the third bijection is distributivity in **Set**. Tracing the identity of K'^K through these bijections shows that the counit is the stated evaluation. □

Proposition 10.26. *Let $\Omega_{\mathsf{Fel}} \coloneqq \langle \Omega, 1 \rangle$ be the two-legged cat with leg set $\{\mathsf{in}, \mathsf{out}\}$, legs Ω at in and 1 at out, and let $\top_{\mathsf{Fel}}\colon \mathbf{1} \to \Omega_{\mathsf{Fel}}$ have gait $* \mapsto \mathsf{in}$ and component $\top\colon 1 \to \Omega$. Then $\top_{\mathsf{Fel}}$ is a subobject classifier of $\mathsf{Fel}(\mathcal{E})$.*

Proof. By Proposition 6.18 (which applies since $\mathcal{E}$ has an initial object), a monomorphism into $K = \langle X_a \rangle_A$ is a morphism (φ, m) with φ injective and each m_c monic. Two monomorphisms represent the same subobject if and only if they have the same image $A' \subseteq A$ of gaits and, for each $a \in A'$, the same subobject of X_a. Thus a subobject of K is a pair $(A', (S_a)_{a\in A'})$ with $A' \subseteq A$ and S_a a subobject of X_a. Given such a pair, define $\chi\colon K \to \Omega_{\mathsf{Fel}}$ with gait $a \mapsto \mathsf{in}$ for $a \in A'$ and $a \mapsto \mathsf{out}$ otherwise, and with component the classifying map $\chi_{S_a}\colon X_a \to \Omega$ for $a \in A'$ and the unique map $X_a \to 1$ otherwise. By Corollary 6.9, the pullback of $\top_{\mathsf{Fel}}$ along χ has legs indexed by $\{a : \chi(a) = \mathsf{in}\} = A'$, with leg $X_a \times_\Omega 1 = S_a$ at a; so χ classifies (A', S).

Conversely, if χ' classifies (A', S), then the pullback computation shows that the gait of χ' sends exactly A' to in, and that for $a \in A'$ the component of χ' classifies S_a; components at legs sent to out are unique. So $\chi' = \chi$. □

THEOREM 10.27 (Feline toposes). *If $\mathcal{E}$ is an elementary topos with small products, then $\mathsf{Fel}(\mathcal{E})$ is an elementary topos, whose subobject classifier is the two-legged cat $\langle \Omega, 1 \rangle$.*

PROOF. $\mathsf{Fel}(\mathcal{E})$ has finite limits by Theorem 6.8, is cartesian closed by Proposition 10.25, and has a subobject classifier by Proposition 10.26. □

Example 10.28. For $\mathcal{E} = \mathbf{Set}$, $\mathsf{Fel}(\mathbf{Set}) \simeq \mathbf{Set}^{\to}$ (Corollary 7.22), the topos of sheaves on the Sierpiński space. Under this equivalence the classifier $\langle 2, 1 \rangle$ corresponds to the function from a three-element set to a two-element set which sends two elements to in and one to out, which is the familiar subobject classifier of $\mathbf{Set}^{\to}$.

Remark 10.29. If $\mathcal{E}$ is a Grothendieck topos, extensivity of $\mathcal{E}$ shows that $\mathsf{Fel}(\mathcal{E})$ is equivalent to the comma category $\mathcal{E} \downarrow \Delta$, where $\Delta \colon \mathbf{Set} \to \mathcal{E}$ sends a set S to the copower $S \cdot 1$: a cat corresponds to the morphism $\Sigma K \to \pi(K) \cdot 1$. Since Δ is the inverse image of the global sections geometric morphism, it preserves finite limits, and $\mathcal{E} \downarrow \Delta$ is the Artin gluing of $\mathbf{Set}$ and $\mathcal{E}$ along Δ, which is a topos [**6**]. We have preferred the direct proof of Theorem 10.27, which uses no extensivity and applies to every elementary topos with small products.

CHAPTER 11

Higher generalizations

This chapter treats three generalizations: iterated cats, whose legs are themselves cats; gaited cats, whose legs form a diagram rather than a family; and two-dimensional cats in a 2-category. In each case the analogue of Proposition 3.9 holds, and the theory reduces to a known one.

1. Iterated cats

Standing Assumption 11.1. In this section $\mathcal{C}$ has an initial object 0 and a terminal object 1.

By Propositions 4.4 and 6.2, $\mathsf{Fel}(\mathcal{C})$ has the initial object $\mathbf{0}$ and the terminal object $\mathbf{1}$, which we take as its chosen extremal objects. Hence $\mathsf{Fel}(\mathsf{Fel}(\mathcal{C}))$ is defined, and inductively we set $\mathsf{Fel}^0(\mathcal{C}) = \mathcal{C}$ and $\mathsf{Fel}^{n+1}(\mathcal{C}) = \mathsf{Fel}(\mathsf{Fel}^n(\mathcal{C}))$. An object of $\mathsf{Fel}^2(\mathcal{C})$ is a *cat of cats*: a family of $\mathcal{C}$-cats, equipped with a legless head and a terminal tail.

Proposition 11.2. *The category* $\mathsf{Fel}(\mathcal{C})$ *has small coproducts, so its body functor*

$$\Sigma_{\mathsf{Fel}(\mathcal{C})} \colon \mathsf{Fel}^2(\mathcal{C}) \longrightarrow \mathsf{Fel}(\mathcal{C}), \qquad \langle K_a \rangle_{a \in A} \longmapsto \coprod_{a \in A} K_a = \langle L_{a,\alpha} \rangle_{(a,\alpha) \in \coprod_a A_a},$$

exists and is left adjoint to the one-legged inclusion $\iota_{\mathsf{Fel}(\mathcal{C})} \colon \mathsf{Fel}(\mathcal{C}) \to \mathsf{Fel}^2(\mathcal{C})$. *We call it* flattening. *It satisfies*

$$\Sigma_{\mathsf{Fel}(\mathcal{C})} \circ \iota_{\mathsf{Fel}(\mathcal{C})} \cong \mathrm{id}_{\mathsf{Fel}(\mathcal{C})} \qquad \textit{and} \qquad \Sigma_{\mathsf{Fel}(\mathcal{C})} \circ \mathsf{Fel}(\iota_{\mathcal{C}}) \cong \mathrm{id}_{\mathsf{Fel}(\mathcal{C})}.$$

Its restriction to $\mathsf{Fel}(\mathsf{Kit}(\mathcal{C}))$ *is the flattening of litters of Definition 4.12.*

Proof. The first assertions are Propositions 6.5 and 7.11 applied to $\mathsf{Fel}(\mathcal{C})$. The body of a unipedal cat of cats $\langle K \rangle_1$ is K reindexed along $1 \times A \cong A$. For a cat $K = \langle L_\alpha \rangle_A$, $\mathsf{Fel}(\iota_{\mathcal{C}})(K) = \langle \iota L_\alpha \rangle_A$, whose flattening is $\langle L_\alpha \rangle$ indexed by $\coprod_\alpha 1 \cong A$. Both isomorphisms have bijective gaits and identity components and are natural. The final statement is a comparison of definitions. □

Remark 11.3. Proposition 11.2 supplies the unit ι and multiplication $\Sigma_{\mathsf{Fel}(\cdot)}$ of what is, on families, the well-known pseudomonad of free coproduct completion, together with the two unit isomorphisms. We do not verify the remaining coherence conditions of a pseudomonad, and we make no use of them.

THEOREM 11.4. *For every $n \geq 1$ there is an equivalence $\mathsf{Fel}^n(\mathbb{1}) \simeq \mathbf{Set}^{[n-1]}$, where $[n-1] = \{0 < 1 < \cdots < n-1\}$. Under it, an object of $\mathsf{Fel}^n(\mathbb{1})$ corresponds to a chain of functions $X_0 \to X_1 \to \cdots \to X_{n-1}$ in which X_{n-1} is the set of legs of the outermost cat.*

PROOF. For $n = 1$ this is Proposition 3.24, as $\mathbf{Set}^{[0]} = \mathbf{Set}$. Suppose $\mathsf{Fel}^n(\mathbb{1}) \simeq \mathbf{Set}^{[n-1]}$. By Corollary 5.14, $\mathsf{Fel}^{n+1}(\mathbb{1}) \simeq \mathsf{Fel}(\mathbf{Set}^{[n-1]})$, and by Theorem 7.21, $\mathsf{Fel}(\mathbf{Set}^{[n-1]}) \simeq \mathbf{Set}^{[n-1]^{\triangleright}} = \mathbf{Set}^{[n]}$, the adjoined terminal object $\top = n$ carrying the leg set. □

THEOREM 11.5. *Let $\mathcal{D}$ be small, and let $\mathcal{D}^{\triangleleft n}$ denote $\mathcal{D}$ with n successively adjoined initial objects, $\mathcal{D}^{\triangleleft 0} = \mathcal{D}$ and $\mathcal{D}^{\triangleleft(n+1)} = (\mathcal{D}^{\triangleleft n})^{\triangleleft}$. Then $\mathsf{Fel}^n(\mathsf{PSh}(\mathcal{D})) \simeq \mathsf{PSh}(\mathcal{D}^{\triangleleft n})$.*

PROOF. Induction on n, using Corollaries 5.14 and 7.23. □

Corollary 11.6. (i) *Let $\mathcal{D}$ be another category with an initial and a terminal object. If $\mathsf{Fel}^n(\mathcal{C}) \simeq \mathsf{Fel}^n(\mathcal{D})$ for some $n \geq 0$, then $\mathcal{C} \simeq \mathcal{D}$.*
(ii) *For $m \neq n$, $\mathsf{Fel}^m(\mathbb{1}) \not\simeq \mathsf{Fel}^n(\mathbb{1})$.*

PROOF. (i) Induction on n, using Theorem 7.17 for the categories $\mathsf{Fel}^{n-1}(\mathcal{C})$ and $\mathsf{Fel}^{n-1}(\mathcal{D})$, which have initial and terminal objects. (ii) Suppose $m > n$ and $\mathsf{Fel}^m(\mathbb{1}) \simeq \mathsf{Fel}^n(\mathbb{1})$. Since $\mathsf{Fel}^m(\mathbb{1}) = \mathsf{Fel}^n(\mathsf{Fel}^{m-n}(\mathbb{1}))$, part (i) applied to $\mathcal{C} = \mathsf{Fel}^{m-n}(\mathbb{1})$ and $\mathcal{D} = \mathbb{1}$ gives $\mathsf{Fel}^{m-n}(\mathbb{1}) \simeq \mathbb{1}$. But for $k \geq 1$ the category $\mathsf{Fel}^k(\mathbb{1})$ has the non-isomorphic objects $\mathbf{0}$ and $\mathbf{1}$, so it is not equivalent to $\mathbb{1}$. □

2. Gaited cats

Standing Assumption 11.7. In this section $\mathcal{C}$ has an initial object 0 and a terminal object 1.

Definition 11.8. Let $\mathbb{G}$ be a small category, called a *gait*. A *$\mathbb{G}$-gaited $\mathcal{C}$-cat* is a tuple (H, T, L) with H initial, T terminal and $L\colon \mathbb{G} \to \mathcal{C}$ a functor, together with the unique morphisms $H \to L(g) \to T$. A morphism $(H, T, L) \to (H', T', L')$ between cats with gaits $\mathbb{G}$ and $\mathbb{G}'$ is a tuple (u, F_H, F_T, θ) with $u\colon \mathbb{G} \to \mathbb{G}'$ a functor, $F_H\colon H \to H'$, $F_T\colon T \to T'$ and $\theta\colon L \Rightarrow L' \circ u$ a natural transformation, such that the squares analogous to (3.2) commute. We write $\mathsf{Fel}_{\mathrm{gait}}(\mathcal{C})$ for the resulting category.

A gaited cat is the same as a functor $(\mathbb{G}^{\triangleleft})^{\triangleright} \to \mathcal{C}$ sending the adjoined initial and terminal objects to an initial and a terminal object of $\mathcal{C}$; for discrete $\mathbb{G}$ this is Proposition 3.6.

Recall the category $\mathsf{Diag}(\mathcal{C})$ of small diagrams in $\mathcal{C}$: objects are pairs $(\mathbb{G}, L)$ with $\mathbb{G}$ small and $L\colon \mathbb{G} \to \mathcal{C}$, and morphisms $(\mathbb{G}, L) \to (\mathbb{G}', L')$ are pairs (u, θ) with $u\colon \mathbb{G} \to \mathbb{G}'$ and $\theta\colon L \Rightarrow L'u$, composed by $(v, \zeta)(u, \theta) = (vu, \zeta u \circ \theta)$. It is the Grothendieck construction (Definition 2.20) of the functor $\mathbf{Cat}^{\mathrm{op}} \to \mathbf{CAT}$, $\mathbb{G} \mapsto \mathsf{Fun}(\mathbb{G}, \mathcal{C})$, $u \mapsto u^*$, and is therefore fibred over $\mathbf{Cat}$.

Proposition 11.9. *The commutativity conditions in Definition 11.8 are automatic, F_H and F_T are unique isomorphisms, and the functor $\mathsf{Fel}_{\mathrm{gait}}(\mathcal{C}) \to \mathsf{Diag}(\mathcal{C})$ forgetting head and tail is an equivalence. Its restriction to discrete gaits is the equivalence $\mathsf{Fel}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C})$ of Theorem 3.13, and $\mathsf{Fam}(\mathcal{C})$ is the full subcategory of $\mathsf{Diag}(\mathcal{C})$ on discrete diagrams.*

Proof. The proof of Proposition 3.9 and Theorem 3.13 applies verbatim. For the last assertion, a functor between discrete categories is a function and a natural transformation between diagrams on a discrete category is a family of morphisms. □

Proposition 11.10. *Let $\pi_{\mathrm{gait}} \colon \mathsf{Fel}_{\mathrm{gait}}(\mathcal{C}) \to \mathbf{Cat}$ send a gaited cat to its gait. Let $\Delta_X\mathbb{G}$ be the gaited cat with gait $\mathbb{G}$ and constant diagram at X, and let $\Gamma_{\mathrm{gait}}(\mathbb{G}, L)$ be the category of elements of the functor $\mathcal{C}(1, L(-)) \colon \mathbb{G} \to \mathbf{Set}$. Then*

$$\Delta_0 \dashv \pi_{\mathrm{gait}} \dashv \Delta_1 \dashv \Gamma_{\mathrm{gait}}.$$

If $\mathcal{C}$ has small colimits, then the functor $\operatorname{colim} \colon \mathsf{Fel}_{\mathrm{gait}}(\mathcal{C}) \to \mathcal{C}$ is left adjoint to the inclusion $X \mapsto (\mathbb{1}, X)$ of unipedal cats, and it restricts on discrete gaits to the body functor.

Proof. Using Proposition 11.9 we compute in $\mathsf{Diag}(\mathcal{C})$. A morphism $(\mathbb{G}, \Delta 0) \to (\mathbb{G}', L')$ is a functor u together with a natural transformation $\Delta 0 \Rightarrow L'u$, of which there is exactly one; so $\mathsf{Diag}(\mathcal{C})(\Delta_0\mathbb{G}, (\mathbb{G}', L')) \cong \mathbf{Cat}(\mathbb{G}, \mathbb{G}')$. Dually $\mathsf{Diag}(\mathcal{C})((\mathbb{G}, L), \Delta_1\mathbb{G}') \cong \mathbf{Cat}(\mathbb{G}, \mathbb{G}')$. A morphism $(\mathbb{G}, \Delta 1) \to (\mathbb{G}', L')$ is a functor u with a family $x_g \in \mathcal{C}(1, L'(ug))$ such that $L'(u(f)) \circ x_g = x_{g'}$ for $f \colon g \to g'$; this is precisely a functor $\mathbb{G} \to \Gamma_{\mathrm{gait}}(\mathbb{G}', L')$, $g \mapsto (ug, x_g)$, $f \mapsto u(f)$. These bijections are natural. Finally, $\mathsf{Diag}(\mathcal{C})((\mathbb{G}, L), (\mathbb{1}, X))$ is the set of cocones from L to X, which is $\mathcal{C}(\operatorname{colim} L, X)$. □

Remark 11.11. Proposition 11.10 specializes to Theorem 7.3 and Proposition 7.11 for discrete gaits, where the category of elements of $\mathcal{C}(1, L(-))$ is the discrete category on the set of points. All statements are standard facts about the category of diagrams.

3. Two-dimensional cats

Standing Assumption 11.12. In this section $\mathcal{K}$ is a strict 2-category. We write $\mathcal{K}(X, Y)$ for its hom-categories, $\circ$ for composition of 1-cells, $\cdot$ for vertical composition of 2-cells, and $*$ for horizontal composition.

Definition 11.13. An object H of $\mathcal{K}$ is *bi-initial* if every hom-category $\mathcal{K}(H, X)$ is equivalent to the terminal category $\mathbb{1}$, and *bi-terminal* if every $\mathcal{K}(X, T)$ is equivalent to $\mathbb{1}$.

Lemma 11.14. *A category is equivalent to $\mathbb{1}$ if and only if it is nonempty and each of its hom-sets is a singleton. In particular, if H is bi-initial, then for any two 1-cells $f, g \colon H \to X$ there is exactly one 2-cell $f \Rightarrow g$, and it is invertible.*

PROOF. A category $\mathcal{A}$ is equivalent to $\mathbb{1}$ if and only if the unique functor $\mathcal{A} \to \mathbb{1}$ is an equivalence, that is, essentially surjective and fully faithful. It is essentially surjective if and only if $\mathcal{A}$ is nonempty, and fully faithful if and only if every map $\mathcal{A}(a, a') \to \mathbb{1}(*, *)$ is a bijection, that is, every hom-set of $\mathcal{A}$ is a singleton. The unique 2-cell $f \Rightarrow g$ has the unique 2-cell $g \Rightarrow f$ as inverse. □

Definition 11.15. A *two-dimensional $\mathcal{K}$-cat* is a tuple

$$K = (H, T, (L_\alpha)_{\alpha \in A}, (h_\alpha), (t_\alpha), \sigma)$$

where H is bi-initial, T is bi-terminal, the L_α are objects, and $h_\alpha \colon H \to L_\alpha$, $t_\alpha \colon L_\alpha \to T$ and $\sigma \colon H \to T$ are chosen 1-cells. By Lemma 11.14 there is a unique invertible 2-cell $\omega_\alpha \colon t_\alpha \circ h_\alpha \Rightarrow \sigma$.

Unlike the one-dimensional case, the canonical 1-cells are now data: they are unique only up to unique invertible 2-cell.

Definition 11.16. A *pseudo-morphism* $F \colon K \to K'$ of two-dimensional $\mathcal{K}$-cats is a tuple $(\varphi, F_H, F_T, (F_\alpha), (\lambda_\alpha), (\rho_\alpha), \kappa)$ consisting of a function $\varphi \colon A \to B$, 1-cells $F_H \colon H \to H'$, $F_T \colon T \to T'$, $F_\alpha \colon L_\alpha \to L'_{\varphi(\alpha)}$, and invertible 2-cells

$$\lambda_\alpha \colon F_\alpha \circ h_\alpha \Rightarrow h'_{\varphi(\alpha)} \circ F_H, \qquad \rho_\alpha \colon t'_{\varphi(\alpha)} \circ F_\alpha \Rightarrow F_T \circ t_\alpha, \qquad \kappa \colon F_T \circ \sigma \Rightarrow \sigma' \circ F_H,$$

such that, for each α, the 2-cell $t'_{\varphi(\alpha)} \circ F_\alpha \circ h_\alpha \Rightarrow \sigma' \circ F_H$ obtained by pasting λ_α and $\omega'_{\varphi(\alpha)}$ equals the one obtained by pasting ρ_α, ω_α and κ. A *2-cell* $F \Rightarrow G$ between pseudo-morphisms with the *same* gait φ is a tuple of 2-cells $\Theta_H \colon F_H \Rightarrow G_H$, $\Theta_T \colon F_T \Rightarrow G_T$, $\Theta_\alpha \colon F_\alpha \Rightarrow G_\alpha$ compatible with the λ, ρ and κ; there are no 2-cells between pseudo-morphisms with different gaits. Composition is by composing 1-cells and pasting 2-cells. We write $\mathsf{Fel}_2(\mathcal{K})$ for the resulting 2-category.

Proposition 11.17 (Automatic pseudo-commutativity)**.** *In Definition 11.16:*

(i) *for any choice of φ, F_H, F_T and (F_α) there exist unique 2-cells λ_α, ρ_α, κ, and they are invertible;*
(ii) *the coherence condition on a pseudo-morphism holds automatically;*
(iii) *for pseudo-morphisms F, G with the same gait and any family of 2-cells $\Theta_\alpha \colon F_\alpha \Rightarrow G_\alpha$, there are unique Θ_H, Θ_T, and the compatibility conditions hold automatically.*

PROOF. The source and target 1-cells of λ_α have the bi-initial domain H; those of ρ_α have the bi-terminal codomain T'; those of κ have domain H. By Lemma 11.14 each of these 2-cells exists, is unique and is invertible. The coherence condition in (ii) is an equation between two 2-cells between the same pair of 1-cells $H \to T'$, which holds by uniqueness. In (iii), Θ_H is a 2-cell between 1-cells $H \to H'$ and Θ_T one between 1-cells $T \to T'$, so both exist uniquely, and each compatibility condition is an equation between 2-cells whose underlying 1-cells have domain H or codomain T'. □

Let $\mathsf{Fam}(\mathcal{K})$ be the 2-category of small families in $\mathcal{K}$: objects $(A, (L_\alpha))$, 1-cells $(\varphi, (F_\alpha))$, and 2-cells $(\varphi, F) \Rightarrow (\varphi, G)$ the families of 2-cells $\Theta_\alpha \colon F_\alpha \Rightarrow G_\alpha$ (none between 1-cells with different gaits).

THEOREM 11.18. *The forgetful* 2*-functor* $U_2 \colon \mathsf{Fel}_2(\mathcal{K}) \to \mathsf{Fam}(\mathcal{K})$ *is a local equivalence: each functor of hom-categories*

$$\mathsf{Fel}_2(\mathcal{K})(K, K') \longrightarrow \mathsf{Fam}(\mathcal{K})(U_2K, U_2K')$$

is surjective on objects and fully faithful. It is surjective on objects if and only if $\mathcal{K}$ *has a bi-initial and a bi-terminal object, in which case* U_2 *is a biequivalence.*

PROOF. Surjectivity on objects of the hom-functors: given $(\varphi, (F_\alpha))$, the hom-categories $\mathcal{K}(H, H')$ and $\mathcal{K}(T, T')$ are nonempty, so F_H and F_T may be chosen, and Proposition 11.17(i),(ii) supply the remaining data. Full faithfulness is Proposition 11.17(iii). For surjectivity on objects, a two-dimensional cat over a given family requires a bi-initial and a bi-terminal object, and conversely, given such objects, the 1-cells h_α, t_α, σ can be chosen since the relevant hom-categories are nonempty. A 2-functor which is a local equivalence and surjective on objects is a biequivalence [**5**]. □

Example 11.19. Let $\mathcal{K} = \mathbf{Cat}$, the 2-category of small categories. The only bi-initial object is the empty category: if H is nonempty then $\mathbf{Cat}(H, \varnothing)$ is empty. The bi-terminal objects are the categories equivalent to $\mathbb{1}$, for instance the indiscrete category on any nonempty small set. Thus a two-dimensional **Cat**-cat may have as its tail the indiscrete category on $\mathbb{N}$, whereas a one-dimensional **Cat**-cat must have a tail with exactly one object and one morphism. By Theorem 11.18 this extra freedom is invisible up to biequivalence.

Remark 11.20. One may similarly consider cats in an $(\infty, 1)$-category, with the head initial and the tail terminal in the ∞-categorical sense. The commutativity data would then be parametrized by contractible spaces, and one expects the ∞-category of such cats to be equivalent to the ∞-category of set-indexed families. We have not written out a proof, and we make no claim to this effect.

CHAPTER 12

Representation and decomposition theorems

Standing Assumption 12.1. Throughout this chapter $\mathcal{C}$ has an initial object 0 and a terminal object 1. A recognition theorem characterizing categories of cats among locally small categories is proved in Chapter 13 (Theorem 13.10), since it is naturally formulated in terms of Gideon Decompositions.

1. The Leg Decomposition Theorem

THEOREM 12.2 (Leg Decomposition). (i) *Every cat K is the coproduct of the unipedal cats $\iota(L_\alpha)$, $\alpha \in \pi(K)$, with coprojections the leg inclusions j_α.*

(ii) (Uniqueness.) *If $(X_i)_{i\in I}$ and $(Y_j)_{j\in J}$ are small families of objects with $\coprod_i \iota(X_i) \cong \coprod_j \iota(Y_j)$, then there is a bijection $\sigma\colon I \to J$ with $X_i \cong Y_{\sigma(i)}$ for all i.*

(iii) (Refinement.) *For a cat K and a small set S, the following are in canonical bijection:*

- (i) *functions $f\colon \pi(K) \to S$;*
- (ii) *morphisms $K \to \Delta_1(S)$;*
- (iii) *coproduct decompositions of K indexed by S, that is, families of morphisms $(\kappa_s\colon K_s \to K)_{s\in S}$ exhibiting K as a coproduct, taken up to isomorphism of families over K.*

PROOF. (i) is Corollary 6.6. (ii) An isomorphism between the two coproducts has, by Propositions 3.23 and 6.5, a bijective gait $\sigma\colon I \to J$ and isomorphisms $X_i \to Y_{\sigma(i)}$ as components.

(iii) The bijection between (a) and (b) is the adjunction $\pi \dashv \Delta_1$. Given f, set $K_s = \langle L_\alpha\rangle_{\alpha\in f^{-1}(s)}$ with κ_s the morphism with gait the inclusion $f^{-1}(s) \hookrightarrow \pi(K)$ and identity components; this is a coproduct decomposition by Proposition 6.5, since $\pi(K) = \coprod_s f^{-1}(s)$. Conversely, let (κ_s) be a coproduct decomposition. Since π preserves coproducts, the gaits of the κ_s are injective with disjoint images covering $\pi(K)$; define $f(\alpha) = s$ if α is in the image of the gait of κ_s. The canonical morphism $\coprod_s K_s \to K$ is an isomorphism, so by Proposition 3.23 each component of each κ_s is an isomorphism, and (κ_s) is isomorphic over K to the decomposition constructed from f. The two constructions are mutually inverse. □

Remark 12.3. Part (i) says that every cat admits a Gideon Decomposition in $\mathsf{Fel}(\mathcal{C})$ (Example 12.13(v) below). Part (ii) has the form of a Krull–Schmidt theorem, but its proof is immediate, because isomorphisms of cats are bijections on legs. Part (iii) is an expression of the extensivity of $\mathsf{Fel}(\mathcal{C})$ (Proposition 6.7).

2. The universal property

THEOREM 12.4 (Universal property of $\mathsf{Fel}(\mathcal{C})$). *Let $\mathcal{E}$ be a category with small coproducts, and let $\mathsf{Fun}_{\sqcup}(\mathsf{Fel}(\mathcal{C}), \mathcal{E})$ be the category of functors $\mathsf{Fel}(\mathcal{C}) \to \mathcal{E}$ preserving small coproducts. Then restriction along ι,*

$$\iota^* \colon \mathsf{Fun}_{\sqcup}(\mathsf{Fel}(\mathcal{C}), \mathcal{E}) \longrightarrow \mathsf{Fun}(\mathcal{C}, \mathcal{E}),$$

is an equivalence of categories, with quasi-inverse $G \mapsto \widehat{G}$, $\widehat{G}(K) = \coprod_\alpha G(L_\alpha)$, the left Kan extension of G along ι.

PROOF. $\widehat{G}$ is a functor, with $\widehat{G}(\varphi, F)$ restricting on the α-th summand to $\mathrm{in}_{\varphi(\alpha)} \circ G(F_\alpha)$. It preserves coproducts because the leg set of $\coprod_i K_i$ is $\coprod_i \pi(K_i)$, and $\widehat{G}\iota \cong G$ since a coproduct of one object is that object. By Proposition 10.13, $\widehat{G} = \mathrm{Lan}_\iota G$. If Φ preserves coproducts then, by Theorem 12.2(i),

$$\Phi(K) \cong \Phi\Big(\coprod_\alpha \iota L_\alpha\Big) \cong \coprod_\alpha \Phi(\iota L_\alpha) = \widehat{\iota^*\Phi}(K),$$

and these isomorphisms are natural in K by (5.1). Finally, a natural transformation $\tau \colon \Phi \Rightarrow \Psi$ between coproduct-preserving functors is determined by its components at unipedal cats, since τ_K is the coproduct of the $\tau_{\iota L_\alpha}$ by naturality with respect to the leg inclusions; and every natural transformation $\rho \colon \iota^*\Phi \Rightarrow \iota^*\Psi$ extends in this way, to the family whose component at K is the coproduct of the ρ_{L_α} under the isomorphisms $\Phi(K) \cong \coprod_\alpha \Phi(\iota L_\alpha)$ and $\Psi(K) \cong \coprod_\alpha \Psi(\iota L_\alpha)$. This family is natural: for $(\psi, F) \colon K \to K'$, both composites in the naturality square restrict on the α-th summand to $\Psi(j_{\psi(\alpha)}) \circ \rho_{L'_{\psi(\alpha)}} \circ \Phi(\iota F_\alpha)$, by (5.1) and naturality of ρ. Hence ι^* is fully faithful, and it is essentially surjective by $\iota^*\widehat{G} \cong G$. □

Corollary 12.5. (i) *If $\mathcal{C}$ has small coproducts, $\Sigma \cong \widehat{\mathrm{id}_{\mathcal{C}}}$.*

(ii) *For any functor $F \colon \mathcal{C} \to \mathcal{D}$, where $\mathcal{D}$ has an initial and a terminal object, $\mathsf{Fel}(F) \cong \widehat{\iota_{\mathcal{D}} F}$.*

(iii) *$\pi \cong \widehat{\Delta}$ where $\Delta \colon \mathcal{C} \to \mathbf{Set}$ is the constant functor at 1, and $\Gamma \cong \widehat{\mathcal{C}(1, -)}$.*

(iv) *$\mathbb{Y} \cong \widehat{y}$ for the Yoneda embedding $y \colon \mathcal{C} \to \mathsf{PSh}(\mathcal{C})$.*

PROOF. Each functor on the left preserves coproducts and has the stated restriction along ι. For π, preservation of coproducts holds because π is a left adjoint (Theorem 7.3). For Γ, it follows from the formula $\Gamma(\coprod_i K_i) = \coprod_{(i,\alpha)} \mathcal{C}(1, L_{i,\alpha}) = \coprod_i \Gamma(K_i)$, or equivalently from $\Gamma \cong \mathsf{Fel}(\mathcal{C})(\mathbf{1}, -)$ and the

connectedness of the unipedal cat $\mathbf{1}$ (Proposition 7.15). For $\mathbb{Y}$, $\mathsf{Fel}(F)$ and Σ it is immediate from the definitions and Proposition 6.5. □

Remark 12.6. Theorem 12.4 is the statement that $\mathsf{Fel}(\mathcal{C})$, like $\mathsf{Fam}(\mathcal{C})$, is the free completion of $\mathcal{C}$ under small coproducts. We emphasize that the universal property makes no reference to heads or tails.

3. The Kitten Approximation Theorem

For a cat K and a subset $F \subseteq \pi(K)$, let $K|_F = \langle L_\alpha \rangle_{\alpha \in F}$ and let $\kappa_F \colon K|_F \to K$ be the morphism with gait the inclusion and identity components. If F is finite, $K|_F$ is a kitten, called a *subkitten* of K.

THEOREM 12.7 (Kitten Approximation). *Let K be a cat, and let $\mathcal{P}_{\mathrm{fin}}$ be the directed poset of finite subsets of $\pi(K)$. The diagram $\mathcal{P}_{\mathrm{fin}} \to \mathsf{Fel}(\mathcal{C})$, $F \mapsto K|_F$, with the evident inclusions, has colimit K, with colimit cocone $(\kappa_F)_F$. In particular every cat is a filtered colimit of kittens, and this colimit is preserved by π.*

PROOF. The κ_F form a cocone. Let $(\mu_F \colon K|_F \to M)_F$ be another cocone, $\mu_F = (\psi_F, G_F)$. For $\alpha \in \pi(K)$, compatibility gives $\psi_F(\alpha) = \psi_{\{\alpha\}}(\alpha)$ and $(G_F)_\alpha = (G_{\{\alpha\}})_\alpha$ whenever $\alpha \in F$. Define $\mu \colon K \to M$ by $\psi(\alpha) = \psi_{\{\alpha\}}(\alpha)$ and $\mu_\alpha = (G_{\{\alpha\}})_\alpha$. Then $\mu\kappa_F = \mu_F$ for all F, and μ is unique because every leg lies in some F. Since $\pi(K)$ is the directed union of its finite subsets, π preserves this colimit. □

Remark 12.8. The colimit in Theorem 12.7 exists in $\mathsf{Fel}(\mathcal{C})$ whether or not $\mathcal{C}$ has any filtered colimits; it is a directed union of subfamilies. We do not claim that $\mathsf{Fel}(\mathcal{C})$ is equivalent to the category of ind-objects of $\mathsf{Kit}(\mathcal{C})$.

4. Slices

THEOREM 12.9 (Slice Decomposition). *For every cat K there is an equivalence of categories*

$$\mathsf{Fel}(\mathcal{C})/K \simeq \prod_{\alpha \in \pi(K)} \mathsf{Fel}(\mathcal{C}/L_\alpha).$$

In particular $\mathsf{Fel}(\mathcal{C})/\mathbf{1} \simeq \mathsf{Fel}(\mathcal{C})$, $\mathsf{Fel}(\mathcal{C})/\Delta_1(S) \simeq \mathsf{Fel}(\mathcal{C})^S$, *and* $\mathsf{Fel}(\mathcal{C})/\mathbf{0} \simeq \mathbb{1}$.

PROOF. Each slice $\mathcal{C}/X$ has the initial object $(0 \to X)$ and the terminal object (id_X), so $\mathsf{Fel}(\mathcal{C}/X)$ is defined. By Theorem 12.2(i) and extensivity (Proposition 6.7), $\mathsf{Fel}(\mathcal{C})/K \simeq \prod_\alpha \mathsf{Fel}(\mathcal{C})/\iota(L_\alpha)$. It remains to show $\mathsf{Fel}(\mathcal{C})/\iota X \simeq \mathsf{Fel}(\mathcal{C}/X)$. An object of $\mathsf{Fel}(\mathcal{C})/\iota X$ is a cat $\langle M_b \rangle_B$ with a morphism to ιX, that is, a family of morphisms $m_b \colon M_b \to X$; this is a family of objects of $\mathcal{C}/X$. A morphism over ιX is a morphism (ψ, G) with $m'_{\psi(b)} G_b = m_b$ for all b, which is a family of morphisms in $\mathcal{C}/X$. This identifies $\mathsf{Fel}(\mathcal{C})/\iota X$ with $\mathsf{Fel}(\mathcal{C}/X)$ up to the choice of heads and tails. The special cases follow from $\mathcal{C}/1 \cong \mathcal{C}$ and from the empty product being $\mathbb{1}$. □

5. Image factorizations

Proposition 12.10. *Suppose $\mathcal{C}$ has small coproducts and every morphism of $\mathcal{C}$ factors as an epimorphism followed by a monomorphism. Then every feline morphism factors as an epimorphism followed by a monomorphism of* $\mathsf{Fel}(\mathcal{C})$.

Proof. Let $(\varphi, F)\colon K \to K'$, and let $B_0 = \varphi(A) \subseteq B$. For $\beta \in B_0$, factor the induced morphism $\tilde{F}_\beta\colon \coprod_{\varphi(\alpha)=\beta} L_\alpha \to L'_\beta$ as an epimorphism $e_\beta\colon \coprod_{\varphi(\alpha)=\beta} L_\alpha \to E_\beta$ followed by a monomorphism $m_\beta\colon E_\beta \to L'_\beta$. Let $E = \langle E_\beta \rangle_{B_0}$. The morphism $(\varphi, (e_{\varphi(\alpha)}\mathrm{in}_\alpha)_\alpha)\colon K \to E$ has surjective gait, and for each β the family $(e_\beta \mathrm{in}_\alpha)_{\alpha\in\varphi^{-1}(\beta)}$ is jointly epimorphic because the coprojections are jointly epimorphic and e_β is epic; it is therefore an epimorphism by Proposition 6.20. The morphism $(B_0 \hookrightarrow B, (m_\beta))\colon E \to K'$ has injective gait and monic components, so it is a monomorphism by Proposition 6.18. Their composite is (φ, F). □

6. Presheaf representation

We restate the results of Chapter 8 as a representation theorem.

Corollary 12.11 (Presheaf Representation)**.** *Assume $\mathcal{C}$ is locally small. The feline Yoneda functor identifies* $\mathsf{Fel}(\mathcal{C})$*, up to equivalence, with the full subcategory of* $\mathsf{PSh}(\mathcal{C})$ *consisting of the presheaves whose category of elements is a small coproduct of categories with terminal objects. Under this identification, the legs of a cat correspond to the connected components of the category of elements, to the elements of the presheaf at the head, and to the summands of its decomposition into representables.*

Proof. This combines Theorems 8.5, 8.6 and 8.10 with Proposition 8.8. □

7. Gideon Decompositions

The Leg Decomposition Theorem decomposes every cat into unipedal cats. The analogous decomposition of an object of $\mathcal{C}$ into connected objects is the subject of this section; its relation to the notions of Section 5 and Section 6 is the content of Chapter 13.

Definition 12.12 (Gideon Decomposition)**.** A *Gideon Decomposition* of an object X of $\mathcal{C}$ is a pair (K, β) where K is a $\mathcal{C}$-cat satisfying the Chloe Condition and $\beta\colon K \to \iota X$ is a cocone exhibiting X as a coproduct of the legs of K; that is, for every object Z the map $\mathcal{C}(X, Z) \to \prod_\alpha \mathcal{C}(L_\alpha, Z)$, $f \mapsto (f\beta_\alpha)_\alpha$, is a bijection. A morphism of Gideon Decompositions $(K, \beta) \to (K', \beta')$ of X is a morphism of cats $\theta\colon K \to K'$ with $\beta'\theta = \beta$. We write $\mathsf{Gid}(X)$ for the category of Gideon Decompositions of X. An object is *Gideon-decomposable* if $\mathsf{Gid}(X)$ is nonempty; we write $\mathcal{C}^{\mathrm{G}}$ for the full subcategory of Gideon-decomposable objects, and say that $\mathcal{C}$ *admits Gideon Decompositions* if $\mathcal{C}^{\mathrm{G}} = \mathcal{C}$.

If $\mathcal{C}$ has small coproducts, (K, β) is a Gideon Decomposition if and only if K is Chloe and the transpose $\beta^\flat \colon \Sigma K \to X$ of β under $\Sigma \dashv \iota$ is an isomorphism. Note that Definition 12.12 makes sense for an arbitrary category $\mathcal{C}$, with the convention of Theorem 4.25 that there are no $\mathcal{C}$-cats when $\mathcal{C}$ lacks an initial or a terminal object; in particular, if some object of $\mathcal{C}$ admits a Gideon Decomposition, then $\mathcal{C}$ has a terminal object. This observation is used in Section 2.

Example 12.13. (i) Every set X has the Gideon Decomposition consisting of the cat $\langle \{x\} \rangle_{x \in X}$ and the inclusions of the points; so **Set** admits Gideon Decompositions.

(ii) A topological space X is Gideon-decomposable if and only if its connected components are open, for instance if X is locally connected. Indeed, if $X \cong \coprod_a X_a$ with each X_a nonempty and connected, the X_a are disjoint clopen connected subsets covering X, hence are its connected components, which are therefore open; conversely, a space is the coproduct of any partition of it into open subsets. In particular $\mathbb{Q}$ is not Gideon-decomposable.

(iii) **Gph** admits Gideon Decompositions: every graph is the coproduct of its connected components.

(iv) In a pointed category, an object is Gideon-decomposable if and only if it is a zero object, since the only Chloe cats are legless (Remark 4.17) and the coproduct of the empty family is initial.

(v) The category $\mathsf{Fel}(\mathcal{C})$ admits Gideon Decompositions: by Theorem 12.2 and Proposition 7.15, the cat of cats $\langle \iota L_\alpha \rangle_{\alpha \in \pi K}$ with the leg inclusions j_α is a Gideon Decomposition of K.

7.1. The body of a Chloe cat.

Proposition 12.14. *Suppose $\mathcal{C}$ has small coproducts. Then:*

(i) *the body functor restricted to Chloe cats, $\Sigma^{\mathrm{Ch}} \colon \mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C}) \to \mathcal{C}$, is fully faithful;*

(ii) *its essential image is $\mathcal{C}^{\mathrm{G}}$, so that $\mathcal{C}^{\mathrm{G}} \simeq \mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C}) \simeq \mathsf{Fam}(\mathcal{C}^\circ)$;*

(iii) *$\mathcal{C}^{\mathrm{G}}$ is closed under small coproducts in $\mathcal{C}$;*

(iv) *$\mathcal{C}$ admits Gideon Decompositions if and only if the functor $\mathsf{Fam}(\mathcal{C}^\circ) \to \mathcal{C}$, $(A, X) \mapsto \coprod_a X_a$, is an equivalence.*

Proof. (i) For Chloe cats K, K',

$$\mathsf{Fel}(\mathcal{C})(K, K') \cong \prod_\alpha \coprod_\beta \mathcal{C}(L_\alpha, L'_\beta) \cong \prod_\alpha \mathcal{C}\Big(L_\alpha, \coprod_\beta L'_\beta\Big) \cong \mathcal{C}(\Sigma K, \Sigma K'),$$

the middle bijection by connectedness of L_α; the composite is the action of Σ. (ii) X is in the essential image if and only if $X \cong \Sigma K$ for a Chloe cat K, that is, if and only if X is Gideon-decomposable; the second equivalence is Proposition 4.16(iv). (iii) A coproduct of coproducts of connected objects is a coproduct of connected objects. (iv) follows from (i), (ii) and Proposition 4.16(iv). □

Example 12.15. Apply Proposition 12.14 to the category $\mathsf{Fel}(\mathcal{C})$, which has small coproducts and admits Gideon Decompositions (Example 12.13(v)). Its Chloe cats are the cats of unipedal cats (Example 4.18(iii)), and its body functor is flattening (Proposition 11.2). Hence flattening restricts to an equivalence between the category of cats of unipedal cats and $\mathsf{Fel}(\mathcal{C})$.

CHAPTER 13

The CMG Correspondence

Standing Assumption 13.1. Throughout this chapter $\mathcal{C}$ is a locally small category with an initial object 0 and a terminal object 1. Small coproducts in $\mathcal{C}$ are not assumed unless stated. We write $\mathcal{C}^\circ$ for the full subcategory of connected objects (Definition 2.9).

This chapter proves the CMG Correspondence Theorem, which relates three notions introduced in earlier chapters:

- the *Chloe Condition* on a cat (Definition 4.15): every leg is connected;
- *Margaret Compatibility* of a cocone $\beta\colon K \to \iota X$ (Definition 8.19): every morphism from a connected object to X factors uniquely through exactly one leg;
- *Gideon Decompositions* of an object X (Definition 12.12): a cat satisfying the Chloe Condition together with a coproduct cocone from its legs to X.

We then deduce a recognition theorem for categories of cats and study the interaction of the three notions with the monoidal structure of Chapter 9. The identification of each notion with a familiar one is recorded in Remark 13.7.

1. The CMG Correspondence Theorem

THEOREM 13.2 (CMG Correspondence). *Let X be an object of $\mathcal{C}$, let K be a $\mathcal{C}$-cat, and let $\beta\colon K \to \iota X$ be a cocone. Consider the conditions:*

(C) *K satisfies the Chloe Condition;*
(M) *β is Margaret compatible;*
(G) *(K, β) is a Gideon Decomposition of X.*

Then:

(i) (G) *implies* (C) *and* (M).
(ii) *Suppose that X admits a Gideon Decomposition (K_0, β_0). If* (C) *and* (M) *hold, then there is a unique morphism of cats $\theta\colon K \to K_0$ with $\beta_0\theta = \beta$; this θ is an isomorphism, and* (G) *holds.*
(iii) *The category $\mathsf{Gid}(X)$ is either empty or equivalent to $\mathbb{1}$: any two Gideon Decompositions of X are related by a unique morphism, and it is an isomorphism.*

PROOF. (i) (C) is part of the definition of a Gideon Decomposition. For (M), let Y be connected. Since X, with the coprojections β_α, is a coproduct

of the family (L_α), the map m_Y^β is precisely the canonical map (2.1) for this family, which is bijective because Y is connected.

(ii) By (i), K_0 satisfies (C) and β_0 satisfies (M). By Proposition 8.20, $\mathbb{Y}^\circ(\beta)\colon \mathbb{Y}^\circ K \to y^\circ X$ and $\mathbb{Y}^\circ(\beta_0)\colon \mathbb{Y}^\circ K_0 \to y^\circ X$ are isomorphisms. Since K satisfies the Chloe Condition, Proposition 8.17 shows that $\mathbb{Y}^\circ\colon \mathsf{Fel}(\mathcal{C})(K, K_0) \to \mathsf{PSh}(\mathcal{C}^\circ)(\mathbb{Y}^\circ K, \mathbb{Y}^\circ K_0)$ is a bijection; let θ be the unique morphism with $\mathbb{Y}^\circ(\theta) = \mathbb{Y}^\circ(\beta_0)^{-1} \circ \mathbb{Y}^\circ(\beta)$. Applying the same proposition to the pair $(K, \iota X)$, the map $\mathbb{Y}^\circ\colon \mathsf{Fel}(\mathcal{C})(K, \iota X) \to \mathsf{PSh}(\mathcal{C}^\circ)(\mathbb{Y}^\circ K, y^\circ X)$ is a bijection. Since $\mathbb{Y}^\circ(\beta_0\theta) = \mathbb{Y}^\circ(\beta)$, we obtain $\beta_0\theta = \beta$. If θ' also satisfies $\beta_0\theta' = \beta$, then $\mathbb{Y}^\circ(\beta_0)\mathbb{Y}^\circ(\theta') = \mathbb{Y}^\circ(\beta)$, so $\mathbb{Y}^\circ(\theta') = \mathbb{Y}^\circ(\theta)$ and $\theta' = \theta$. Since K and K_0 both satisfy the Chloe Condition, Proposition 8.17 shows that $\mathbb{Y}^\circ$ is fully faithful on the full subcategory spanned by K and K_0, and so reflects isomorphisms there; as $\mathbb{Y}^\circ(\theta)$ is an isomorphism, so is θ. Finally, write $\theta = (\varphi, (\theta_\alpha))$. By Proposition 3.23, φ is bijective and each θ_α is invertible, so the family $\beta_\alpha = \beta_{0,\varphi(\alpha)} \circ \theta_\alpha$ is a coproduct cocone, being obtained from the coproduct cocone β_0 by reindexing along a bijection and precomposing with isomorphisms. Hence (G) holds.

(iii) Given Gideon Decompositions (K, β) and (K_0, β_0), part (i) shows that (K, β) satisfies (C) and (M), and part (ii) provides a unique morphism $(K, \beta) \to (K_0, \beta_0)$, which is an isomorphism. □

Corollary 13.3. *If X is Gideon-decomposable, then a cocone $\beta\colon K \to \iota X$ from a cat satisfying the Chloe Condition is a Gideon Decomposition if and only if it is Margaret compatible. In symbols: under the hypothesis that $\mathsf{Gid}(X) \neq \varnothing$,*

$$\text{(G)} \iff \text{(C)} \wedge \text{(M)}.$$

Proof. Combine Theorem 13.2(i),(ii). □

Sharpness. Each hypothesis in Theorem 13.2 and Corollary 13.3 is necessary.

Counterexample 13.4 (Existence cannot be dropped)**.** In each of the following, the object X admits no Gideon Decomposition.

(i) In **Top**, the cocone of Example 8.21(iii) from the cat $K = \langle \{q\} \rangle_{q \in \mathbb{Q}}$ to $\mathbb{Q}$ satisfies (C) and (M) but not (G). Here $\mathbb{Q}$ admits no Gideon Decomposition (Example 12.13(ii)).

(ii) In **Ab**, for any nonzero abelian group X, the unique cocone from the legless cat to X satisfies (C) vacuously and (M) vacuously (Example 8.21(ii)), but not (G), since X is not initial.

Counterexample 13.5 (The Chloe Condition cannot be dropped)**.** In **Set** let Y be a set with at least one element and $X = Y \sqcup Y$. The unipedal cat ιX with the identity cocone, and the two-legged cat $\langle Y, Y \rangle$ with the coprojections, are both Margaret compatible cocones exhibiting X as a coproduct of their legs. They are not isomorphic over X, since their leg sets have different cardinalities. Without (C), therefore, neither the implication

(M)⇒(G) nor the uniqueness of Theorem 13.2(iii) survives. In this example both cocones are coproduct cocones; the following shows that without (C), Margaret Compatibility does not even force a coproduct cocone, although X is Gideon-decomposable. In **Ab**, the object $X = 0$ has the legless Gideon Decomposition. The cocone from $\iota(\mathbb{Z})$ to 0 is Margaret compatible, vacuously, since **Ab** has no connected objects, but it is not a coproduct cocone, since $\mathbb{Z}$ is not a zero object.

Counterexample 13.6 (Margaret Compatibility cannot be dropped)**.** In **Set**, the unipedal cat $\iota(1)$ with the cocone $1 \to 2$ picking out one element satisfies (C), and 2 is Gideon-decomposable, but the cocone is neither Margaret compatible nor a Gideon Decomposition.

Remark 13.7. We identify the three notions and the theorem with classical ones.

(i) The Chloe Condition on a cat is the condition that its legs be connected objects; Chloe cats are, up to equivalence, families of connected objects (Proposition 4.16(iv)).
(ii) Margaret Compatibility of a cocone is the invertibility of the induced map of restricted Yoneda presheaves on connected objects (Proposition 8.20).
(iii) A Gideon Decomposition is a decomposition of an object as a coproduct of connected objects, i.e. a decomposition into connected components.
(iv) Theorem 13.2(iii) is the uniqueness of decompositions into connected components, and Theorem 13.2(ii) is the observation that, between coproducts of connected objects, a morphism is determined by, and is invertible if and only if it induces a bijection on, morphisms out of connected objects.

We are not aware of any application of the CMG Correspondence Theorem outside this monograph. Within it, the theorem is used only to show that the Gideon number (Definition 13.8) is well defined and in the proof of Theorem 13.10.

Definition 13.8. For a Gideon-decomposable object X, its *Gideon number* $\gamma(X)$ is the cardinality of the leg set of any Gideon Decomposition of X. It is well defined by Theorem 13.2(iii) and Proposition 3.23.

Proposition 13.9. (i) *$\gamma(X) = 0$ if and only if X is initial, and $\gamma(X) = 1$ if and only if X is connected.*
(ii) *If $\mathcal{C}$ has small coproducts and the X_i are Gideon-decomposable, then $\gamma(\coprod_i X_i) = \sum_i \gamma(X_i)$.*
(iii) *$\gamma(X) = |X|$ for a set X; $\gamma(X)$ is the number of connected components of a space whose components are open; and, in $\mathsf{Fel}(\mathcal{C})$, the Gideon number of a cat is its number of legs.*

Proof. (i) A legless Gideon Decomposition exhibits X as an empty coproduct; a unipedal one exhibits X as isomorphic to a connected object;

conversely an initial object has the legless Gideon Decomposition and a connected object X has $(\iota X, \mathrm{id})$. (ii) Coproducts of Gideon Decompositions are Gideon Decompositions, by Proposition 4.16(iii) and the associativity of coproducts. (iii) follows from Example 12.13. □

2. Recognition of categories of cats

Let $\mathcal{E}$ be a locally small category with small coproducts. Then $\mathcal{E}$ has an initial object (the empty coproduct). Recall that $\mathcal{E}^\circ$ is the full subcategory of connected objects and $y^\circ\colon \mathcal{E} \to \mathsf{PSh}(\mathcal{E}^\circ)$ the restricted Yoneda functor. As noted after Definition 12.12, the notions of $\mathcal{E}$-cat and of Gideon Decomposition in $\mathcal{E}$ make sense without assuming a terminal object, but there are no $\mathcal{E}$-cats unless $\mathcal{E}$ has one. Consequently, if some object of $\mathcal{E}$ admits a Gideon Decomposition, or is the vertex of a cocone from an $\mathcal{E}$-cat, then $\mathcal{E}$ has a terminal object.

THEOREM 13.10 (Recognition). *For a locally small category $\mathcal{E}$ the following are equivalent.*

(i) *$\mathcal{E} \simeq \mathsf{Fel}(\mathcal{D})$ for some category $\mathcal{D}$ with an initial and a terminal object.*
(ii) *$\mathcal{E}$ has small coproducts, $\mathcal{E}$ admits Gideon Decompositions, and $\mathcal{E}^\circ$ has an initial object.*
(iii) *$\mathcal{E}$ has small coproducts, $\mathcal{E}^\circ$ has an initial object, the restricted Yoneda functor $y^\circ\colon \mathcal{E} \to \mathsf{PSh}(\mathcal{E}^\circ)$ is conservative, and every object of $\mathcal{E}$ is the vertex of a Margaret compatible cocone from an $\mathcal{E}$-cat satisfying the Chloe Condition.*

In this case $\mathcal{D} \simeq \mathcal{E}^\circ$, and $\mathcal{E}$ has a terminal object.

PROOF. (i)⇒(ii). Each property in (ii) is invariant under equivalence, so we may take $\mathcal{E} = \mathsf{Fel}(\mathcal{D})$. It has small coproducts (Proposition 6.5) and the terminal object $\mathbf{1}$ (Proposition 6.2); it admits Gideon Decompositions (Example 12.13(v)); and $\mathsf{Fel}(\mathcal{D})^\circ$ is the full subcategory of unipedal cats (Proposition 7.15), which is equivalent to $\mathcal{D}$ via ι and so has the initial object $\mathbf{h}$.

(ii)⇒(i). Let $\mathcal{D} = \mathcal{E}^\circ$. By the remarks preceding the theorem, $\mathcal{E}$ has a terminal object, so Proposition 12.14 applies, and by Proposition 12.14(iv), $\Sigma\colon \mathsf{Fam}(\mathcal{D}) \to \mathcal{E}$ is an equivalence. By hypothesis $\mathcal{D}$ has an initial object $0_\mathcal{D}$. The terminal object of $\mathcal{E}$ corresponds to a terminal object (S, Z) of $\mathsf{Fam}(\mathcal{D})$. Since $\mathcal{D}$ has an initial object, $\mathsf{Fam}(\mathcal{D})((1, 0_\mathcal{D}), (S, Z)) \cong S$ as in Lemma 6.3, and this set is a singleton; so $S = 1$, and Z is terminal in $\mathcal{D}$ because $\mathcal{D}(W, Z) \cong \mathsf{Fam}(\mathcal{D})((1, W), (1, Z))$ is a singleton for every W. Thus $\mathcal{D}$ has an initial and a terminal object, and $\mathcal{E} \simeq \mathsf{Fam}(\mathcal{D}) \simeq \mathsf{Fel}(\mathcal{D})$ by Theorem 3.13.

(ii)⇒(iii). By Theorem 13.2(i), a Gideon Decomposition of X is a Margaret compatible cocone from a Chloe $\mathcal{E}$-cat. For conservativity: by Proposition 12.14, every object is isomorphic to ΣK for a Chloe cat K, and

$y^\circ(\Sigma K) \cong \mathbb{Y}^\circ K$ naturally in K, by connectedness of the objects of $\mathcal{E}^\circ$. By Propositions 8.17 and 12.14(i), y° is therefore fully faithful, and in particular conservative.

(iii)⇒(ii). As before, $\mathcal{E}$ has a terminal object. Let X be an object and $\beta\colon K \to \iota X$ a Margaret compatible cocone with K Chloe. Let $\beta^\flat\colon \Sigma K \to X$ be its transpose. For connected Y,

$$\coprod_\alpha \mathcal{E}(Y, L_\alpha) \xrightarrow{\kappa} \mathcal{E}(Y, \Sigma K) \xrightarrow{\beta^\flat \circ -} \mathcal{E}(Y, X)$$

composes to m_Y^β, which is bijective, and κ is bijective by connectedness of Y; hence $y^\circ(\beta^\flat)$ is an isomorphism. Since y° is conservative, $\beta^\flat$ is an isomorphism, and (K, β) is a Gideon Decomposition.

The final assertion follows from Theorem 7.17 or directly from the proof of (ii)⇒(i). □

Remark 13.11. The equivalence of (i) and (ii), with Fam in place of Fel and without the conditions on initial and terminal objects, is the well-known and elementary characterization of free coproduct completions: a category with small coproducts is a free coproduct completion if and only if every object is a coproduct of connected objects. We have given a complete proof and have not attempted to trace the statement to its origin. The additional conditions record the existence of a head (an initial connected object) and of a tail (a terminal object, which is built into the notion of a Gideon Decomposition).

Counterexample 13.12 (Independence of the conditions)**.** Each of the three conditions in Theorem 13.10(ii) is independent of the other two. In each example below, all conditions but the named one hold.

(i) *Small coproducts.* $\mathcal{E} = \mathsf{Kit}(\mathcal{C})$. The leg functor $\pi\colon \mathsf{Kit}(\mathcal{C}) \to \mathbf{FinSet}$ is left adjoint to Δ_1, so it preserves every coproduct that exists in $\mathsf{Kit}(\mathcal{C})$. Hence a family of kittens has a coproduct in $\mathsf{Kit}(\mathcal{C})$ only if all but finitely many of its members are legless, and in that case the coproduct of Proposition 6.5 is a kitten and is a coproduct in $\mathsf{Kit}(\mathcal{C})$. In particular $\mathsf{Kit}(\mathcal{C})$ has finite coproducts but lacks the coproduct of countably many copies of $\mathbf{1}$, which would be sent to a coproduct of countably many copies of 1 in **FinSet**. With coproducts so described, the proof of Proposition 7.15 shows that the connected objects of $\mathsf{Kit}(\mathcal{C})$ are the unipedal kittens, which have the initial object $\mathbf{h}$; and every kitten is a finite coproduct of unipedal kittens, so $\mathsf{Kit}(\mathcal{C})$ admits Gideon Decompositions.

(ii) *Initial connected object.* $\mathcal{E} = \mathbf{Gph}$. It has small coproducts, the terminal object consisting of one vertex with a loop, and Gideon Decompositions. A connected graph G_0 admitting a morphism to the one-vertex graph without loop has no edges, hence is a single vertex, and then it admits two morphisms to the connected graph with two vertices and one edge. So $\mathbf{Gph}^\circ$ has no initial object.

(iii) *Gideon Decompositions.* $\mathcal{E} = \mathsf{Fel}(\mathcal{D}) \times [1]$, where $[1] = \{0 < 1\}$ and $\mathcal{D}$ has initial and terminal objects. Coproducts are computed componentwise (in $[1]$ they are suprema), and $(\mathbf{1}, 1)$ is terminal. One checks from the definitions that the connected objects are the objects isomorphic to $(\iota X, 0)$: for these, $\mathcal{E}((\iota X, 0), \coprod_i (K_i, e_i)) \cong \mathsf{Fel}(\mathcal{D})(\iota X, \coprod_i K_i) \cong \coprod_i \mathcal{E}((\iota X, 0), (K_i, e_i))$; if K is not legless, the identity of $(K, 1) \cong (K, 0) \sqcup (\mathbf{0}, 1)$ factors through neither coprojection, because $\mathcal{E}((K, 1), (K, 0)) = \varnothing$ (as $1 \not\leq 0$) and $\mathcal{E}((K, 1), (\mathbf{0}, 1)) = \varnothing$; the object $(\mathbf{0}, 1)$ fails connectedness for the family $((\mathbf{0}, 1), (\mathbf{0}, 1))$, whose coproduct is $(\mathbf{0}, 1)$ and which yields two elements on the left of (2.1) but one on the right; and since $\mathcal{E}((K, 0), (M, e)) = \mathsf{Fel}(\mathcal{D})(K, M)$ for every (M, e), an object $(K, 0)$ is connected in $\mathcal{E}$ if and only if K is connected in $\mathsf{Fel}(\mathcal{D})$, that is, unipedal (Proposition 7.15). Thus $\mathcal{E}^\circ \simeq \mathcal{D}$ has the initial object $(\mathbf{h}, 0)$. But $(K, 1)$ is not a coproduct of connected objects, since such coproducts have second coordinate 0.

The last example also shows that conservativity in Theorem 13.10(iii) cannot be omitted: the cocone from the Chloe $\mathcal{E}$-cat $\langle(\iota L_\alpha, 0)\rangle_\alpha$ to $(K, 1)$, with components $(j_\alpha, 0 \leq 1)$, is Margaret compatible, but it is not a Gideon Decomposition, and the morphism $(K, 0) \to (K, 1)$ is inverted by y° without being an isomorphism.

Remark 13.13. The category $\mathsf{Fam}(\mathbb{N}, \leq)$ has small coproducts, every object is a coproduct of connected objects (one-element families), and its connected objects form a category equivalent to $(\mathbb{N}, \leq)$, which has an initial object. It is nevertheless not equivalent to any $\mathsf{Fel}(\mathcal{D})$, since it has no terminal object. It does not contradict Theorem 13.10, because, having no terminal object, it has no cats and hence no Gideon Decompositions. This is one point at which the requirement that a cat have a tail affects the formulation of a theorem.

3. CMG and the tensor product

Proposition 13.14. *Let $\mathcal{C}$ be monoidal, with $\otimes$ preserving those small coproducts that exist in each variable, and suppose that the tensor product of two connected objects is connected. If (K, β) and (K', β') are Gideon Decompositions of X and X', then $K \otimes K'$, with the cocone $(\beta_\alpha \otimes \beta'_{\alpha'})_{(\alpha,\alpha')}$, is a Gideon Decomposition of $X \otimes X'$. Consequently $\mathcal{C}^{\mathrm{G}}$ is closed under $\otimes$, and $\gamma(X \otimes X') = \gamma(X)\gamma(X')$.*

Proof. $K \otimes K'$ is Chloe by hypothesis. Preservation of coproducts in each variable gives $X \otimes X' \cong (\coprod_\alpha L_\alpha) \otimes X' \cong \coprod_\alpha (L_\alpha \otimes \coprod_{\alpha'} L'_{\alpha'}) \cong \coprod_{\alpha,\alpha'} L_\alpha \otimes L'_{\alpha'}$, with coprojections $\beta_\alpha \otimes \beta'_{\alpha'}$. □

Example 13.15. The hypotheses of Proposition 13.14 hold for $(\mathbf{Set}, \times)$ and for $(\mathbf{Top}, \times)$, products of connected spaces being connected and products distributing over disjoint unions. They fail for $(\mathbf{Gph}, \times)$: the product of the connected graph K_2 (two vertices joined by an edge, no loops) with itself

has vertices (i, j) with $(i, j) \sim (i', j')$ if and only if $i \neq i'$ and $j \neq j'$, and so consists of the two disjoint edges $\{(0, 0), (1, 1)\}$ and $\{(0, 1), (1, 0)\}$. Thus $\gamma(K_2 \times K_2) = 2 \neq 1 = \gamma(K_2)^2$, and the conclusion of Proposition 13.14 fails, although $\times$ distributes over coproducts in **Gph**.

CHAPTER 14

Concluding remarks

1. Summary

We have defined $\mathcal{C}$-cats for a category $\mathcal{C}$ with an initial object, defined their morphisms by commutative diagrams, and developed the resulting theory through the topics customary in a monograph on a new categorical structure: subclasses of objects, functors and natural transformations, limits and colimits, adjunctions, Yoneda-type theorems, monoidal and enriched structure, fibrations, higher-dimensional variants, and representation and decomposition theorems. We have also introduced the Chloe Condition, Margaret Compatibility and Gideon Decompositions, and proved the CMG Correspondence Theorem and a recognition theorem for categories of cats.

Every result in the theory has been proved, and we know of no inconsistency in it. It is also, in a precise sense, empty. By the Automatic Commutativity Proposition (Proposition 3.9), the diagrams that define a morphism of cats impose no conditions, and by the Fundamental Theorem (Theorem 3.13), the category of cats is equivalent to the category of families. Every subsequent result is therefore a statement about families, and in most cases we have identified the classical statement in question; the remaining results (for instance Theorem 12.9, Proposition 12.10, and Corollary 11.6) are elementary consequences of the same identification.

The head and the tail of a particular cat never carry information. What does have a visible effect is the existence, or the relation between, the initial and terminal objects of $\mathcal{C}$. The existence of an initial object makes the leg functor representable by the cephalic cat (Lemma 6.3), with consequences for limits and monomorphisms (Theorem 6.8 and Proposition 6.18) that fail for headless cats (Counterexamples 6.13 and 6.19). The existence of a terminal object makes the leg functor a left adjoint, which is used in the analysis of colimits (Theorem 6.14), and is built into the notion of a Gideon Decomposition (Remark 13.13). Whether the initial and terminal objects coincide governs the length of the feline adjoint string (Propositions 7.5, 7.6 and 7.9) and the existence of natural transformations from legs to points (Proposition 8.15).

2. Points at which the theory is not trivial

For the reader who wishes to know which of the results above required any thought, we list them. None is new.

(i) The necessity halves of Theorems 6.8 and 6.14, which use the representability of the leg functor and its left adjointness respectively.
(ii) The equivalence $\mathsf{Fel}(\mathbf{Set}^{\mathcal{J}}) \simeq \mathbf{Set}^{\mathcal{J}^{\triangleright}}$ of Theorem 7.21 and its consequences, in particular that $\mathsf{Fel}(\mathsf{PSh}(\mathcal{D}))$ is a presheaf topos.
(iii) The direct proof that $\mathsf{Fel}(\mathcal{E})$ is a topos (Theorem 10.27), whose subobject classifier has two legs.
(iv) The adjoint string of Theorem 7.3 and the precise conditions under which it extends (Propositions 7.5, 7.6 and 7.13).
(v) The behaviour of the additive enrichment (Theorem 9.18), in which a leg equal to the head can be removed without changing the isomorphism type of the cat.
(vi) The counterexample $\mathsf{Fel}(\mathcal{D}) \times [1]$ of Counterexample 13.12, which shows that the Gideon condition in the recognition theorem is independent of the others.

3. Questions not pursued

We list some directions that we have not pursued. We do not regard any of them as important, and we expect each to be routine.

(i) Cats in an $(\infty, 1)$-category (Remark 11.20). We expect the ∞-category of such cats to be equivalent to that of set-indexed families, by the same argument as Theorem 3.13, with contractible spaces in place of singletons.
(ii) Enrichment over a general monoidal base $\mathcal{V}$ (Remark 9.20). The additive case shows that the answer depends on how $\otimes$ interacts with coproducts in $\mathcal{V}$, and that it is not obtained by enriching $\mathsf{Fel}(\mathcal{C})$.
(iii) Head-absorbing morphisms (Remark 10.12) and retrograde cats (Remark 7.26). The latter are objects of the free product completion of $\mathcal{C}$, and their theory is the dual of the theory of $\mathsf{Fel}(\mathcal{C}^{\mathrm{op}})$.
(iv) Gaited cats with structured gaits, for instance gaits that are posets or groupoids. By Proposition 11.9, these are diagrams of the relevant shapes.

4. A final remark

The reader may ask what has been gained by recording the head and the tail at all. Formally, nothing: Theorem 4.25 shows that, up to equivalence, removing them changes the category of cats only by deciding whether it is empty.

APPENDIX A

Dictionary

The following table records, for each feline notion, the classical notion to which it is equivalent or of which it is an instance. Here $\mathcal{C}$ has an initial and a terminal object, and "family" means small family of objects of $\mathcal{C}$.

Feline term	**Classical counterpart**
$\mathcal{C}$-cat (Definition 3.2)	a family, together with a choice of initial and terminal object
head, tail	a chosen initial, terminal object; no information
shoulders, hips, spine	the unique morphisms out of an initial object and into a terminal object
legs, leg set	the members of the family and its index set
gait of a morphism	the reindexing function of a morphism of families
$\mathsf{Fel}(\mathcal{C})$	$\mathsf{Fam}(\mathcal{C})$, the free completion of $\mathcal{C}$ under small coproducts (Theorem 3.13)
normalized cat	an object of $\mathsf{Fam}(\mathcal{C})$ (Proposition 3.20)
feline shape $\mathfrak{F}_A$	the poset $(A^{\triangleleft})^{\triangleright}$
kitten	a finite family
litter	a family of finite families
legless cat $\mathbf{0}$	the empty family; the initial object of $\mathsf{Fam}(\mathcal{C})$
cephalic cat $\mathbf{h}$	the one-member family (0); representing object of the leg functor
caudal cat $\mathbf{1}$	the one-member family (1); the terminal object of $\mathsf{Fam}(\mathcal{C})$
headless, tailless cat	a family, with a chosen terminal or initial object
one-legged inclusion ι	the unit $\mathcal{C} \to \mathsf{Fam}(\mathcal{C})$ of the coproduct completion
leg functor π	the family fibration $\mathsf{Fam}(\mathcal{C}) \to \mathbf{Set}$
reindexing morphism	cartesian morphism of the family fibration (Theorem 10.3)

Feline term	**Classical counterpart**
Δ_0, Δ_1	sections of the family fibration by fibrewise initial and terminal objects
points functor Γ	the global-sections functor $\mathsf{Fam}(\mathcal{C})(\mathbf{1}, -)$
body Σ	the coproduct of the family; left adjoint of ι
felinization $\mathsf{Fel}(F)$	$\mathsf{Fam}(F)$
fibred feline functor	fibred functor between family fibrations (Theorem 5.9)
feline Yoneda functor $\mathbb{Y}$	the embedding of $\mathsf{Fam}(\mathcal{C})$ in $\mathsf{PSh}(\mathcal{C})$ as coproducts of representables
feline presheaf	a presheaf on $\mathsf{Fam}(\mathcal{C})$ sending coproducts to products
monoidal cat	graded monoid (Theorem 9.10)
self-enrichment	the leg functor, regarded as an identity-on-objects functor
$\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$	column-finite matrices over $\mathcal{C}$
tail-absorbing morphism	Kleisli morphism for the monad $- \sqcup \mathbf{1}$
cat of cats, flattening	a family of families; multiplication of the family pseudomonad
gaited cat	a diagram in $\mathcal{C}$; an object of the category of diagrams
two-dimensional cat	a family in a 2-category, up to biequivalence
$\mathsf{Fel}(\mathbf{Set})$	the arrow category $\mathbf{Set}^{\to}$ (Corollary 7.22)
$\mathsf{Fel}(\mathsf{PSh}(\mathcal{D}))$	$\mathsf{PSh}(\mathcal{D}^{\triangleleft})$ (Corollary 7.23)
$\mathsf{Fel}^n(\mathbb{1})$	$\mathbf{Set}^{[n-1]}$ (Theorem 11.4)
census, automorphism group	multiplicity functions; products of wreath products
kitten rig	the monoid semiring $\mathbb{N}[\mathsf{Iso}(\mathcal{C})]$
Chloe Condition	every leg is a connected object
Margaret Compatibility	the induced map of presheaves on connected objects is invertible
Gideon Decomposition	decomposition into connected components
Gideon number	number of connected components
CMG Correspondence Theorem	uniqueness of decomposition into connected components; detection of such decompositions by connected objects
Recognition Theorem	characterization of free coproduct completions

APPENDIX B

Index of notation

Symbol	Meaning	Ref.
$K = (H, T, (L_\alpha)_{\alpha\in A})$	a $\mathcal{C}$-cat with head, tail and legs	3.2
h_α, t_α, σ	shoulders, hips, spine	3.2
$\mathfrak{F}_A$	feline shape on A	3.5
(φ, F)	a feline morphism with gait φ	3.10
$\mathsf{Fel}(\mathcal{C})$	category of $\mathcal{C}$-cats	3.8
$\mathsf{Fam}(\mathcal{C})$	category of small families	2.12
$\mathsf{Fel}^{\mathrm{nrm}}(\mathcal{C})$	normalized cats	3.19
$\langle L_\alpha\rangle_{\alpha\in A}$	the normalized cat with legs L_α	3.21
ι	one-legged inclusion	3.21
$\mathbf{0}$, $\mathbf{h}$, $\mathbf{1}$	legless, cephalic and caudal cats	3.21
j_α	leg inclusion	3.21
π	leg functor	3.21
$\mathsf{Fel}_n(\mathcal{C})$	n-legged cats	4.6
$\mathsf{Kit}(\mathcal{C})$	kittens	4.8
$\mathsf{Fel}^{\mathrm{hl}}(\mathcal{C})$, $\mathsf{Fel}^{\mathrm{tl}}(\mathcal{C})$	headless and tailless cats	4.19, 4.22
$\mathsf{Fel}(F)$, $\mathsf{Fel}(\tau)$	felinization	5.2
Δ_X	$S \mapsto \langle X\rangle_{s\in S}$	5.4
$\mathsf{FelFun}(\mathcal{C}, \mathcal{D})$	fibred feline functors	5.7
$\Phi^\natural$	unipedal restriction	5.8
Γ	points functor	7.2
Σ	body functor	7.10
$\mathbb{Y}$, $\mathbb{Y}^\circ$	feline Yoneda functors	8.2, 8.16
$\mathsf{PSh}_\sqcup(\mathsf{Fel}(\mathcal{C}))$	feline presheaves	8.11
$K \otimes K'$, $\mathbf{I}$	feline tensor product and unit	9.2
$[K, K']$	self-enriched hom	9.13
$\mathsf{Fel}_{\mathbf{Ab}}(\mathcal{C})$	additively enriched cats	9.16
$\varphi^* K'$, $\varphi_! K$	reindexing, opreindexing	10.2, 10.6
$\mathsf{Fel}^{\mathrm{ta}}(\mathcal{C})$	tail-absorbing morphisms	10.8
m_K	multiplicity function	10.18
$\mathcal{R}(\mathcal{C})$	kitten rig	10.22
Ω_{Fel}	feline subobject classifier	10.26

Symbol	Meaning	Ref.
$\mathsf{Fel}^n(\mathcal{C})$	iterated cats	1
$\mathsf{Fel}_{\mathrm{gait}}(\mathcal{C})$	gaited cats	11.8
$\mathsf{Fel}_2(\mathcal{K})$	two-dimensional cats	11.16
$\mathcal{C}^\circ$	connected objects	2.9
$\mathsf{Fel}^{\mathrm{Ch}}(\mathcal{C})$	cats satisfying the Chloe Condition	4.15
m_Y^β	Margaret comparison map	8.19
$\mathsf{Gid}(X)$, $\mathcal{C}^{\mathrm{G}}$	Gideon Decompositions; decomposable objects	12.12
$\gamma(X)$	Gideon number	13.8

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Note on references. The works listed are general sources for the classical material on which this monograph depends. None of them is concerned with cats in the sense of Definition 3.2, and no result of this monograph is attributed to them except where the text says so explicitly.